

claude-windows / d8d334d9-032d-4216-9bb2-1e940c8d1de0

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\d8d334d9-032d-4216-9bb2-1e940c8d1de0.jsonl`
- SHA-256: `a700614eaad711d33b1994329209f9042a0b199f425f0610c939caad52b2029c`
- Source modified: `2026-06-29T14:03:55+00:00`
- Imported at: `2026-07-05T16:48:19+00:00`
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`
- session_id: `d8d334d9-032d-4216-9bb2-1e940c8d1de0`
```

Transcript

1. user (2026-06-22T12:15:43.027Z)

? START HERE ? Golden Regression Fixtures (optional next items)

****You are here:**** the track is ****implementation-COMPLETE**** (8 PRs: `#186 #189 #191 #192 #193 #194 #314 #317`). A clean clone runs rally-seg regression ? synthetic ****+ 6 de-attributed real fixtures `fx\_r01?r06`**** ? with no video/GPU/DB, cross-OS enforced, leak-guarded, audited. Tracker `M1·M2·M3·MX·P1·P2·D0C·P0·P3` = all ?.

****Ramp-up kit (read in order):**** `docs/GOLDEN\_REGRESSION\_FIXTURES.md` (\$7 tracker = source of truth) ? `backend/eval/golden\_fixtures.py` + `golden\_real\_fixtures.py` ? `tests/test\_golden\_fixtures.py` + `test\_golden\_real\_fixtures.py` ? `eval\_baselines/fixtures/` ? `ci.yml` (`golden-crossos` + `leak-scan` jobs) ? `tools/leak\_scan.py`.

Next steps ? prioritized (all optional, none blocking)

1. ****M4 ? quality-floor gate**** *(recommended first; now meaningful on real data).* Assert per-fixture + mean ****F1 ? a committed floor**** in `eval\_baselines/regression\_baseline.json` (reserved-but-absent path). ***First move:*** the F1 is already frozen in each `fx\_\*/expected.json` 'score' ? seed the per-slug floor from it (the real fixtures sit at ****0.0?0.59****, so floor = 'current ? tolerance', not a global bar). ****Gotchas:**** include a ****windowing fingerprint**** in the floor record (or a SERVED retune silently passes), keep it ****CONTROL-only**** (learned/numpy-BLAS = non-deterministic across CPUs), add it to the `golden-crossos` matrix.
2. ****Deferred video-name rename**** *(do before any open-sourcing).* Rename the functional names (`eval\_baselines/\*.json` keys, `rally\_seg\_eval.MULTICOURT\_CLIPS`, quality logs) ? opaque ids. ***Why it's hard:*** the keys feed `training/gen0/harness.py`'s LOVO floor, matched to the ****gitignored `output/human\_lovo\_manifest.json` + data files**** ? rename them together or you break the floor; also collides with the hot quality track. ***First move:*** put the names in a gitignored `leakscan\_private`, run `python -m tools.leak\_scan --names .leakscan\_private` to enumerate every occurrence, then plan the coordinated rename.
3. ****01 protobuf / 02 Tier-1 key-gated**** ? only if their specific need (byte-stable snapshots / person-stream sharing) arises.

Locked decisions ? don't relitigate

Option **** (a) **** = commit the de-attributed ****per-frame**** `trajectory.csv` (non-attributability rests on the source staying unpublished). ****CONTROL-only**** cross-machine gate. ****Pinned**** windowing 3-tuple per fixture (never the named `SERVED`). ****Parse-compare****, never byte. P0 = safe-scope only.

Facts to carry

- ****Golden data:**** `?\Annotation Setup\Collect\{Trajectories,Golden Labelled}\` + repo `output\\*.rallies.csv` (manifest = 15 videos); F:\ has source videos.
- ****Off-git `fx\_` map**** (keep in `leakscan\_private`): r01=mahadevpura_single, r02=kushagra, r03=mahadevpura_1, r04=largetest_doubles, r05=GX010128, r06=Boxhill_Doubles.
- ****Shared checkout:**** branch from `origin/master` in an isolated worktree; the `gh` "master used by worktree" merge error is benign. Repo is on the ****Khelsutra**** org (CI bills there).

```
DIFF: backend/worker/__main__.py ==
diff --git a/backend/worker/__main__.py b/backend/worker/__main__.py
index 081d8f5..08d8d84 100644
--- a/backend/worker/__main__.py
+++ b/backend/worker/__main__.py
@@ -20,6 +20,7 @@ import csv
import json
import logging
import os
+import re
import sys
import tempfile
from typing import Any, List
@@ -264,6 +265,29 @@ def cmd_infer(args: argparse.Namespace) -> int:
#: per-video sidecar ? .srt/.json/thumbnail ? can't shadow the real video on a stem collision).
_VIDEO_EXTS = ("mp4", "mov", "mkv", ".avi", ".webm", ".m4v")

+ #: Leading ISO-date prefix on a canonical label name (`labels/<YYYY-MM-DD>_<name>.rallies.csv`,
+ #: docs/DATA_IN_GCS.md §7.1). It disambiguates recycled GoPro names by authoring date but is NOT
+ #: part of the clip's join key, so it's stripped when deriving the videos/ stem.
+_LABEL_DATE_PREFIX = re.compile(r"^\\d{4}-\\d{2}-\\d{2}_")
+
+def _label_clip_name(fname: str) -> str:
+    """Derive the clip name (the `videos/<name>.<ext>` join key) from a label filename.
+
+    Strips both the `.rallies.csv` / `.csv` suffix and an optional leading ISO-date prefix,
so a
+    canonical dated label and a legacy bare label resolve to the SAME bare clip stem:
+
+    * `2026-06-22_GX020094.rallies.csv` -> `GX020094` (canonical, date-stamped)
+    * `GX020094.rallies.csv` -> `GX020094` (legacy bare ? same stem)
+    * `GX020094_proxy.rallies.csv` -> `GX020094_proxy` (`_proxy` is part of the
name, kept)
+    * `2026-06-21_mahadevpura_1.rallies.csv` -> `mahadevpura_1` (underscores in <name>
survive)
+
+    Without the date strip, `--trajectory-source detector` can't match a dated label to its
bare
+    video stem, so every clip is skipped and train aborts with `<2 videos` (DATA_IN_GCS §7b
#1).
+
+    """
+    name = fname.replace(".rallies.csv", "").replace(".csv", "")
+    return _LABEL_DATE_PREFIX.sub("", name)
+
def _probe_video_fps_width(path: str):
    """Robust ffprobe (fps, frame_width) for a real video. Returns None on ANY failure.
@@ -380,8 +404,8 @@ def cmd_train(args: argparse.Namespace) -> int:
print(f"[worker] trajectory source: {args.trajectory_source}")
specs: List[dict] = []
for uri in label uris:
```

```

-         fname = storage.parse_uri(uri).name                    # e.g.
GX020094_proxy.rallies.csv
-         name = fname.replace(".rallies.csv", "").replace(".csv", "")
+         fname = storage.parse_uri(uri).name                    # e.g.
2026-06-22_GX020094.rallies.csv
+         name = _label_clip_name(fname)                        # -> GX020094 (date prefix
stripped)
        local_label = storage.fetch(uri, os.path.join(labels_dir, fname), config=storage_cfg)
        if args.trajectory_source == "detector":
            vuri = video_by_stem.get(name)
=== DIFF: tests/test_worker.py ===
diff --git a/tests/test_worker.py b/tests/test_worker.py
index 642e46a..c48fa02 100644
--- a/tests/test_worker.py
+++ b/tests/test_worker.py
@@ -191,6 +191,66 @@ def test_cmd_infer_validation_failure_returns_2(tmp_path, monkeypatch):

    # --- cmd_train: pull labels -> synth trajectories -> manifest -> harness -> push bundle ---

+@pytest.mark.parametrize("fname,expected", [
+    ("2026-06-22_GX020094.rallies.csv", "GX020094"),            # canonical dated label
+    (DATA_IN_GCS $7.1)
+    ("GX020094.rallies.csv", "GX020094"),                       # legacy bare label -> SAME
+    clip_stem
+    ("GX020094_proxy.rallies.csv", "GX020094_proxy"),          # _proxy is part of the name,
+    kept
+    ("2026-06-21_mahadevpura_1.rallies.csv", "mahadevpura_1"), # underscores in <name> survive
+    ("v1.csv", "v1"),                                           # bare .csv suffix
+    ("2026-06-22_GX020094.csv", "GX020094"),                   # dated + bare .csv
+])
+def test_label_clip_name_strips_date_prefix(fname, expected):
+    """A dated label and a bare label must both resolve to the bare videos/<name> stem."""
+    assert wmain._label_clip_name(fname) == expected
+
+def test_cmd_train_detector_joins_dated_labels_to_bare_videos(tmp_path, monkeypatch):
+    """Regression (DATA_IN_GCS $7b blocker #1): canonical labels carry an ISO-date prefix
+    (labels/<date>_<name>.rallies.csv) while videos are bare (videos/<name>.mp4). The label-
+    derived
+    clip name must strip the date so the video join succeeds ? otherwise every clip is skipped
+    and
+    train aborts with <2 videos."""
+    import os
+    import subprocess

+    monkeypatch.setenv("RALLY_DETECTOR_IMPL", "stub") # CPU stub runner (no GPU/WSL)
+    monkeypatch.setattr(wmain, "_probe_video_fps_width", lambda p: (30.0, 1280.0))
+    corpus = tmp_path / "corpus"
+    labels_dir = corpus / "labels"
+    videos_dir = corpus / "videos"
+    labels_dir.mkdir(parents=True)
+    videos_dir.mkdir(parents=True)
+    header = "rally_number,start_time,end_time,ending_reason,sport,shots_count\n"
+    # canonical DATED labels ...
+    (labels_dir / "2026-06-22_GX020094.rallies.csv").write_text(header +
"1,2.0,5.0,winner,badminton,\n")
+    (labels_dir / "2026-06-21_mahadevpura_1.rallies.csv").write_text(header +
"1,3.0,6.0,winner,badminton,\n")
+    # ... joining to BARE-stem videos
+    (videos_dir / "GX020094.mp4").write_bytes(b"FAKEVIDEO1")
+    (videos_dir / "mahadevpura_1.mp4").write_bytes(b"FAKEVIDEO2")
+    out_dir = tmp_path / "out"
+    scratch = tmp_path / "scratch"
+    cfg = tmp_path / "config.json"
+    cfg.write_text("{}")

```

```

+
+ def fake_run(cmd, *a, **k):
+     out = cmd[cmd.index("--out") + 1]
+     ver = cmd[cmd.index("--version") + 1]
+     bdir = os.path.join(out, ver)
+     os.makedirs(bdir, exist_ok=True)
+     for fn in ("weights.json", "manifest.json"):
+         with open(os.path.join(bdir, fn), "w") as f:
+             f.write("{}")
+     return subprocess.CompletedProcess(cmd, 0)
+ monkeypatch.setattr(subprocess, "run", fake_run)
+
+ rc = wmain.main(["train", "--corpus-uri", str(corpus), "--out-uri", str(out_dir),
+                 "--version", "0.0.4", "--config", str(cfg), "--scratch", str(scratch),
+                 "--trajectory-source", "detector", "--segmenter", "wasb_hybrid"])
+ assert rc == 0 # before the fix: every clip skipped -> rc 2 ("<2 videos")
+ specs = json.loads((scratch / "manifest.json").read_text())
+ assert len(specs) == 2 # both dated labels joined
+ assert {s["name"] for s in specs} == {"GX020094", "mahadevpura_1"} # downstream name is
date-stripped
+
+
+ def test_cmd_train_orchestration(tmp_path, monkeypatch):
+     import os
+     import subprocess

```

4. user (2026-06-22T12:16:06.187Z)

On branch fix/worker-train-dated-label-join
Your branch is up to date with 'origin/master'.

Changes not staged for commit:

```

(use "git add <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
    modified:   backend/worker/__main__.py
    modified:   tests/test_worker.py

```

Untracked files:

```

(use "git add <file>..." to include in what will be committed)
    artifacts/
    docs/KHELSUTRA_PRODUCT_OVERVIEW.md

```

no changes added to commit (use "git add" and/or "git commit -a")

=== BRANCH/HEAD ===

fix/worker-train-dated-label-join

d301a7f fix(nightly): GCP-L4-validated fixes ? gs:// fetch dst, WASB env, weights path, pure-engine (#318)

fa2342c feat(nightly): E2E engine reel-count regression on a GCP GPU VM (emailed) + cost (#315)

7a40434 feat(eval): golden fixtures P3 ? 6 curated, trimmed, de-attributed REAL Tier-0 fixtures

+ --max-frames trim (#317)

5c2eec8 chore(security): P0 (safe scope) ? scrub personal paths from code + add leak-scan CI guard (#314)

689c738 docs(pkg): mark P2c ? + the loop closed ? v1 LIGHT is feature-complete (#313)

5. assistant (2026-06-22T12:16:17.756Z)

The in-flight work is clean and self-contained: a fix in

[`backend/worker/\_\_main\_\_.py`](backend/worker/__main__.py) adding a `\_label\_clip\_name()` helper that strips the ISO-date prefix from canonical label filenames so `train --trajectory-source detector` can join dated labels (`labels/2026-06-22\_GX020094.rallies.csv`) to bare videos (`videos/GX020094.mp4`) ? with a parametrized unit test and a full regression test. The branch has **no commits yet** (working-tree only). Let me check the two untracked items to see if they belong with this change.

6. user (2026-06-22T12:16:20.457Z)

```
=== artifacts/ ===
total 1084
drwxr-xr-x 1 avidu 197609      0 Jun 20 23:51 .
drwxr-xr-x 1 avidu 197609      0 Jun 22 17:44 ..
-rw-r--r-- 1 avidu 197609 16460 Jun 20 14:42 appliance_design.json
drwxr-xr-x 1 avidu 197609      0 Jun 20 00:44 landing
-rw-r--r-- 1 avidu 197609 18149 Jun 18 13:25 long_rally_lens_n15.json
-rw-r--r-- 1 avidu 197609  3870 Jun 18 13:25 long_rally_lens_n15.txt
drwxr-xr-x 1 avidu 197609      0 Jun 20 01:12 multishuttle_142
-rw-r--r-- 1 avidu 197609  5602 Jun 19 22:23 pr245_code_review.md
-rw-r--r-- 1 avidu 197609  4028 Jun 19 22:23 pr245_fixes_plan.md
-rw-r--r-- 1 avidu 197609  8520 Jun 19 22:25 pr245_review.json
-rw-r--r-- 1 avidu 197609  5909 Jun 19 22:22 pr245_review.md
drwxr-xr-x 1 avidu 197609      0 Jun 21 06:33 promo
drwxr-xr-x 1 avidu 197609      0 Jun 10 15:54 rally_tal_gen0
drwxr-xr-x 1 avidu 197609      0 Jun 20 23:52 review
drwxr-xr-x 1 avidu 197609      0 Jun 20 07:59 site_deploy
-rw-r--r-- 1 avidu 197609 15148 Jun 20 14:40 so.json
-rw-r--r-- 1 avidu 197609 15485 Jun 20 14:40 so.py
-rw-r--r-- 1 avidu 197609 912462 Jun 16 08:50 tridet.pdf
-rw-r--r-- 1 avidu 197609 46340 Jun 16 08:50 tridet.txt
```

```
=== git check-ignore artifacts ===
```

```
.gitignore:68: artifacts/
```

```
=== docs/KHELSUTRA_PRODUCT_OVERVIEW.md (head) ===
```

KhelSutra ? Product Overview & Early-User Guide

What it does today, how to get the best out of it, and where it's going.

For field associates and for invited alpha / beta academies, coaches, and players.

In one line

****KhelSutra turns ordinary badminton video into useful pieces ? automatically.**** Put a phone or GoPro

on a tripod behind the court, record a session, and KhelSutra automatically finds the rallies, stitches

a ****highlight reel****, and gives you a ****simple summary**** (how many rallies, how much real play time).

No special hardware, no editing, no manual scrubbing.

The problem we're solving

Academies and serious players already record a lot of video ? and then almost never watch it.

Hours of

footage pile up on phones and memory cards because cutting it into useful clips by hand is tedious and

nobody has the time. The good rallies, the moments worth reviewing, stay buried.

Most academy halls also run ****several courts at once****, so a single recording is a busy, noisy scene

that ordinary video tools choke on. ****Pulling the real rallies out of busy, multi-court footage is the**

hard problem KhelSutra is built to crack** ? it already works very well on clean, single-court recordings, and the multi-court case is exactly what we're sharpening now.

Who it's for

```
...
203 docs/KHELSUTRA_PRODUCT_OVERVIEW.md
```

7. assistant (2026-06-22T12:16:50.394Z)

```
`artifacts/` is gitignored scratch (not part of any commit).
`docs/KHELSUTRA_PRODUCT_OVERVIEW.md` is a separate untracked 203-line product doc ? unrelated to
this branch.
```

So the actual in-flight change is exactly the worker date-prefix-strip fix + its tests. No user task is pending, so rather than guess, let me verify the fix is sound and green (non-destructive), then I'll lay out the options. Running the worker tests, lint, and type-check now:

8. user (2026-06-22T12:16:55.831Z)

```
..... [100%]
26 passed in 2.04s
```

9. user (2026-06-22T12:16:59.875Z)

```
=== ruff ===
All checks passed!
=== mypy ===
backend\sports\__init__.py:7: note: By default the bodies of untyped functions are not checked,
consider using --check-untyped-defs [annotation-unchecked]
backend\pipeline\validators\base.py:36: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend\providers\__init__.py:7: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 1 source file
```

10. assistant (2026-06-22T12:17:05.118Z)

Tests pass (26), ruff clean, mypy clean. Let me confirm the fix's logic against the surrounding code ? specifically that `video\_by\_stem` is keyed by bare stems so the date-stripped name actually joins.

11. user (2026-06-22T12:17:05.917Z)

```
380         os.makedirs(traj_dir, exist_ok=True)
381
382         corpus = args.corpus_uri.rstrip("/")
383         label_uris = sorted(u for u in storage.list_uris(f"{corpus}/labels/",
config=storage_cfg)
384                             if u.lower().endswith(".csv"))
385         if len(label_uris) < 2:
386             print(f"[worker] need >=2 labeled videos for leave-one-video-out; found
{len(label_uris)} "
387                   f"under {corpus}/labels/")
388             return 2
389
390         # Real-detector source: resolve the runner once + index the bucket's videos/ by
stem.
391         runner = None
392         video_by_stem: dict = {}
393         if args.trajectory_source == "detector":
394             os.makedirs(videos_dir, exist_ok=True)
395             runner = _resolve_train_detector(args, config)
396             if runner is None:
397                 return 2
398             for vuri in storage.list_uris(f"{corpus}/videos/", config=storage_cfg):
399                 vname = storage.parse_uri(vuri).name
400                 if not vname.lower().endswith(_VIDEO_EXTS):
401                     continue # skip sidecars (.srt/.json/thumbnails) so they can't shadow
```

```

the video
402         video_by_stem[os.path.splitext(vname)[0]] = vuri
403
404     print(f"[worker] trajectory source: {args.trajectory_source}")
405     specs: List[dict] = []
406     for uri in label_uris:
407         fname = storage.parse_uri(uri).name           # e.g.
2026-06-22_GX020094.rallies.csv
408         name = _label_clip_name(fname)               # -> GX020094 (date
prefix stripped)
409         local_label = storage.fetch(uri, os.path.join(labels_dir, fname),
config=storage_cfg)
410         if args.trajectory_source == "detector":
411             vuri = video_by_stem.get(name)
412             if not vuri:
413                 print(f"[worker] no video for {name!r} under {corpus}/videos/ ?
skipping.")
414                 continue
415             local_video = storage.fetch(vuri, os.path.join(videos_dir,
storage.parse_uri(vuri).name),
416                                     config=storage_cfg)
417             video_id = compute_video_id(local_video)
418             # Real fps/width drive the trajectory frame->time mapping; PROBE (don't
guess) ? a
419             # wrong fps silently mis-aligns every label, so skip the video if ffprobe
can't read it.
420             meta = _probe_video_fps_width(local_video)
421             if meta is None:
422                 print(f"[worker] could not probe fps/width for {name!r} (ffprobe) ?
skipping (won't guess).")
423                 continue
424             fps, frame_width = meta
425             traj = runner.run_predict(local_video, traj_dir, video_id=video_id)
426             if not traj:
427                 print(f"[worker] detector produced no trajectory for {name!r} ?
skipping.")
428                 continue
429             specs.append({"name": name, "trajectory": traj, "labels": local_label,
430                          "fps": fps, "frame_width": frame_width})
431         else:
432             from backend.pipeline.detectors.stub_runner import
synth_trajectory_from_labels
433             traj = os.path.join(traj_dir, f"{name}_traj.csv")
434             synth_trajectory_from_labels(local_label, traj, fps=args.fps,
frame_width=args.frame_width)
435             specs.append({"name": name, "trajectory": traj, "labels": local_label,
436                          "fps": args.fps, "frame_width": args.frame_width})
437
438     if len(specs) < 2:
439         print(f"[worker] need >=2 videos with trajectories for leave-one-video-out; got
{len(specs)}.")
440         return 2
441     manifest_path = os.path.join(scratch, "manifest.json")
442     with open(manifest_path, "w", encoding="utf-8") as f:
443         json.dump(specs, f, indent=2)
444     src_label = "real detector" if args.trajectory_source == "detector" else "CPU-mock
stub"
445     print(f"[worker] built manifest: {len(specs)} videos ({src_label} trajectories)")
446
447     out_dir = os.path.join(scratch, "artifacts")
448     cmd = [sys.executable, "-m", "training.gen0.harness",
449           "--manifest", manifest_path, "--out", out_dir, "--version", args.version]

```

Logic confirmed: `video\_by\_stem` is keyed by the bare stem (`os.path.splitext(vname)[0]` ? `GX020094`), and the label name is now date-stripped to match ? so the join works. The fix is minimal, correct, and the changed files are green. Let me run the full offline suite to confirm nothing else regressed, since this fix is otherwise commit-ready.

13. user (2026-06-22T12:19:09.998Z)

```
pts, fps=fps, frame_width=fw, velocity_thresh=_PROPOSAL[0],
merge_gap=_PROPOSAL[1], min_window_duration=_PROPOSAL[2])
iv = [(w["start"], w["end"]) for w in wins]
est = rally_gate.estimate_play_axis(pts)
gated = rally_gate.gate_windows(pts, iv, fps, min_crossings=0, estimate=est)
on = [(g.start, g.end) for g in gated if g.crossings >= 2]
gts = load_golden_gts(s["labels"])
d_on.append(score(on, gts).f1)
d_off.append(score(iv, gts).f1)
seg_on.append(_segm.segment_count_ratio(on, gts))
seg_off.append(_segm.segment_count_ratio(iv, gts))
n = len(d_on)
mean_delta = (sum(d_on) - sum(d_off)) / n
mean_seg_off, mean_seg_on = sum(seg_off) / n, sum(seg_on) / n
# Durable direction-of-effect (corpus-independent): the gate only drops windows, so
under the
# over-segmenting proposal windowing it lowers the over-seg ratio.
assert mean_seg_on < mean_seg_off, (mean_seg_on, mean_seg_off)
# Honest measured snapshot (2026-06-18, present 9-clip multicourt set): mean ?F1 ?
-0.001,
# over-seg 1.16 ? 0.64 (was +0.074 / 1.68 ? 1.01 on the retired original-6). ~F1-neutral
band:
assert mean_delta == pytest.approx(0.0, abs=0.03), mean_delta
> assert mean_seg_off == pytest.approx(1.16, abs=0.10), mean_seg_off
E AssertionError: 1.3639692020429994
E assert 1.3639692020429994 == 1.16 ± 0.1
E
E comparison failed
E Obtained: 1.3639692020429994
E Expected: 1.16 ± 0.1
```

```
tests\test_served_gate_a_regression.py:179: AssertionError
===== short test summary info =====
FAILED tests/test_served_gate_a_regression.py::test_proposal_windowing_gate_a_is_f1_neutral_and_
cuts_over_seg
1 failed, 1426 passed in 99.79s (0:01:39)
=====
RESULT: FAIL - pytest exit 1
=====
```

14. assistant (2026-06-22T12:19:29.379Z)

Full suite: **1426 passed, 1 failed in ~100s.** The single failure is in
[`tests/test\_served\_gate\_a\_regression.py`](tests/test_served_gate_a_regression.py) ? an over-
segmentation **snapshot** assertion (`mean\_seg\_off` expected `1.16 ±0.10`, observed `1.36`).
That test exercises the rally-gate path and never touches `backend/worker`, so it can't be
caused by this fix. Let me confirm it's pre-existing by stashing my two changes and re-running
just that test.

15. user (2026-06-22T12:19:38.243Z)

```
Exit code 1
No local changes to save
=== stashed; running gate-A test on clean tree ===
# Honest measured snapshot (2026-06-18, present 9-clip multicourt set): mean ?F1 ?
-0.001,
# over-seg 1.16 ? 0.64 (was +0.074 / 1.68 ? 1.01 on the retired original-6). ~F1-neutral
```



```

band:
    assert mean_delta == pytest.approx(0.0, abs=0.03), mean_delta
>
    assert mean_seg_off == pytest.approx(1.16, abs=0.10), mean_seg_off
E
    AssertionError: 1.3639692020429994
E
    assert 1.3639692020429994 == 1.16 ± 0.1
E
E
    comparison failed
E
    Obtained: 1.3639692020429994
E
    Expected: 1.16 ± 0.1

tests\test_served_gate_a_regression.py:179: AssertionError
===== short test summary info =====
FAILED tests/test_served_gate_a_regression.py::test_proposal_windowing_gate_a_is_f1_neutral_and_cuts_over_seg
1 failed in 6.01s
=== restoring ===
No stash entries found.

```

16. assistant (2026-06-22T12:19:45.273Z)

The stash said "No local changes to save" ? that's unexpected; I need to verify my working-tree state wasn't disturbed (shared checkout). Checking now:

17. user (2026-06-22T12:19:46.491Z)

On branch fix/worker-train-dated-label-join
Your branch is up to date with 'origin/fix/worker-train-dated-label-join'.

Untracked files:
(use "git add <file>..." to include in what will be committed)
artifacts/
docs/KHELSUTRA_PRODUCT_OVERVIEW.md

nothing added to commit but untracked files present (use "git add" to track)
=== worker diff present? ===

18. assistant (2026-06-22T12:19:53.870Z)

The working tree changed underneath me ? a concurrent session committed the worker fix and the branch now tracks `origin/fix/worker-train-dated-label-join` (it tracked `origin/master` minutes ago). This is the shared-checkout / concurrent-session case my instructions warn about. Let me see exactly what landed.

19. user (2026-06-22T12:19:56.871Z)

```

=== local log ===
a2c8be0 fix(worker): strip ISO-date prefix when joining labels to videos in train
d301a7f fix(nightly): GCP-L4-validated fixes ? gs:// fetch dst, WASB env, weights path, pure-engine (#318)
fa2342c feat(nightly): E2E engine reel-count regression on a GCP GPU VM (emailed) + cost (#315)
7a40434 feat(eval): golden fixtures P3 ? 6 curated, trimmed, de-attributed REAL Tier-0 fixtures + --max-frames trim (#317)
5c2eec8 chore(security): P0 (safe scope) ? scrub personal paths from code + add leak-scan CI guard (#314)
=== commits on this branch vs master ===
a2c8be0 fix(worker): strip ISO-date prefix when joining labels to videos in train
=== remote branch log ===
a2c8be0 fix(worker): strip ISO-date prefix when joining labels to videos in train
d301a7f fix(nightly): GCP-L4-validated fixes ? gs:// fetch dst, WASB env, weights path, pure-engine (#318)
fa2342c feat(nightly): E2E engine reel-count regression on a GCP GPU VM (emailed) + cost (#315)

```

20. assistant (2026-06-22T12:20:16.473Z)

No work was lost ? that exact verified change is now committed as `a2c8be0` (the sole commit over `master`) and pushed to `origin/fix/worker-train-dated-label-join`. Let me quickly check whether the failing gate-A test is local-data-gated (so I know whether it's a CI-affecting break or just local-corpus drift).

21. user (2026-06-22T12:20:17.590Z)

```
1      """SERVED Gate-A regression litmus ? the *production* FusionSegmenter path on real
footage.
2
3      The flagged gap (#146): the served default flip (`default_segmenter=fusion_hybrid` +
4      `min_crossings=2`) is **mock-tested only** on real video (`test_fusion_hybrid.py`
drives a
5      MagicMock runner). The offline ablation litmus
6      (`test_ablation.test_litmus_reproduces_gate_a`)
7      validates the Gate-A *signal* ? but on the **permissive proposal windowing**
8      (`distill_local.PROPOSAL`), NOT the production operating point. This module closes the
gap: it
9      runs the real `FusionSegmenter` decision seams (`enrich_candidate_stats` net-
crossings +
10     `select_for_handoff` Gate A) over windows from the **production**
11     `trajectory_to_action_windows` config, on the cached golden WASB trajectories, and
asserts
12     the served path's measured behaviour.
13
14     GPU-free / Gemini-free (person cue OFF) ? it stops at the AI-handoff boundary. Skips
cleanly in
15     CI where the golden trajectories are absent; runs on the owner box where they are
present, over
16     **whatever golden clips are currently on disk** (`golden_specs` requires  $\geq 6$ ).
17
18     CORPUS NOTE ? re-grounded 2026-06-18 (the golden set is mutable and grows over time).
The
19     original 6 *single-court* golden clips this litmus was first calibrated on (2026-06-14)
have been
20     retired from the box ? their labels are gone ? so the present golden set is the 9
**multicourt**
21     clips now in `output/` (mahadevpura_1/2, GX0101xx, GX0x0094, Badminton_BXH_2,
Boxhill_Doubles).
22     On that corpus the served path is **recall-starved**: WASB tracks a single shuttle while
several
23     courts rally at once, so `trajectory_to_action_windows` emits FEWER windows than there
are
24     rallies (served over-seg ratio  $< 1$ , i.e. under-segmentation). Because Gate A only ever
DROPS
25     windows, on this under-segmented corpus it is mildly **F1-negative** (mean served ?F1 ?
-0.034),
26     not the ~neutral -0.008 the original 6 showed. Under the permissive PROPOSAL windowing
the same
27     gate is ~F1-neutral (?F1 ? -0.001) while still cutting over-segmentation (1.16?0.64).
28
29     The canonical **+0.074 / 6-of-6 / p=0.031** Gate-A win was a property of the
*original-6* corpus
30     and is NOT reproduced here (those clips are no longer on the box); it is preserved in
the archive
31     (`docs/archives/quality-iterations/2026-06-first-ab-experiment/`). These tests now pin
the
32     **present-corpus** behaviour so a future windowing/corpus change that silently shifts it
trips the
33     litmus. The pinned magnitudes are a date-stamped snapshot of the present golden set and
will need
34     re-measurement as the corpus grows.
35     """
36     from __future__ import annotations
```

```

37     import json
38     import os
39
40     import pytest
41
42     # cv2 is imported transitively by trajectory_hybrid (the served engine). On a box
without it
43     # (CI), skip the whole module rather than erroring at import.
44     pytest.importorskip("cv2")
45
46     from backend.eval import served_gate_a as sga # noqa: E402
47     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner # noqa: E402
48     from backend.pipeline.detectors import rally_gate # noqa: E402
49     from backend.eval.calibrate_wasb import load_golden_gts # noqa: E402
50     from backend.eval.metrics import score # noqa: E402
51     from backend.eval import segmentation_metrics as _segm # noqa: E402
52
53     # The golden manifest lives in the MAIN checkout's output/ (gitignored; absent in a
fresh
54     # worktree / CI). Resolve it relative to this repo root first, with an env override for
an
55     # out-of-tree checkout.
56     _REPO_ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
57     MANIFEST = os.environ.get(
58         "SERVED_GATE_A_MANIFEST",
59         os.path.join(_REPO_ROOT, "output", "human_lovo_manifest.json"),
60     )

```

22. assistant (2026-06-22T12:20:56.540Z)

Now I have the full picture. Here's where things stand.

Status

****1. The in-flight worker fix is committed, pushed, and verified green.****

While I was verifying, a concurrent session committed it (shared checkout ? exactly the case the branching-safety convention warns about). Nothing was lost: the commit `a2c8be0 fix(worker): strip ISO-date prefix when joining labels to videos in train` is the ****sole commit over `master`****, pushed to `origin/fix/worker-train-dated-label-join`, and contains exactly the change I verified:

- ? `tests/test\_worker.py` ? 26 passed (incl. the new `\_label\_clip\_name` param test + the dated-label?bare-video join regression)
- ? `ruff check` clean, `mypy` clean on `backend/worker/\_\_main\_\_.py`
- ? Logic confirmed: `video\_by\_stem` is keyed by bare stem (`splitext(name)[0]`), and the date-stripped label name now joins correctly.

****2. One full-suite failure ? unrelated to this fix, and local-only (not a CI break).****

[`test\_served\_gate\_a\_regression.py`](tests/test_served_gate_a_regression.py) trips on an over-seg ****snapshot**** assertion: `mean\_seg\_off` expected `1.16 ±0.10`, observed `1.36`. I confirmed it:

- reproduces on the ****clean tree**** (stashing the worker change makes no difference ? it never touches `backend/worker`), and
- the test itself documents that its pinned magnitudes are ****"a date-stamped snapshot of the present golden set [that] will need re-measurement as the corpus grows"**** and ****skips cleanly in CI**** (no `cv2` / no `output/human\_lovo\_manifest.json`).

Your local golden corpus has grown since the 2026-06-18 snapshot (the 6 new `fx\_r0X` real fixtures via #317, etc.), so the litmus is firing exactly as designed ? local-corpus drift, not a regression. The other ****1426 tests pass****.

****3. The two untracked items are unrelated:**** `artifacts/` is gitignored scratch; `docs/KHELSUTRA\_PRODUCT\_OVERVIEW.md` is a standalone 203-line product doc, not part of this branch.

What I'd suggest next

You didn't leave a specific task, so I held at verification rather than guess. The natural forks:

- ****Open the PR**** for `a2c8be0` ? it's verified-green and self-contained (I'll do this only on your go-ahead, per the one-PR-per-item / "don't open a PR unless asked" convention).
- ****Re-ground the gate-A snapshot**** ? re-measure on the current local golden corpus and update the pinned `1.16` ? `~1.36` (this is an owner-gated quality/corpus call, and per the Launch-Report convention a default/quality-snapshot flip should be recorded honestly ? I shouldn't silently edit a measured-regression test).
- ****Something else entirely**** (e.g. the golden-fixtures M4 quality-floor next item, or the product doc).

Which direction do you want?

23. user (2026-06-22T12:20:56.896Z)

please merge the green PRs are ensuring impvred test coverage and green CI
please complete 1. and 2. and mark 3 as a backlog and can fix the docs and archive any if they are completed merging all the green PRs with improved test coverage numebrs

24. assistant (2026-06-22T12:22:01.259Z)

Understood. Let me consolidate your asks into a concrete plan:

1. ****Merge the green PRs**** (with improved coverage + green CI) ? including opening/merging the verified worker fix (`a2c8be0`).
2. ****Re-ground the failing gate-A snapshot**** so it's green again.
3. ****Backlog the "item 3" optional work**** (golden-fixtures M4 etc.).
4. ****Fix docs + archive any that describe completed work**** per the archive policy.

First I need to survey the real state across both repos before mutating anything. Let me fetch fresh and enumerate open PRs, CI status, the full gate-A test, and the doc/backlog registers.

25. user (2026-06-22T12:22:05.611Z)

Already up to date.

=== OPEN PRs (engine) ===

```
[{"headRefName":"feat/nightly-scheduler-fix","isDraft":false,"mergeable":"MERGEABLE","number":320,"reviewDecision":"","title":"feat(nightly): make the Cloud Run scheduler path work E2E + turnkey email/trigger"}, {"headRefName":"fix/worker-train-dated-label-join","isDraft":false,"mergeable":"MERGEABLE","number":319,"reviewDecision":"","title":"fix(worker): strip ISO-date prefix when joining labels to videos in train"}, {"headRefName":"work/golden-corpus-hygiene","isDraft":false,"mergeable":"MERGEABLE","number":316,"reviewDecision":"","title":"docs(data): GCS corpus source-of-truth map + Phase-1 metadata migration"}]
```

26. user (2026-06-22T12:22:07.931Z)

Already up to date.

=== OPEN PRs (khelsutra) ===

```
[]
```

27. assistant (2026-06-22T12:22:23.593Z)

Three open PRs in the engine repo (none in khelsutra). Notably ****#319 already exists**** for the worker fix (the concurrent session opened it too). Let me pull CI status + change stats + bodies for all three so I can classify each (engineering-fix vs product/strategy/licensing) and confirm green before merging.

28. user (2026-06-22T12:22:32.191Z)

```
##### PR #319 #####
```

Traceback (most recent call last):

```
File "<string>", line 1, in <module>
```

```

import json,sys; d=json.load(sys.stdin); print('TITLE:',d['title']);
print('BRANCH:',d['headRefName']);
print('FILES:',d['changedFiles'],'({}/{-}}'.format(d['additions'],d['deletions']));
print('MERGEABLE:',d['mergeable'],'/',d['mergeStateStatus']); print('--- BODY (first 1500 chars)
---'); print((d['body'] or '')[:1500])
~~~~~AAAAAAAAAAAAA

File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\json\__init__.py", line 298,
in load
    return loads(fp.read(),
        cls=cls, object_hook=object_hook,
        parse_float=parse_float, parse_int=parse_int,
        parse_constant=parse_constant, object_pairs_hook=object_pairs_hook, **kw)
File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\json\__init__.py", line 352,
in loads
    return _default_decoder.decode(s)
    ~~~~~AAA

File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\json\decoder.py", line 345,
in decode
    obj, end = self.raw_decode(s, idx=_w(s, 0).end())
    ~~~~~AAAAAAAAAAAAAAAAAAAAAAAAAAAAA

File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\json\decoder.py", line 363,
in raw_decode
    raise JSONDecodeError("Expecting value", s, err.value) from None
json.decoder.JSONDecodeError: Expecting value: line 1 column 1 (char 0)

```

--- CI checks ---

GraphQL: Could not resolve to a PullRequest with the number of 319. (repository.pullRequest)

PR #320

Traceback (most recent call last):

```

File "<string>", line 1, in <module>
    import json,sys; d=json.load(sys.stdin); print('TITLE:',d['title']);
print('BRANCH:',d['headRefName']);
print('FILES:',d['changedFiles'],'({}/{-}}'.format(d['additions'],d['deletions']));
print('MERGEABLE:',d['mergeable'],'/',d['mergeStateStatus']); print('--- BODY (first 1500 chars)
---'); print((d['body'] or '')[:1500])
~~~~~AAAAAAAAAAAAA

File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\json\__init__.py", line 298,
in load
    return loads(fp.read(),
        cls=cls, object_hook=object_hook,
        parse_float=parse_float, parse_int=parse_int,
        parse_constant=parse_constant, object_pairs_hook=object_pairs_hook, **kw)
File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\json\__init__.py", line 352,
in loads
    return _default_decoder.decode(s)
    ~~~~~AAA

File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\json\decoder.py", line 345,
in decode
    obj, end = self.raw_decode(s, idx=_w(s, 0).end())
    ~~~~~AAAAAAAAAAAAAAAAAAAAAAAAAAAAA

File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\json\decoder.py", line 363,
in raw_decode
    raise JSONDecodeError("Expecting value", s, err.value) from None
json.decoder.JSONDecodeError: Expecting value: line 1 column 1 (char 0)

```

--- CI checks ---

GraphQL: Could not resolve to a PullRequest with the number of 320. (repository.pullRequest)

PR #316

Traceback (most recent call last):

```

File "<string>", line 1, in <module>
    import json,sys; d=json.load(sys.stdin); print('TITLE:',d['title']);
print('BRANCH:',d['headRefName']);
print('FILES:',d['changedFiles'],'({}/{-}}'.format(d['additions'],d['deletions']));

```



```

Golden fixtures cross-OS (macos-latest) pass    21s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952167599/job/82711480516
Golden fixtures cross-OS (windows-latest)      pass    59s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952167599/job/82711480557
Leak scan (PII / attribution guard)            pass    7s      https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952167599/job/82711480482
Mypy (lenient baseline) pass                    29s      https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952167599/job/82711480599
Ruff lint                                       pass    10s      https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952167599/job/82711480551
Syntax sweep + import smoke (no GPU stack)      pass    22s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952167599/job/82711480477
Full test suite (offline, mocked)               pending 0      https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952167599/job/82711480614
Built wheel smoke (install + launch)            pass    1m6s      https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952167599/job/82711480680
Dependency license scan (no copyleft in runtime deps) pass    39s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952167599/job/82711480637
Golden fixtures cross-OS (ubuntu-latest)        pass    28s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952167599/job/82711480547

```

PR #316

docs(data): GCS corpus source-of-truth map + Phase-1 metadata migration

files=2 +250/-0 mergeable=MERGEABLE/CLEAN

--- CI checks ---

```

Built wheel smoke (install + launch)            pass    1m12s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27946559018/job/82692684255
Dependency license scan (no copyleft in runtime deps) pass    36s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27946559018/job/82692684239
Full test suite (offline, mocked)               pass    2m32s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27946559018/job/82692684230
Golden fixtures cross-OS (macos-latest) pass    18s      https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27946559018/job/82692684247
Golden fixtures cross-OS (ubuntu-latest)        pass    24s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27946559018/job/82692684242
Golden fixtures cross-OS (windows-latest)      pass    1m8s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27946559018/job/82692684250
Leak scan (PII / attribution guard)             pass    8s      https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27946559018/job/82692684264
Mypy (lenient baseline) pass                    36s      https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27946559018/job/82692684505
Ruff lint                                       pass    9s      https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27946559018/job/82692684515
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27946559018/job/82692684432

```

31. assistant (2026-06-22T12:23:27.459Z)

CI status is clear:

- **##319** (worker fix): ****all 10 checks pass, CLEAN**** ? ready.
- **##316** (docs(data) GCS map + migration): ****all pass, CLEAN**** ? green.
- **##320** (nightly scheduler feat): ****UNSTABLE**** ? "Full test suite (offline, mocked)" is still ****pending****; everything else passes.

Before merging anything I need to (a) check file overlap so an earlier merge doesn't break a later PR, (b) confirm each actually adds/maintains test coverage, and (c) confirm none touch product/strategy/licensing. Let me pull file lists + the diffs I haven't seen.

32. user (2026-06-22T12:23:31.832Z)

```
=== PR #319 files ===
backend/worker/__main__.py (+26/-2)
tests/test_worker.py (+60/-0)
=== PR #320 files ===
deploy/gcp/nightly_regression/README.md (+21/-14)
deploy/gcp/nightly_regression/launch.sh (+22/-6)
deploy/gcp/nightly_regression/register_scheduler.sh (+25/-9)
deploy/gcp/nightly_regression/setup_email.sh (+49/-0)
=== PR #316 files ===
docs/DATA_IN_GCS.md (+249/-0)
docs/README.md (+1/-0)
```

33. assistant (2026-06-22T12:24:48.548Z)

No file overlap between the three PRs ? they're independent, so merge order won't create conflicts. Coverage-wise: ****#319 adds +60 lines of tests**** (coverage up); ****#320 is shell-only**** (deploy/gcp scripts, no Python) and ****#316 is docs-only**** ? both coverage-neutral, neither regresses.

One logical dependency I noticed: #319's code comments cite `docs/DATA\_IN\_GCS.md §7.1` / `§7b` ? sections that ****#316 creates****. Let me confirm that file doesn't yet exist on master (? merge #316 first), quick-review the two diffs I haven't seen, and read the full gate-A test to prep the re-grounding.

34. user (2026-06-22T12:24:50.384Z)

```
=== DATA_IN_GCS.md on disk (working tree)? ===
ls: cannot access 'docs/DATA_IN_GCS.md': No such file or directory
=== exists in git HEAD? ===
fatal: path 'docs/DATA_IN_GCS.md' does not exist in 'HEAD'
NOT in HEAD (new file from #316)

===== PR #316 DIFF =====
diff --git a/docs/DATA_IN_GCS.md b/docs/DATA_IN_GCS.md
new file mode 100644
index 0000000..795e418
--- /dev/null
+++ b/docs/DATA_IN_GCS.md
@@ -0,0 +1,249 @@
+# Data in GCS ? the corpus source-of-truth map
+
+**Status:** LIVE / in-progress (migration not complete). **Owner-gated** decisions in §7.
+**Last inventoried:** 2026-06-22 against `gs://khelsutra-rally-corpus`.
+**Goal (owner, 2026-06-22):** run **all** training + inference from the cloud with **all data in GCS**,
+so the local Windows box (and the intermittently-mounted external drive) is **no longer a blocker**.
+
+> **? Other sessions: this is where the data lives. **Pull from `gs://khelsutra-rally-corpus`, not
+from anyone's `C:` / `F:` / Google Drive. If something you need isn't in the bucket yet, it's in the
+gap table (§5) ? flag it / migrate it, don't hard-code a local path.
+
+This doc is the **data map**. The **machine** runbooks are
+[`SETUP_NEW_MACHINE.md`](SETUP_NEW_MACHINE.md)
+(spin up a box) and [`deploy/gcp/README.md`](../deploy/gcp/README.md) (provision GPU + train/infer).
+The cloud engine itself was built by the (archived) DECOUPLED_COMPUTE project
+(`docs/archives/past_projects/DECOUPLED_COMPUTE/`); this doc does **not** redesign it ? it records
+where the bytes are* and *what's still missing*.
+
```



```

+---
+
+## 1. The bucket
+
+| | |
+|---|---|
+| Bucket | **`gs://khelsutra-rally-corpus`** (single corpus bucket) |
+| Project | `gen-lang-client-0306026826` (billing *Khelsutra Billing*) |
+| Region | `asia-south1` (Mumbai), UBLA, co-located with the GPU VMs |
+| Auth | SA `rally-worker@gen-lang-client-0306026826.iam.gserviceaccount.com`
+|       | (`roles/storage.objectAdmin`); key at `~/gcp/rally-worker-sa.json` (local only, **never
+|       | committed**). Interactive: `gcloud auth login`. |
+| Size today | ~2.06 GB (mostly past dry-run reels/outputs ? **not** the full corpus) |
+
+The engine is **backend-agnostic by URI** (`backend/storage/ref.py`): `gs://` (or `gs://`),
+`s3://`, `gdrive://`, or a bare local path. Pick the backend with **config + creds, no code
+change**.
+
+---
+
+## 2. Canonical layout (the target contract)
+
+The cloud worker's input/output contract (`backend/worker/__main__.py`; lineage:
+`DECOUPLED_COMPUTE/04`):
+
+| Prefix | Holds | Who reads/writes it |
+|---|---|---|
+| `videos/` | source videos / proxies (`<clip>.mp4`) | **input** to `worker infer` & `worker
+|       | train` |
+| `labels/` | golden rally labels (`<clip>.rallies.csv`) | **input** ground truth to `worker
+|       | train` + all eval |
+| `trajectories/` | cached WASB trajectory CSVs | ? **eval/litmus input** (GPU-free), consumed
+|       | via `manifests/golden.json`. **NOT** read by `worker train` (it synths or re-extracts) |
+| `weights/<version>/` | trained Gen-0 bundles (`weights.json` + a run `manifest.json`) |
+|       | **output** of `worker train` |
+| `models/` | upstream pretrained weights (`wasb_badminton_best.pth.tar`) | **staged to a local
+|       | path** (`$WASB_WEIGHTS` / `~/models/?`) for the detector ? not read from `gs://` in place |
+| `outputs/<video_id>/` | per-video inference outputs (intervals, etc.) | **output** of `worker
+|       | infer` |
+| `highlights/`, `promo/` | rendered reels + promo artifacts | output / demo |
+
+> **Note on the manifest.** `worker train` does **not** read a pre-staged corpus
+`manifest.json` ? it
+> *builds* one on-box at train time from `labels/` (+`videos/`), extracting trajectories
+**synthetically**
+> by default** (`--trajectory-source synth`, CPU-mock) or for real via opt-in `--trajectory-
+source
+> detector` (needs `videos/`). A `manifest.json` only appears as an **output** inside each
+> `weights/<version>/` bundle. The **eval/litmus** path is what wants a stable, path-resolvable
+manifest
+> (§7.3) ? and `manifests/golden.json` is currently consumed by **no code yet** (eval is local-
+FS only;
+> see Known blockers).
+
+---
+
+## 3. What's actually in the bucket today (2026-06-22)
+
+> **Phase-1 landed 2026-06-22:** added `trajectories/` (15), 15 canonical
+`labels/<date>_<name>.rallies.csv`
+> (= original-6 + `GX010137`/`GX010141` + 7 new; the 7 old `_proxy`/bare dupes were deleted),
+and
+> `manifests/golden.json`. The table below is the **pre-Phase-1** snapshot; the remaining gap
+is **videos**
+> (14 of 15) ? see §6/§7b/§8.

```

```

+
+| Prefix | Present | Drift / notes |
+|---|---|---|
+| `labels/` | 7 files: `GX010128`, `mahadevpura-1`, + `Badminton BXH 2_proxy`, `Boxhill
Doubles_proxy`, `GX020094_proxy`, `GX030094_proxy`, `mahadevpura-2_proxy` | ? **inconsistent
naming** (`_proxy` on some, not others); ? **missing** the original-6 labels + `GX010137`,
`GX010141` |
+| `videos/` | 3: `GX030094_proxy.mp4`, `mahadevpura-1.MP4`, `mahadevpura_smoke_180s.mp4` | ?
corpus videos/proxies almost entirely absent |
+| `trajectories/` | ? | ? **does not exist** (every trajectory is local-only) |
+| `models/` | `wasb_badminton_best.pth.tar` | ? (? WASB-C3 weight-provenance still an open
legal item before any external share) |
+| `weights/` | `0.1.0-l4/` | ? first real L4 train bundle |
+| `outputs/` | one `/`, `parity/` | ? dry-run outputs |

```

35. user (2026-06-22T12:24:51.268Z)

```

===== PR #320 DIFF =====
diff --git a/deploy/gcp/nightly_regression/README.md b/deploy/gcp/nightly_regression/README.md
index 37bfd0e..c9a6704 100644
--- a/deploy/gcp/nightly_regression/README.md
+++ b/deploy/gcp/nightly_regression/README.md
@@ -26,8 +26,9 @@ Cloud Scheduler (cron, 03:00 IST)
 | `../../scripts/nightly_email.py` | SMTP email (creds from Secret Manager/env; graceful no-
creds skip) |
 | `run_on_vm.sh` | VM startup-script: env ? run ? email ? upload report ? **self-delete** (even
on failure) |
 | `launch.sh` | driver-agnostic: stage repo ? create the self-deleting VM. **Dry-run this
first.** |
-| `Dockerfile.launcher` + `register_scheduler.sh` | the Cloud Scheduler ? Cloud Run job wiring
|
+| `Dockerfile.launcher` + `register_scheduler.sh` | the Cloud Scheduler ? Cloud Run job wiring
(`--run-now` to trigger immediately; auto-grants SA IAM + bakes the git SHA) |
 | `create_nightly_bucket.sh` | create the **ephemeral** nightly bucket (30-day TTL), isolated
from golden |
+| `setup_email.sh` | one-time: store the Gmail app password in Secret Manager + grant the
worker SA read |
 | `../../eval_baselines/reelcount_manifest.json` | the corpus (add a video = one row) |
 | `../../eval_baselines/reelcount_baseline.json` | frozen expected counts (owner-frozen via
`--update-baseline`) |

@@ -45,26 +46,32 @@ A bucket-wide 30-day TTL on the *dedicated* nightly bucket can never touch
golde
`gcloud storage cp clip.mp4 gs://khelsutra-rally-corpus/nightly-inputs/videos/` and the
matching rally labels
(a *_golden_features.json* with a `gts` key, or a `start,end` CSV) to ../../nightly-
inputs/labels/. (Inputs are
golden ? they live in the no-TTL corpus bucket, NOT the 30d nightly bucket.)
-4. **WASB weights** ? `gcloud storage cp wasb_badminton_best.pth.tar gs://khelsutra-rally-
corpus/weights/`.
-5. **SMTP secret** (Gmail app password is simplest):
+4. **WASB weights** at gs://khelsutra-rally-corpus/models/wasb_badminton_best.pth.tar`
(already there; that's the path `launch.sh` defaults to).
+5. **Email** (Gmail app password ? Secret Manager, one command ? needs 2-Step Verification on
the account, then https://myaccount.google.com/apppasswords):
`bash
- printf '%s' '<gmail-app-password>' | gcloud secrets create nightly-smtp-pass --data-file=-
--project=gen-lang-client-0306026826
- # the launcher passes NIGHTLY_SMTP_SECRET + NIGHTLY_SMTP_USER(=your gmail) +
NIGHTLY_EMAIL_TO to the VM
+ printf '%s' '<16-char-app-password>' | bash deploy/gcp/nightly_regression/setup_email.sh
+ # stores nightly-smtp-pass + grants the worker SA read access; emails AS that Gmail to
NIGHTLY_EMAIL_TO.
`

```

```

-## Enable
+## Enable (all from a LOCAL `master` checkout ? NOT the droplet; needs gcloud auth, no local GPU)
+```bash
-# 0) create the ephemeral nightly bucket (30d TTL) + point the manifest at your gs:// clips:
-bash deploy/gcp/nightly_regression/create_nightly_bucket.sh # edit
eval_baselines/reelcount_manifest.json too
-# 1) DRY-RUN the whole VM flow once and watch it (it self-deletes; freezes nothing yet):
-NIGHTLY SMTP_USER=<your-gmail> bash deploy/gcp/nightly_regression/launch.sh
+# 0) bucket (30d TTL) + email + point the manifest at your gs:// clips:
+bash deploy/gcp/nightly_regression/create_nightly_bucket.sh # + edit
eval_baselines/reelcount_manifest.json
+printf '%s' '<gmail-app-pw>' | bash deploy/gcp/nightly_regression/setup_email.sh
+# 1) DRY-RUN the whole VM flow once + watch it (it self-deletes), and freeze the first baseline:
+RALLY_REF=origin/master bash deploy/gcp/nightly_regression/launch.sh
gcloud compute instances get-serial-port-output <printed-name> --zone=<printed-zone> # tail it
-# 2) FREEZE the baseline from the configured engine ? run on a GPU box (the dry-run VM, or a manual VM):
-# python scripts/nightly_reelcount.py --update-baseline # then commit
reelcount_baseline.json
-# 3) WIRE the nightly schedule (cloud-native, scale-to-zero):
-bash deploy/gcp/nightly_regression/register_scheduler.sh
+# on that VM (or any GPU box): python scripts/nightly_reelcount.py --update-baseline ?
commit reelcount_baseline.json
+# 2) WIRE the cloud-native nightly (scale-to-zero) AND trigger one run right now:
+bash deploy/gcp/nightly_regression/register_scheduler.sh --run-now
+```

+### ? Start a scheduled run on demand (same path the 03:00 cron uses)
+```bash
+gcloud run jobs execute nightly-regression-launcher --region asia-south1 --project gen-lang-client-0306026826
+```
+`register_scheduler.sh` auto-grants the worker SA the IAM it needs (`compute.instanceAdmin.v1` + `iam.serviceAccountUser` + `run.invoker`) and bakes the git SHA into the job, so the launcher container (which has no `.git`) stages the baked tree correctly. Smoke-test the launcher without a GPU VM: `RALLY_DRY_RUN=1 bash launch.sh` (stages the tarball, skips the VM create).
+
+### Trigger options (pick one)
+ - **Cloud Scheduler ? Cloud Run job (recommended)** ? `register_scheduler.sh`. Fully cloud, scale-to-zero, machine-independent. The only always-on cost is $0; the GPU VM is ephemeral.
diff --git a/deploy/gcp/nightly_regression/launch.sh b/deploy/gcp/nightly_regression/launch.sh
index 71f555a..840bd4d 100644
--- a/deploy/gcp/nightly_regression/launch.sh
+++ b/deploy/gcp/nightly_regression/launch.sh
@@ -6,9 +6,12 @@
#
# bash deploy/gcp/nightly_regression/launch.sh # L4, stages HEAD, kicks the VM
# RALLY_GPU=t4 RALLY_REF=$(git rev-parse HEAD) bash deploy/gcp/nightly_regression/launch.sh
+# RALLY_DRY_RUN=1 bash deploy/gcp/nightly_regression/launch.sh # stage only, DON'T create the VM (smoke test)
#
-# ? Run this MANUALLY once as a dry-run (watch the serial log) before wiring a scheduler ? the VM-side
-# flow is NOT covered by CI. Tail: gcloud compute instances get-serial-port-output <name> --zone=<z>.
+# Works BOTH from a local git checkout (archives REF) AND from the Cloud Run launcher container, which has
+# no .git (it tars the baked tree + takes the SHA from $RALLY_GIT_SHA ? set by register_scheduler.sh).
+#

```

```

## ? The VM-side flow is NOT covered by CI. Tail: gcloud compute instances get-serial-port-
output <name> --zone=<z>.
set -uo pipefail

PROJECT="${RALLY_GCP_PROJECT:-gen-lang-client-0306026826}"
@@ -24,23 +27,36 @@ DO_EMAIL="${RALLY_EMAIL:-true}"
SMTP_SECRET="${NIGHTLY_SMTP_SECRET:-projects/$PROJECT/secrets/nightly-smtp-
pass/versions/latest}"
SMTP_USER="${NIGHTLY_SMTP_USER:-}"
EMAIL_TO="${NIGHTLY_EMAIL_TO:-avi.dullu@gmail.com}"
+DRY_RUN="${RALLY_DRY_RUN:-}"
HERE="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
+ROOT="$(cd "$HERE/../../.." && pwd)" # repo root (deploy/gcp/nightly_regression ? 3 up)
STARTUP="$HERE/run_on_vm.sh"
export CLOUDSDK_CORE_DISABLE_PROMPTS=1

ZONES="${RALLY_GCP_ZONES:-asia-south1-a asia-south1-b asia-south1-c asia-southeast1-a asia-
southeast1-b asia-southeast1-c}"
if [ "$GPU" = "l4" ]; then MACHINE="g2-standard-4"; ACCEL=(); else MACHINE="n1-standard-8";
ACCEL=(--accelerator=type=nvidia-tesla-t4,count=1); fi

-# 1) Stage the repo @ REF to the EPHEMERAL nightly bucket (30d TTL; no GitHub auth on the VM).
-# Create it first if missing: bash deploy/gcp/nightly_regression/create_nightly_bucket.sh
-SHA="$(git rev-parse "$REF")"
+# 1) Stage the repo to the EPHEMERAL nightly bucket (30d TTL; no GitHub auth on the VM).
+# Create the bucket first if missing: bash
deploy/gcp/nightly_regression/create_nightly_bucket.sh
+# Two modes: a git checkout archives the tracked tree @ REF; the Cloud Run container (no
.git) tars
+# the baked tree and takes the SHA from $RALLY_GIT_SHA / a baked GIT_SHA file.
+SHA="${RALLY_GIT_SHA:-$( (cd "$ROOT" && git rev-parse "$REF" 2>/dev/null) || cat
"$ROOT/GIT_SHA" 2>/dev/null || echo unknown )}"
TGZ_URI="gs://$NIGHTLY_BUCKET/repo/${SHA}.tgz"
echo "== staging repo $SHA -> $TGZ_URI =="
TMP_TGZ="$(mktemp --suffix=.tgz)"
-git archive --format=tar.gz -o "$TMP_TGZ" "$REF"
+if (cd "$ROOT" && git rev-parse --git-dir >/dev/null 2>&1); then
+ ( cd "$ROOT" && git -c core.autocrlf=false archive --format=tar.gz -o "$TMP_TGZ" "$REF" ) #
local checkout
+else
+ tar czf "$TMP_TGZ" -C "$ROOT" --exclude='./.git' . #
container: baked tree
+fi
gcloud storage cp "$TMP_TGZ" "$TGZ_URI" --project="$PROJECT"
rm -f "$TMP_TGZ"

+if [ -n "$DRY_RUN" ]; then
+ echo "[dry-run] staged $TGZ_URI OK; would create $GPU VM '$NAME' with run_on_vm.sh +
metadata. Skipping VM create."
+ exit 0
+fi
+
# 2) Zone-sweep create the VM (mirrors create_gpu_vm.sh) with run_on_vm.sh + the config
metadata.
ERR="$(mktemp)"
for Z in $ZONES; do
diff --git a/deploy/gcp/nightly_regression/register_scheduler.sh
b/deploy/gcp/nightly_regression/register_scheduler.sh
index e5bf588..0f8a965 100644
--- a/deploy/gcp/nightly_regression/register_scheduler.sh
+++ b/deploy/gcp/nightly_regression/register_scheduler.sh
@@ -9,9 +9,11 @@
# gs://<bucket>, and the SMTP secret (see README "Enable").
#
# bash deploy/gcp/nightly_regression/register_scheduler.sh # build + deploy +

```

```

schedule (03:00 IST)
+# bash deploy/gcp/nightly_regression/register_scheduler.sh --run-now # ...and trigger one run
immediately
# RALLY_CRON='0 3 * * *' RALLY_TZ='Asia/Kolkata' bash ../register_scheduler.sh
set -euo pipefail

+RUN_NOW=""; [ "${1:-}" = "--run-now" ] && RUN_NOW=1
PROJECT="${RALLY_GCP_PROJECT:-gen-lang-client-0306026826}"
REGION="${RALLY_RUN_REGION:-asia-south1}"
JOB="${RALLY_RUN_JOB:-nightly-regression-launcher}"
@@ -21,6 +23,7 @@ TZ="${RALLY_TZ:-Asia/Kolkata}"
SA="rally-worker@${PROJECT}.iam.gserviceaccount.com"
IMAGE="${RALLY_LAUNCHER_IMAGE:-${REGION}-docker.pkg.dev/${PROJECT}/rally/nightly-
launcher:latest}"
REPO_ROOT="$(git rev-parse --show-toplevel)"
+GIT_SHA="$(git -C "$REPO_ROOT" rev-parse HEAD)" # baked into the job so the container's
launch.sh (no .git) has it
export CLOUDSDK_CORE_DISABLE_PROMPTS=1

echo "== enable APIs (run, scheduler, build, artifact registry) =="
@@ -36,10 +39,17 @@ echo "== build + push the launcher image from the repo root =="
--gcs-source-staging-dir="gs://${RALLY_NIGHTLY_BUCKET:-khelsutra-rally-nightly}/cloudbuild"
\
-f deploy/gcp/nightly_regression/Dockerfile.launcher . )

-echo "== create/update the Cloud Run JOB (launcher needs compute-admin to create the GPU VM)
=="
+echo "== grant the launcher SA the IAM it needs (idempotent) =="
+# create/delete the ephemeral GPU VM + attach itself to it + be invocable by Cloud Scheduler.
+for ROLE in roles/compute.instanceAdmin.v1 roles/iam.serviceAccountUser roles/run.invoker; do
+ gcloud projects add-iam-policy-binding "$PROJECT" --member="serviceAccount:$SA"
--role="$ROLE" \
+ --condition=None >/dev/null
+done
+
+echo "== create/update the Cloud Run JOB (bakes the repo image + the git SHA) =="
gcloud run jobs deploy "$JOB" --project="$PROJECT" --region="$REGION" \
--image="$IMAGE" --service-account="$SA" --max-retries=0 --task-timeout=900s \
- --set-env-vars="RALLY_GPU=${RALLY_GPU:-
l4},RALLY_EMAIL=true,NIGHTLY_EMAIL_TO=${NIGHTLY_EMAIL_TO:-avi.dullu@gmail.com}"
+ --set-env-vars="RALLY_GPU=${RALLY_GPU:-
l4},RALLY_EMAIL=true,NIGHTLY_EMAIL_TO=${NIGHTLY_EMAIL_TO:-
avi.dullu@gmail.com},RALLY_GIT_SHA=${GIT_SHA}"

echo "== schedule it (Cloud Scheduler ? Run job admin API) =="
RUN_URI="https://${REGION}-
run.googleapis.com/apis/run.googleapis.com/v1/namespaces/${PROJECT}/jobs/${JOB}:run"
@@ -52,11 +62,17 @@ gcloud scheduler jobs create http "$SCHED" --project="$PROJECT"
--location="$REG

cat <<EOF

-? Scheduled '$SCHED' ($CRON $TZ) ? Cloud Run job '$JOB' ? launch.sh ? ephemeral GPU VM.
- IAM the launcher SA needs (grant once if missing):
- roles/compute.instanceAdmin.v1 (create/delete the GPU VM)
- roles/iam.serviceAccountUser (attach $SA to the VM)
- roles/run.invoker + roles/cloudscheduler.jobRunner (Scheduler ? Run)
- Trigger now to test: gcloud run jobs execute $JOB --region=$REGION --project=$PROJECT
- Remove: gcloud scheduler jobs delete $SCHED --location=$REGION; gcloud run
jobs delete $JOB --region=$REGION
+? Scheduled '$SCHED' ($CRON $TZ) ? Cloud Run job '$JOB' ? launch.sh ? ephemeral GPU VM (SA IAM
granted above).
+ ? START A RUN NOW (same as the nightly cron does):
+ gcloud run jobs execute $JOB --region=$REGION --project=$PROJECT
+ Watch it: gcloud run jobs executions list --job=$JOB --region=$REGION --project=$PROJECT

```

```

+          gcloud beta run jobs executions logs read <execution> --region=$REGION #
launcher logs
+          gcloud compute instances get-serial-port-output <vm> --zone=<zone> # the
GPU run
+   Email:      bash deploy/gcp/nightly_regression/setup_email.sh   (Gmail app password ? Secret
Manager)
+   Remove:     gcloud scheduler jobs delete $SCHED --location=$REGION; gcloud run jobs delete
$JOB --region=$REGION
EOF
+
+if [ -n "$RUN_NOW" ]; then
+  echo "== --run-now: triggering one run immediately =="
+  gcloud run jobs execute "$JOB" --region="$REGION" --project="$PROJECT"
+fi
diff --git a/deploy/gcp/nightly_regression/setup_email.sh
b/deploy/gcp/nightly_regression/setup_email.sh
new file mode 100644
index 00000000..db483da
--- /dev/null
+++ b/deploy/gcp/nightly_regression/setup_email.sh
@@ -0,0 +1,49 @@
+#!/usr/bin/env bash
+## One-time email setup for the nightly report: store a Gmail APP PASSWORD in Secret Manager +
let the
+## worker SA read it. The nightly emails the report to NIGHTLY_EMAIL_TO (default
avi.dullu@gmail.com),
+## sending AS that Gmail (smtp lib STARTTLS, cert-verified) ? so it needs an app password, not
your
+## normal password.
+##
+## 1. Enable 2-Step Verification on the Google account, then create an app password:
+##      https://myaccount.google.com/apppasswords   (pick "Mail" / "Other")
+## 2. Feed the 16-char password on STDIN (never as an argv/history-leaking flag):
+##      printf '%s' 'xxxxxxxxxxxxxxxx' | bash deploy/gcp/nightly_regression/setup_email.sh
+##
+## Idempotent: re-running adds a new secret version. The secret name matches launch.sh's default
+## (projects/<proj>/secrets/nightly-smtp-pass/versions/latest).
+set -euo pipefail
+
+PROJECT="${RALLY_GCP_PROJECT:-gen-lang-client-0306026826}"
+SA="rally-worker@${PROJECT}.iam.gserviceaccount.com"
+SECRET="${RALLY_SMTP_SECRET_NAME:-nightly-smtp-pass}"
+export CLOUDSDK_CORE_DISABLE_PROMPTS=1
+
+if [ -t 0 ]; then
+  echo "ERROR: pipe the Gmail app password on STDIN, e.g.:" >&2
+  echo "  printf '%s' '<16-char-app-password>' | bash $0" >&2
+  exit 2
+fi
+PW="$(cat)"
+PW="${PW//['\r\n ']/}" # strip stray whitespace/newlines (Gmail shows the pw with spaces)
+[ -n "$PW" ] || { echo "ERROR: empty password on STDIN." >&2; exit 2; }
+
+gcloud services enable secretmanager.googleapis.com --project="$PROJECT" >/dev/null
+
+if gcloud secrets describe "$SECRET" --project="$PROJECT" >/dev/null 2>&1; then
+  echo "== adding a new version to existing secret '$SECRET' =="
+  printf '%s' "$PW" | gcloud secrets versions add "$SECRET" --project="$PROJECT" --data-file=-
+else
+  echo "== creating secret '$SECRET' =="
+  printf '%s' "$PW" | gcloud secrets create "$SECRET" --project="$PROJECT" --replication-
policy=automatic --data-file=-
+fi
+
+echo "== granting the worker SA read access =="

```

```
+gcloud secrets add-iam-policy-binding "$SECRET" --project="$PROJECT" \
+  --member="serviceAccount:$SA" --role=roles/secretmanager.secretAccessor >/dev/null
+
+cat <<EOF
+? Email ready. The nightly resolves projects/$PROJECT/secrets/$SECRET/versions/latest at send
time and
+  emails NIGHTLY_EMAIL_TO (default avi.dullu@gmail.com), sending as that Gmail.
+  To send to a different address / send-as a different account, set NIGHTLY_EMAIL_TO /
NIGHTLY_SMTP_USER
+  on the Cloud Run job (register_scheduler.sh) or in launch.sh's env.
+EOF
```

36. assistant (2026-06-22T12:25:28.985Z)

Both diffs are clean and safe to merge:

```
- **#316** ? pure docs: new `docs/DATA_IN_GCS.md` (the GCS corpus map, marked LIVE/in-progress,
owner-gated $7) + 1 line in `docs/README.md`. No code, no leaks (leak-scan green). Confirmed
`DATA_IN_GCS.md` is **not** in HEAD ? must merge before #319 so the worker fix's `$7.1`/`$7b`
comment references resolve.
- **#320** ? pure deploy shell scripts (`deploy/gcp/nightly_regression/`): `--run-now`, dry-run
mode, SA IAM auto-grant, `setup_email.sh` (reads the app password from STDIN, not argv ? good
hygiene). Infra plumbing for the nightly track, no product/strategy/licensing surface. Only
blocker: full suite still pending (can't fail ? zero Python changed).
```

No file overlap among the three ? safe to merge independently. **Merge order: #316 ? #319 ? #320 (when green).** Coverage: #319 adds tests (?), #316/#320 add zero Python (neutral); all pass CI's coverage-gated suite.

Let me re-verify HEAD is clean/mine, move to master, and merge the two green PRs (#316 then #319).

37. user (2026-06-22T12:25:33.680Z)

```
=== pre-flight: HEAD + cleanliness ===
fix/worker-train-dated-label-join
a2c8be0 fix(worker): strip ISO-date prefix when joining labels to videos in train
  M tests/test_worker.py
=== checkout master + ff pull ===
fatal: 'master' is already used by worktree at 'C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/tier1-windowing'
Already up to date.
```

38. assistant (2026-06-22T12:26:04.179Z)

Two concurrency signals I must handle before mutating anything:

1. `git status` now shows `M tests/test\_worker.py` ? an **uncommitted change that isn't mine** (the worker change was already committed as `a2c8be0`). A sibling session is likely editing it live.
2. `git checkout master` failed ? **master is checked out in another worktree**
`.claude/worktrees/tier1-windowing` (a concurrent session, and its name suggests it's working the *windowing/gate-A* area I was about to touch).

I must not clobber sibling WIP, and I should not duplicate gate-A work if it's already in progress. Let me investigate before doing anything.

39. user (2026-06-22T12:26:07.017Z)

```
=== uncommitted change to tests/test_worker.py (NOT mine) ===
diff --git a/tests/test_worker.py b/tests/test_worker.py
index c48fa02..8bd3af 100644
--- a/tests/test_worker.py
+++ b/tests/test_worker.py
@@ -251,6 +251,137 @@ def test_cmd_train_detector_joins_dated_labels_to_bare_videos(tmp_path,
```

monkeypa

```
    assert {s["name"] for s in specs} == {"GX020094", "mahadevpura_1"} # downstream name is
date-stripped
```

```
+def test_cmd_train_detector_skips_label_with_no_matching_video(tmp_path, monkeypatch):
+    """A dated label whose <name> has no videos/<name>.<ext> is SKIPPED (the join-miss branch),
+    while the labels that DO join are kept ? so one orphan label can't abort an otherwise-valid
+    run."""
+    import os
+    import subprocess
+
+    monkeypatch.setenv("RALLY_DETECTOR_IMPL", "stub")
+    monkeypatch.setattr(wmain, "_probe_video_fps_width", lambda p: (30.0, 1280.0))
+    corpus = tmp_path / "corpus"
+    labels_dir = corpus / "labels"
+    videos_dir = corpus / "videos"
+    labels_dir.mkdir(parents=True)
+    videos_dir.mkdir(parents=True)
+    header = "rally_number,start_time,end_time,ending_reason,sport,shots_count\n"
+    row = "1,2.0,5.0,winner,badminton,\n"
+    (labels_dir / "2026-06-22_GX020094.rallies.csv").write_text(header + row)
+    (labels_dir / "2026-06-21_mahadevpura_1.rallies.csv").write_text(header + row)
+    (labels_dir / "2026-06-20_orphan.rallies.csv").write_text(header + row) # NO
+    videos/orphan.*
+    (videos_dir / "GX020094.mp4").write_bytes(b"FAKEVIDEO1")
+    (videos_dir / "mahadevpura_1.mp4").write_bytes(b"FAKEVIDEO2")
+    out_dir = tmp_path / "out"
+    scratch = tmp_path / "scratch"
+    cfg = tmp_path / "config.json"
+    cfg.write_text("{}")
+
+    def fake_run(cmd, *a, **k):
+        out = cmd[cmd.index("--out") + 1]
+        ver = cmd[cmd.index("--version") + 1]
+        bdir = os.path.join(out, ver)
+        os.makedirs(bdir, exist_ok=True)
+        for fn in ("weights.json", "manifest.json"):
+            with open(os.path.join(bdir, fn), "w") as f:
+                f.write("{}")
+        return subprocess.CompletedProcess(cmd, 0)
+    monkeypatch.setattr(subprocess, "run", fake_run)
+
+    rc = wmain.main(["train", "--corpus-uri", str(corpus), "--out-uri", str(out_dir),
+        "--version", "0.0.5", "--config", str(cfg), "--scratch", str(scratch),
+        "--trajectory-source", "detector", "--segmenter", "wasb_hybrid"])
+    assert rc == 0 # the orphan label was skipped; the two joinable clips carried the run
+    specs = json.loads((scratch / "manifest.json").read_text())
+    assert {s["name"] for s in specs} == {"GX020094", "mahadevpura_1"} # 'orphan' absent
+
+def test_cmd_train_detector_skips_unprobeable_and_trajless_videos(tmp_path, monkeypatch):
+    """cmd_train's detector loop must SKIP (never crash, never guess) a video whose fps/width
+    won't
+    probe and a video the detector yields no trajectory for, keeping the healthy clips."""
+    import os
+    import subprocess
+
+    corpus = tmp_path / "corpus"
+    labels_dir = corpus / "labels"
```

=== worktrees ===

C:/Users/avidu/Projects/badminton-highlight-indexer

a2c8be0 [fix/worker-train-dated-label-join]

C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-a771ff1d85e18f2ed


```

7e0881a [perf/wasb-fast-frames]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-execpolicy
5cb7bbf [bhi-execpolicy]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/boxhill17
8736e2d [run/boxhill17-highlights]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gap-doc
f04f3f3 [docs/rally-gap-closure]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/hardchop
614325d [feat/w1-hard-cap-fallback]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/pr-stitch
a6adcc5 [fix/runner-stitch-watermark]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/recdoc
9fd4215 [docs/recording-single-match]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/resweep
fe34013 [docs/lovo-resweep-n13]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/sched-fix
4051ed5 [feat/nightly-scheduler-fix]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing
a3b209a [master]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-safeguard
5ba6955 [feat/w2-never-empty-safeguard]

```

=== all local branches ===

```

+ bhi-execpolicy                                5cb7bbf (C:/Users/avidu/Projects/badminton-
highlight-indexer/.claude/worktrees/bhi-execpolicy) [origin/bhi-execpolicy: gone] feat(jobs):
worker self-scheduling drain loop honoring WAIT_AND_RESUME (EXECUTION_POLICY P3a)
  chore/consolidate-fusion-schema-predicate e268972 [origin/chore/consolidate-fusion-schema-
predicate: gone] refactor(eval): single source of truth for the _is_fusion_schema predicate
  docs/adr-012-gpu-cloudrun                    118d9b2 [origin/docs/adr-012-gpu-cloudrun: gone]
docs(adr): ADR-012 ? GPU-native compute; TPU deferred; Cloud Run+GPU is the serving model
  docs/archive-decoupled-compute               e6110ce [origin/master: behind 48] feat(api): GET
/api/health + return download_url from /api/compile (M1/P6) (#268)
  docs/archive-decoupled-v2                   fb51cdc [origin/docs/archive-decoupled-v2: gone]
docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP)
  docs/compute-decoupled-serving               a60da27 [origin/docs/compute-decoupled-serving:
gone] docs: spin up COMPUTE_DECOUPLED_SERVING subproject + ADR-014
  docs/court-floor-rally-end                  430de76 [origin/docs/court-floor-rally-end: gone]
docs(rally): capture the shuttle-floor-contact rally-END signal (owner note)
  docs/decouple-scope-defer                   4b2c202 [origin/docs/decouple-scope-defer: gone]
docs(decoupled): narrow DECOUPLED_COMPUTE scope (M0?M2 + M3.5); defer M3/M4; D0=train, GCP=serve
(ADR-011)
  docs/deploy-report-polish                   7cc2300 [origin/docs/deploy-report-polish: gone]
docs(deploy): report polish ? highlight reel smoke-test result + VM-deleted final state
  docs/desktop-app-plan                       b9ff235 [origin/docs/desktop-app-plan: gone]
docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port
  docs/doc-status-register                   b1c0f37 [origin/docs/doc-status-register: gone]
docs: add DOC_STATUS register + archival due-diligence convention
  docs/drop-digitalocean-gcp-only             5225bc6 [origin/docs/drop-digitalocean-gcp-only:
gone] docs: drop DigitalOcean ? GCP is the sole compute stack (ADR-013)
  docs/findings-1-2-pervideo-nextsteps       190a2c8 [origin/docs/findings-1-2-pervideo-
nextsteps: gone] docs: resolve DOC_STATUS findings #1 (per-video supersede) + #2 (NEXT_STEPS
refresh)
  docs/gcp-deploy-report                     76ae710 [origin/docs/gcp-deploy-report: gone]
docs(deploy): first real GCP-L4 run report + gpu_setup conda-ToS/deps fix + Ops Agent runbook
  docs/gcp-pivot-runbook                     e2e9754 [origin/docs/gcp-pivot-runbook: gone]
docs(gcp): add Spot-T4 fast path (likely skips the GPUS_ALL_REGIONS wait)
  docs/lock-m0-decisions                     21c93d1 [origin/docs/lock-m0-decisions: gone]
docs(serving): lock the 5 M0 gating decisions (D7-D12) + add the data-flywheel harness (M11)
+ docs/lovo-resweep-n13                     fe34013 (C:/Users/avidu/Projects/badminton-
highlight-indexer/.claude/worktrees/resweep) [origin/docs/lovo-resweep-n13: gone] docs(archive):
make lovo_resweep.py lint-clean (def/rename/noqa E402)
  docs/m0-approved-d13-d15                   93d2188 [origin/docs/m0-approved-d13-d15: gone]
docs(serving): M0 APPROVED + lock D13-D15 (switchable profiles, Drive Northstar, consent-not-
surprise)
  docs/m2-finalized-report                   8d8d230 [origin/docs/m2-finalized-report: gone]

```

```
docs(deploy): M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed; both VMs deleted ($0)
docs(packaging-plan-tracked) 173a305 [origin/docs/packaging-plan-tracked: gone]
docs(pkg): track the batteries-included app plan (PACKAGING_PLAN.md) + index pointers
docs(pkg-p2c-done) d0d9049 [origin/docs/pkg-p2c-done: gone] docs(pkg):
mark P2c ? + the loop closed ? v1 LIGHT is feature-complete
docs(point-i18n-ui-to-khelsutra) 8bd1b4a [origin/docs/point-i18n-ui-to-khelsutra:
gone] docs: point I18N_PLAN + UI_PROPOSAL (+ DOC_STATUS) at the khelsutra UX/l10n project
+ docs/rally-gap-closure f04f3f3 (C:/Users/avidu/Projects/badminton-
highlight-indexer/.claude/worktrees/gap-doc) [origin/docs/rally-gap-closure: gone] docs(rally):
fold GX010137 (golden #13) into the gap-closure doc
+ docs/recording-single-match 9fd4215 (C:/Users/avidu/Projects/badminton-
highlight-indexer/.claude/worktrees/recdoc) [origin/docs/recording-single-match: gone]
docs(recording): assert single-match capture as the #1 rule (now MEASURED)
docs(restructure-field-roora-data) 1f8d854 [origin/docs/restructure-field-roora-data:
gone] docs: isolate field-ops/PMF + Roora into archives/; carve a no-ML docs/data_pipeline/
docs(serving-design-picks-p3-reconcile) 31213e9 [origin/docs/serving-design-
picks-p3-reconcile: gone] docs(serving): resolve $8 design picks (D16/D17) + reconcile per-video
P3 vs M3
feat(api-m1-health-downloadurl) 6b5fd17 [origin/feat/api-m1-health-downloadurl:
gone] feat(api): GET /api/health + return download_url from /api/compile (M1/P6)
feat(gcp-ops-agent-and-dryrun-guide) c3dbd00 [origin/feat/gcp-ops-agent-and-dryrun-guide:
gone] feat(gcp): always-install Ops Agent on GPU VMs + a cloud-GPU dry-run guide
feat(gdrive-storage-backend) 58a56e8 [origin/feat/gdrive-storage-backend: gone]
feat(storage): actionable gdrive:// errors ? install + sync-the-folder remedy
feat(highlights-runner) bd3c8b5 [origin/feat/highlights-runner: gone]
feat(tools): batch highlight RUNNER ? clean, non-destructive, Drive-safe reel generation
feat(local-no-ai-segmenter) 3ea7ad4 [origin/feat/local-no-ai-segmenter: gone]
feat(runner): fully-local no-AI segmenter mode (+ WASB stream flag, .env load)
feat(locale-analytics) b6245ec [origin/feat/locale-analytics: gone]
feat(db): v7 ? record the UI locale per job (PMF analytics) + count_jobs_by_locale
feat(localized-watermark) 5132e14 [origin/feat/localized-watermark: gone]
feat(stitch): localize the reel watermark to the UI locale (+ OFL Noto fonts)
feat(long-rally-lens) cbb2494 [origin/feat/long-rally-lens: gone]
feat(eval): P7 long-rally (>5s) quality lens + reel knob (both default-OFF)
feat(native-wasb-runner) 72f3736 [origin/feat/native-wasb-runner: gone]
feat(detectors): Linux-native WASB runner (M3.5) + worker real-trajectory extraction
+ feat(nightly-scheduler-fix) 4051ed5 (C:/Users/avidu/Projects/badminton-
highlight-indexer/.claude/worktrees/sched-fix) [origin/feat/nightly-scheduler-fix]
feat(nightly): make the Cloud Run scheduler path work end-to-end + turnkey email/trigger
feat(packaging-p0a-app-home) 7c81c4d [origin/feat/packaging-p0a-app-home: gone]
feat(pkg): P0a ? resolved app-data HOME (RALLY_APP_HOME) for config/.env/DB/output (default-OFF)
feat(packaging-p0b-config-redact) 07157cd [origin/feat/packaging-p0b-config-redact:
gone] feat(pkg): P0b ? redact secret-shaped fields from the unauthenticated GET /api/config
feat(packaging-p0c-bind-host) 6b78c0a [origin/feat/packaging-p0c-bind-host: gone]
feat(pkg): P0c ? config/profile-driven bind address + DNS-rebinding Host-header guard (default-
OFF)
feat(packaging-p0d-scipy-ci) 86d023a [origin/feat/packaging-p0d-scipy-ci: gone]
feat(pkg): P0d ? declare scipy (undeclared serving dep) + license-scan & built-wheel CI lanes

=== any branch/PR mentioning gate / windowing / snapshot? ===
feat/prominent-court-gate
fix/served-gate-a-corpus-regrounding
```

40. assistant (2026-06-22T12:27:05.457Z)

Now I see the real picture: this checkout is ****shared by ~12 concurrent sessions**** (worktrees + the main checkout). Critically:

- The `M tests/test\_worker.py` is a ****sibling actively adding more worker tests**** to the #319 branch ***right now*** (two new skip-branch coverage tests, uncommitted). Merging #319 at `a2c8be0` would strand that in-flight work.
- ****Task 2 is already in progress on a sibling branch**** `fix/served-gate-a-corpus-regrounding` (and the `tier1-windowing` worktree holds `master`) ? I should not duplicate it.
- `master` is checked out in another worktree, so I can't switch to it here (I'll merge server-side via `gh` instead).

Let me get the fresh PR list and the state of the gate-A re-grounding branch before deciding what's genuinely safe to merge.

41. user (2026-06-22T12:27:11.156Z)

```
=== FRESH open PR list ===
[{"headRefName":"feat/nightly-scheduler-
fix","mergeStateStatus":"CLEAN","mergeable":"MERGEABLE","number":320,"title":"feat(nightly):
make the Cloud Run scheduler path work E2E + turnkey email/trigger"},"{"headRefName":"fix/worker-
train-dated-label-
join","mergeStateStatus":"UNSTABLE","mergeable":"MERGEABLE","number":319,"title":"fix(worker):
strip ISO-date prefix when joining labels to videos in train"},"{"headRefName":"work/golden-
corpus-
hygiene","mergeStateStatus":"CLEAN","mergeable":"MERGEABLE","number":316,"title":"docs(data):
GCS corpus source-of-truth map + Phase-1 metadata migration"}]

=== gate-A re-grounding branch: pushed? ahead/behind master? ===
de4240f60f65769630890474863698210fd38970
de4240f docs(eval): re-ground the +0.074 "standing guard" claims in QUALITY_ITERATIONS
e59b2fd test(eval): re-ground served Gate-A litmus on the present (multicourt) golden set
--- remote tracking for gate-A branch ---
fix/served-gate-a-corpus-regrounding -> origin/fix/served-gate-a-corpus-regrounding [[gone]]

=== prominent-court-gate branch ===
8b13a43 feat(eval): prominent-court window gate ? drop multicourt background-court rallies
(default-OFF)
feat/prominent-court-gate -> origin/feat/prominent-court-gate [[gone]]
```

42. assistant (2026-06-22T12:27:53.466Z)

The picture is shifting live (siblings are committing as I work):

- **#320** is now **CLEAN** (full suite finished green).
- **#319** flipped to **UNSTABLE** ? the sibling likely just pushed those new worker tests, re-triggering CI.
- **#316** stays **CLEAN**.
- The **gate-A re-grounding (task 2)** already exists as 2 commits on `fix/served-gate-a-corpus-regrounding`(`e59b2fd test(eval): re-ground served Gate-A litmus?` + `de4240f docs(eval): re-ground the +0.074 standing-guard claims`), but its upstream shows [gone].`

I need three more facts before merging: whether #320's worktree has uncommitted WIP (stranding risk), whether the gate-A branch is pushed/has a PR, and whether gate-A's fix is already on master.

43. user (2026-06-22T12:28:01.794Z)

```
=== #320 worktree (sched-fix) uncommitted? ===
(clean ? no tracked WIP)

=== gate-A: any PR (any state)? ===
[{"mergeStateStatus":"UNKNOWN","number":236,"state":"MERGED","title":"test(eval): re-ground
served Gate-A litmus on the present (multicourt) golden set"}]

=== is gate-A fix already in origin/master? (search the re-grounded value) ===
origin/master:tests/test_served_gate_a_regression.py:26:gate is ~F1-neutral (?F1 ? -0.001) while
still cutting over-segmentation (1.16?0.64).
origin/master:tests/test_served_gate_a_regression.py:177:      # over-seg 1.16 ? 0.64 (was +0.074
/ 1.68 ? 1.01 on the retired original-6). ~F1-neutral band:
origin/master:tests/test_served_gate_a_regression.py:179:      assert mean_seg_off ==
pytest.approx(1.16, abs=0.10), mean_seg_off
--- on the gate-A branch: ---
178:      assert mean_delta == pytest.approx(0.0, abs=0.03), mean_delta
179:      assert mean_seg_off == pytest.approx(1.16, abs=0.10), mean_seg_off
180:      assert mean_seg_on == pytest.approx(0.64, abs=0.10), mean_seg_on
```

```
=== #319 latest remote head vs local ===
b3b495212b8608d44c45b89ac08349eabe549802      refs/heads/fix/worker-train-dated-label-join
local a2c8be0; sibling WIP uncommitted in main tree
```

44. assistant (2026-06-22T12:29:41.844Z)

Now everything is clear:

- **#320's worktree is clean** (no stranding risk) ? safe to merge.
- **Gate-A: PR #236 already merged** that re-grounding ? but it pinned `1.16`, and `origin/master` still has `1.16` while the **local corpus** has since drifted to `1.36`. So task 2 is genuinely stale *again* (corpus grew) and needs re-grounding for the current corpus. The local `fix/served-gate-a-corpus-regrounding` branch is just the dead remnant of merged #236 (squash artifact), not new work.
- **#319's remote advanced to `b3b4952`** ? the sibling pushed the two new coverage tests; that's why it's UNSTABLE (CI re-running). Better coverage incoming ? I'll merge it once green.

Phase 1 ? merge the two ready green PRs. Merging #316 first (so #319's `DATA\_IN\_GCS §7` comment refs resolve), then #320 (no `--delete-branch` since its branch is live in a worktree).

45. user (2026-06-22T12:29:50.812Z)

```
=== MERGE #316 (docs: DATA_IN_GCS) ===
```

```
=== post-merge: origin/master head ===
d301a7f..e221cb9 master    -> origin/master
e221cb9 docs(data): GCS corpus source-of-truth map + Phase-1 metadata migration (#316)
```

46. assistant (2026-06-22T12:29:59.617Z)

#316 merged ? `origin/master` now at `e221cb9`. Now #320 (squash, no `--delete-branch` since its branch is live in the `sched-fix` worktree):

47. user (2026-06-22T12:30:08.404Z)

```
=== MERGE #320 (nightly scheduler) ===
```

```
=== post-merge: origin/master head ===
e221cb9..48f9d20 master    -> origin/master
48f9d20 feat(nightly): make the Cloud Run scheduler path work E2E + turnkey email/trigger (#320)
```

```
=== #319 CI status now ===
Ruff lint      fail    10s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952535184/job/82712711633
Built wheel smoke (install + launch)    pass    1m10s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952535184/job/82712711912
Dependency license scan (no copyleft in runtime deps)    pass    34s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952535184/job/82712711911
Full test suite (offline, mocked)    pass    2m29s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952535184/job/82712711460
Golden fixtures cross-OS (macos-latest)    pass    14s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952535184/job/82712711590
Golden fixtures cross-OS (ubuntu-latest)    pass    26s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952535184/job/82712711690
Golden fixtures cross-OS (windows-latest)    pass    1m4s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952535184/job/82712711385
Leak scan (PII / attribution guard)    pass    5s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952535184/job/82712711362
Mypy (lenient baseline)    pass    33s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952535184/job/82712711522
Syntax sweep + import smoke (no GPU stack)    pass    18s
```

<https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952535184/job/82712711622>

48. assistant (2026-06-22T12:30:17.775Z)

#320 merged ? `origin/master` at `48f9d20`.

***#319 is red on Ruff lint** (full suite passes ? the sibling's two new tests are functionally correct but have a lint violation). The working tree holds exactly that pushed version, so I can reproduce locally. Let me see the violation.

49. user (2026-06-22T12:30:22.313Z)

```
=== ruff on working-tree tests/test_worker.py (== #319 b3b4952) ===
All checks passed!
=== confirm working tree matches remote b3b4952 ===
* branch          fix/worker-train-dated-label-join -> FETCH_HEAD
tests/test_worker.py | 1 -
1 file changed, 1 deletion(-)
(empty diff above = working tree == b3b4952)
```

50. assistant (2026-06-22T12:30:50.605Z)

Ruff passes on my working tree but it ****differs from `b3b4952` by one deleted line**** ? the sibling is ***actively fixing the exact ruff error right now*** (deleted the offending line, not yet committed). So #319 is mid-repair by its owner. I'll let them finish and poll it.

Before I touch task 2 (gate-A), I need to check the `tier1-windowing` sibling ? its name suggests it works the same windowing/gate area, and re-grounding the snapshot could collide with or be invalidated by their work.

51. user (2026-06-22T12:30:53.276Z)

```
=== tier1-windowing worktree: branch + commits ahead of origin/master ===
a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge (?F1 +0.139, 6/0) (#187)
90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline (#185)
960c00d docs: rally-quality research + eval-framework roadmap (foundation doc) (#184)
--- ahead of origin/master? ---
=== tier1-windowing uncommitted files (is it touching the gate-A test or windowing?) ===
```

52. assistant (2026-06-22T12:31:36.187Z)

`tier1-windowing` is ****dormant**** ? checked out at old master (`a3b209a` = #187), zero commits ahead, no WIP. The name was a red herring; no collision risk. So task 2 (gate-A re-grounding) has no active sibling ? I'll do it.

Key constraint: the gate-A test needs the local golden corpus (`output/human\_lovo\_manifest.json`, gitignored, only in the ****main**** checkout) to measure ? a fresh worktree would just skip it. Let me read the full measurement block so I can re-ground all three pinned numbers consistently from one clean measurement.

53. user (2026-06-22T12:31:37.118Z)

```
60      )
61
62
63      def _golden_specs():
64          """Return the manifest specs whose trajectory + label files are all present, else
65          []."""
66          if not os.path.isfile(MANIFEST):
67              return []
68          with open(MANIFEST, encoding="utf-8") as f:
```

```

68         specs = json.load(f)
69         ok = [s for s in specs
70               if os.path.isfile(str(s.get("trajectory", "")))
71               and os.path.isfile(str(s.get("labels", "")))]
72         return ok if len(ok) >= 6 else []
73
74
75     _SPECS = _golden_specs()
76     _skip = pytest.mark.skipif(
77         not _SPECS,
78         reason=f"golden trajectories absent (manifest {MANIFEST!r}); runs on the owner box",
79     )
80
81
82     # ----- served-path
litmus
83
84     @_skip
85     def test_served_path_runs_on_all_golden_videos():
86         """The production FusionSegmenter decision path runs end-to-end (no GPU/Gemini) on
EVERY
87         present golden trajectory and yields a handoff window list for each, for both gate
settings.
88         Asserts against the present corpus count (>= 6), not a stale fixed 6 ? the golden
set grows."""
89         results = sga.evaluate_manifest(_SPECS)
90         assert len(results) == len(_SPECS)
91         assert len(results) >= 6
92         for r in results:
93             assert r.num_gt > 0
94             assert r.n_handoff_off >= r.n_handoff_on # Gate A only ever drops windows,
never adds
95
96
97     @_skip
98     def test_served_gate_a_does_not_materially_regress_f1():
99         """SERVED no-material-regression guard: on the production windowing Gate A must not
MATERIALLY
100         hurt F1. On the present recall-starved multicourt corpus the served path is under-
segmented
101         (over-seg ratio < 1), so Gate A ? which only DROPS windows ? trims a few borderline-
true
102         windows and is mildly negative, NOT the ~neutral it was on the original single-court
6. We pin
103         a band that admits this measured mild loss but trips on a material regression."""
104         results = sga.evaluate_manifest(_SPECS)
105         agg = sga.aggregate(results)
106         # Honest measured value (2026-06-18, present 9-clip multicourt golden set): mean
served
107         # ?F1 ? -0.034 (was ? -0.008 on the retired original-6). Band re-measured to admit
the mild
108         # recall-starved loss while still tripping if Gate A materially destroys served F1.
109         assert -0.06 <= agg["mean_delta_f1"] <= 0.03, agg
110         # And Gate A must never collapse served F1 to ~half the OFF baseline (measured ratio
? 0.83).
111         assert agg["mean_f1_on"] >= 0.5 * agg["mean_f1_off"] - 1e-9, agg
112
113
114     @_skip
115     def test_served_gate_a_does_not_increase_over_segmentation():
116         """Gate A only ever DROPS windows, so the served over-seg ratio must not rise when
it's on
117         (the direction-of-effect invariant ? even where the absolute lift is ~neutral)."""
118         results = sga.evaluate_manifest(_SPECS)
119         agg = sga.aggregate(results)

```

```

120     assert agg["mean_seg_ratio_on"] <= agg["mean_seg_ratio_off"] + 1e-9, agg
121     # Per-video too: the gate is monotone (kept windows ? all windows).
122     for r in results:
123         assert r.seg_ratio_on <= r.seg_ratio_off + 1e-9, r
124
125
126     @_skip
127     def test_served_uses_production_windowing_operating_point():
128         """Pin the served operating point to config.json's indexing defaults ? if someone
retunes
129         production windowing, this litmus's premise (and the divergence it documents) is
revisited
130         deliberately, not silently."""
131         assert sga.PROD_VELOCITY_THRESH == 0.02
132         assert sga.PROD_MERGE_GAP == 2.0
133         assert sga.PROD_MIN_WINDOW == 2.0
134
135
136     # ----- Gate A under the permissive PROPOSAL
windowing
137
138     # distill_local.PROPOSAL ? the permissive, recall-oriented candidate windowing the
OFFLINE
139     # ablation/fusion_compare harness builds on (low velocity thresh, short merge/min-dur ?
many
140     # small windows). On the original-6 single-court corpus the Gate-A lift here was +0.074
(6/6,
141     # p=0.031; archived). On the present multicourt corpus that lift is GONE ? the gate is
~F1-neutral
142     # but still cuts over-segmentation. We pin the present-corpus behaviour, not the retired
+0.074.
143     _PROPOSAL = (0.005, 1.0, 0.5)
144
145
146     @_skip
147     def test_proposal_windowing_gate_a_is_f1_neutral_and_cuts_over_seg():
148         """CONTRAST pin: under the OFFLINE harness's permissive proposal windowing, the SAME
149         ``rally_gate`` net-crossing gate the served seam uses behaves DIFFERENTLY than under
the
150         production windowing. The durable invariant (any corpus): the gate reduces over-
segmentation
151         (seg_on < seg_off), since it only drops windows. On the present multicourt corpus it
is
152         ~F1-neutral (?F1 ? -0.001, was +0.074 on the now-retired original-6) ? proving the
served
153         behaviour is a property of the operating point + corpus, not a broken gate."""
154         d_on, d_off, seg_on, seg_off = [], [], [], []
155         for s in _SPECS:
156             fps, fw = float(s["fps"]), float(s["frame_width"])
157             pts = TrackNetRunner.parse_trajectory_csv(s["trajectory"])
158             wins = TrackNetRunner.trajectory_to_action_windows(
159                 pts, fps=fps, frame_width=fw, velocity_thresh=_PROPOSAL[0],
160                 merge_gap=_PROPOSAL[1], min_window_duration=_PROPOSAL[2])
161             iv = [(w["start"], w["end"]) for w in wins]
162             est = rally_gate.estimate_play_axis(pts)
163             gated = rally_gate.gate_windows(pts, iv, fps, min_crossings=0, estimate=est)
164             on = [(g.start, g.end) for g in gated if g.crossings >= 2]
165             gts = load_golden_gts(s["labels"])
166             d_on.append(score(on, gts).f1)
167             d_off.append(score(iv, gts).f1)
168             seg_on.append(_segm.segment_count_ratio(on, gts))
169             seg_off.append(_segm.segment_count_ratio(iv, gts))
170             n = len(d_on)
171             mean_delta = (sum(d_on) - sum(d_off)) / n
172             mean_seg_off, mean_seg_on = sum(seg_off) / n, sum(seg_on) / n

```

```

173         # Durable direction-of-effect (corpus-independent): the gate only drops windows, so
under the
174         # over-segmenting proposal windowing it lowers the over-seg ratio.
175         assert mean_seg_on < mean_seg_off, (mean_seg_on, mean_seg_off)
176         # Honest measured snapshot (2026-06-18, present 9-clip multicourt set): mean ?F1 ?
-0.001,
177         # over-seg 1.16 ? 0.64 (was +0.074 / 1.68 ? 1.01 on the retired original-6).
~F1-neutral band:
178         assert mean_delta == pytest.approx(0.0, abs=0.03), mean_delta
179         assert mean_seg_off == pytest.approx(1.16, abs=0.10), mean_seg_off
180         assert mean_seg_on == pytest.approx(0.64, abs=0.10), mean_seg_on
181

```

54. assistant (2026-06-22T12:32:09.702Z)

I now have the full test. Only **one** test fails (the proposal-windowing snapshot, lines 178?180); the other corpus-dependent tests in this file still pass on the current corpus. I need the current measured `mean_delta`, `mean_seg_off`, `mean_seg_on` to re-ground lines 178?180 + the comments/docstring consistently. Let me check the manifest path format and write a small scratch measurement (scratch/ is gitignored, sanctioned) that replicates the test's exact loop.

55. user (2026-06-22T12:32:12.399Z)

```

=== manifest exists + path format (absolute?) ===
specs: 15
sample keys: ['name', 'trajectory', 'fps', 'frame_width', 'labels']
traj: C:/Users/avidu/Projects/Annotation Setup/Collect/Trajectories/adarsh_avi_full_traj.csv
labels: C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/Adarsh and
Avi.rallies.csv

```

56. assistant (2026-06-22T12:32:27.311Z)

Manifest has **15** specs now (grew from 9 ? that's the drift source), and paths are **absolute** ? so an isolated worktree run with `SERVED_GATE_A_MANIFEST` pointing at this manifest will resolve the data. Let me write a faithful scratch measurement (exact copy of the test's loop) and run it in the main checkout to get the current numbers.

57. assistant (2026-06-22T12:32:37.924Z)

```

"""Scratch (gitignored): re-measure the proposal-windowing Gate-A snapshot on the CURRENT local
golden corpus, replicating tests/test_served_gate_a_regression.py lines 154-172 exactly, so the
pinned values (mean_delta / mean_seg_off / mean_seg_on) can be re-grounded honestly."""
import json
import os

from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
from backend.pipeline.detectors import rally_gate
from backend.eval.calibrate_wasb import load_golden_gts
from backend.eval.metrics import score
from backend.eval import segmentation_metrics as _segm

MANIFEST = os.path.join("output", "human_lovo_manifest.json")
with open(MANIFEST, encoding="utf-8") as f:
    specs = json.load(f)
_SPECS = [s for s in specs
            if os.path.isfile(str(s.get("trajectory", "")))
            and os.path.isfile(str(s.get("labels", "")))]
print(f"present specs (traj+labels both on disk): {len(_SPECS)} / {len(specs)}")
print("names:", sorted(s["name"] for s in _SPECS))

_PROPOSAL = (0.005, 1.0, 0.5)
d_on, d_off, seg_on, seg_off = [], [], [], []
for s in _SPECS:
    fps, fw = float(s["fps"]), float(s["frame_width"])

```



```

pts = TrackNetRunner.parse_trajectory_csv(s["trajectory"])
wins = TrackNetRunner.trajectory_to_action_windows(
    pts, fps=fps, frame_width=fw, velocity_thresh=_PROPOSAL[0],
    merge_gap=_PROPOSAL[1], min_window_duration=_PROPOSAL[2])
iv = [(w["start"], w["end"]) for w in wins]
est = rally_gate.estimate_play_axis(pts)
gated = rally_gate.gate_windows(pts, iv, fps, min_crossings=0, estimate=est)
on = [(g.start, g.end) for g in gated if g.crossings >= 2]
gts = load_golden_gts(s["labels"])
d_on.append(score(on, gts).f1)
d_off.append(score(iv, gts).f1)
seg_on.append(_segm.segment_count_ratio(on, gts))
seg_off.append(_segm.segment_count_ratio(iv, gts))
n = len(d_on)
mean_delta = (sum(d_on) - sum(d_off)) / n
mean_seg_off, mean_seg_on = sum(seg_off) / n, sum(seg_on) / n
print(f"n                = {n}")
print(f"mean_delta        = {mean_delta:.4f}    (pinned: approx 0.0, abs=0.03)")
print(f"mean_seg_off        = {mean_seg_off:.4f}    (pinned: approx 1.16, abs=0.10)")
print(f"mean_seg_on         = {mean_seg_on:.4f}    (pinned: approx 0.64, abs=0.10)")
print(f"direction (on<off): {mean_seg_on < mean_seg_off}")

```

58. user (2026-06-22T12:32:37.949Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\measure_gate_a.py (file state is current in your context ? no need to Read it back)

59. user (2026-06-22T12:32:43.770Z)

Traceback (most recent call last):

File "C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\measure_gate_a.py", line 7, in <module>

```

    from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
ModuleNotFoundError: No module named 'backend'

```

60. user (2026-06-22T12:32:54.065Z)

```

present specs (traj+labels both on disk): 15 / 15
names: ['Badminton_BXH_2', 'Boxhill_Doubles', 'GX010128', 'GX010137', 'GX010141', 'GX020094',
'GX030094', 'adarsh_avi_singles', 'gbaaddy', 'kushagra_singles', 'largetest_doubles',
'mahadevpura_1', 'mahadevpura_2', 'mahadevpura_singles', 'testlarge_short']
n                = 15
mean_delta       = 0.0287    (pinned: approx 0.0, abs=0.03)
mean_seg_off     = 1.3640    (pinned: approx 1.16, abs=0.10)
mean_seg_on      = 0.7919    (pinned: approx 0.64, abs=0.10)
direction (on<off): True

```

61. assistant (2026-06-22T12:33:35.851Z)

Measured on the **current 15-clip corpus** (was 9 at the 2026-06-18 snapshot ? that's the drift):

```

- `mean_delta` = 0.0287 (old pin `0.0 ±0.03` ? now marginal)
- `mean_seg_off` = 1.364 (old pin `1.16` ? fails)
- `mean_seg_on` = 0.792 (old pin `0.64`)
- direction invariant `on < off` still holds ?

```

The docstring also prose-pins the **served-path** numbers (``9-clip``, ``F1`` ? `-0.034``, ratio ``0.83``) for the passing tests ? I'll re-ground those too for an honest, complete update. Let me measure the served aggregate.

62. user (2026-06-22T12:33:51.468Z)

```

mean_delta       = 0.0287    (pinned: approx 0.0, abs=0.03)

```

```

mean_seg_off  = 1.3640  (pinned: approx 1.16, abs=0.10)
mean_seg_on   = 0.7919  (pinned: approx 0.64, abs=0.10)
direction (on<off): True
--- SERVED path aggregate ---
served n      = 15
mean_delta_f1 = -0.0233  (band -0.06..0.03; docstring says ~-0.034)
mean_f1_on    = 0.2134
mean_f1_off   = 0.2366
f1_on/off ratio = 0.9017  (docstring says ~0.83)
mean_seg_ratio_off = 0.5825
mean_seg_ratio_on  = 0.4696

```

63. assistant (2026-06-22T12:34:20.492Z)

All current numbers captured (15-clip corpus, 2026-06-22):

```

| metric | old pin | measured now |
|---|---|---|
| proposal `mean_delta` | ?0.0 | **+0.029** |
| proposal `mean_seg_off` | 1.16 | **1.36** |
| proposal `mean_seg_on` | 0.64 | **0.79** |
| served `mean_delta_f1` | ?-0.034 | **-0.023** |
| served f1 on/off ratio | ?0.83 | **0.90** |
| corpus size | 9 | **15** |

```

All the passing tests' bands still hold; only the proposal snapshot trips. Now I'll create an ****isolated worktree off `origin/master`**** (per the shared-checkout convention) to make the edit safely, since the main checkout is on a sibling-occupied branch.

64. user (2026-06-22T12:34:25.869Z)

```

* branch          master      -> FETCH_HEAD
Preparing worktree (new branch 'fix/gate-a-litmus-reground-n15')
branch 'fix/gate-a-litmus-reground-n15' set up to track 'origin/master'.
HEAD is now at 48f9d20 feat(nightly): make the Cloud Run scheduler path work E2E + turnkey
email/trigger (#320)
=== worktree HEAD ===
48f9d20 feat(nightly): make the Cloud Run scheduler path work E2E + turnkey email/trigger (#320)
=== sanity: stale numbers present in the test on this fresh base? ===
19:retired from the box ? their labels are gone ? so the present golden set is the 9
**multicourt**
24:windows, on this under-segmented corpus it is mildly **F1-negative** (mean served ?F1 ?
-0.034),
26:gate is ~F1-neutral (?F1 ? -0.001) while still cutting over-segmentation (1.16?0.64).
106:  # Honest measured value (2026-06-18, present 9-clip multicourt golden set): mean served
107:  # ?F1 ? -0.034 (was ? -0.008 on the retired original-6). Band re-measured to admit the
mild
110:  # And Gate A must never collapse served F1 to ~half the OFF baseline (measured ratio ?
0.83).
176:  # Honest measured snapshot (2026-06-18, present 9-clip multicourt set): mean ?F1 ?
-0.001,
177:  # over-seg 1.16 ? 0.64 (was +0.074 / 1.68 ? 1.01 on the retired original-6). ~F1-neutral
band:
179:  assert mean_seg_off == pytest.approx(1.16, abs=0.10), mean_seg_off
180:  assert mean_seg_on == pytest.approx(0.64, abs=0.10), mean_seg_on

```

65. assistant (2026-06-22T12:34:34.661Z)

Worktree ready on a clean ``origin/master`` base. Let me read the exact regions to edit in the worktree's copy.

66. user (2026-06-22T12:34:36.061Z)

14 CI where the golden trajectories are absent; runs on the owner box where they are

```

present, over
15     **whatever golden clips are currently on disk** (`_golden_specs` requires >= 6).
16
17     CORPUS NOTE ? re-grounded 2026-06-18 (the golden set is mutable and grows over time).
The
18     original 6 *single-court* golden clips this litmus was first calibrated on (2026-06-14)
have been
19     retired from the box ? their labels are gone ? so the present golden set is the 9
**multicourt**
20     clips now in `output/` (mahadevpura_1/2, GX0101xx, GX0x0094, Badminton_BXH_2,
Boxhill_Doubles).
21     On that corpus the served path is **recall-starved**: WASB tracks a single shuttle while
several
22     courts rally at once, so `trajectory_to_action_windows` emits FEWER windows than there
are
23     rallies (served over-seg ratio < 1, i.e. under-segmentation). Because Gate A only ever
DROPS
24     windows, on this under-segmented corpus it is mildly **F1-negative** (mean served ?F1 ?
-0.034),
25     not the -neutral -0.008 the original 6 showed. Under the permissive PROPOSAL windowing
the same
26     gate is -F1-neutral (?F1 ? -0.001) while still cutting over-segmentation (1.16?0.64).
27
28     The canonical **+0.074 / 6-of-6 / p=0.031** Gate-A win was a property of the
*original-6* corpus
29     and is NOT reproduced here (those clips are no longer on the box); it is preserved in
the archive
30     (`docs/archives/quality-iterations/2026-06-first-ab-experiment/`). These tests now pin
the
31     **present-corpus** behaviour so a future windowing/corpus change that silently shifts it
trips the
32     litmus. The pinned magnitudes are a date-stamped snapshot of the present golden set and
will need
33     re-measurement as the corpus grows.
34     ""

```

67. assistant (2026-06-22T12:35:23.267Z)

Now I'll make the re-grounding edits in the worktree ? updating the docstring CORPUS NOTE, the served-test comment, and the proposal-test snapshot pins, all to the measured 15-clip values, honestly noting this is the second re-grounding.

68. user (2026-06-22T12:35:37.275Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gate-a-reground\tests\test_served_gate_a_regression.py has been updated successfully. (file state is current in your context ? no need to Read it back)

69. user (2026-06-22T12:35:45.187Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gate-a-reground\tests\test_served_gate_a_regression.py has been updated successfully. (file state is current in your context ? no need to Read it back)

70. user (2026-06-22T12:35:52.119Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gate-a-reground\tests\test_served_gate_a_regression.py has been updated successfully. (file state is current in your context ? no need to Read it back)

71. assistant (2026-06-22T12:35:57.940Z)

Now let me check whether `QUALITY\_ITERATIONS.md` carries the same stale numbers (PR #236 also re-grounded it), then verify the edited test passes in the worktree against the main checkout's

corpus.

72. user (2026-06-22T12:36:24.491Z)

```
=== QUALITY_ITERATIONS.md stale proposal-windowing numbers? ===
337:**MEASURED ? golden n=6 (W1 cap=12 ? +hard-cap):** ?F1 **+0.006** [?0.019, +0.033], **CI
spans 0,
373:### Tier 1 (W2) ? net-crossings rally-state gate on W1: ?F1 +0.019, 6/0 *(2026-06-16)*
398:### Tier 1 (W1) ? bounded windowing fixes the over-merge: ?F1 +0.139, 6/0 *(2026-06-16)*
477: | min_crossings | mean F1 | precision | recall | real-rally windows lost vs off | ?F1 vs
off |
501:served ?F1 ?0.008, ~12 rallies dropped, 2/6 losses). If Gate A is ever opted back in on the
served path, `mc=1` is the
540:| Variant (honest LOVO, IoU ? 0.5) | F1 | ?F1 vs prior | 95% CI | sign test | CI clears 0 |
559:**neutral-to-negative, mean served ?F1 ?0.008**, dropping ~12 real rallies across the golden
6 (2 losses, 0
568:left to cut and the gate instead occasionally trims a borderline true window ? slightly
negative ?F1. The gate
590:> retired from the box (their labels are gone); the present golden set is **9 multicourt
clips**. On that
592:> served Gate-A ?F1 is now **?0.034** (was ?0.008), and under the permissive proposal
windowing the gate is
593:> **~F1-neutral** (?F1 ? ?0.001) while still cutting over-seg (1.16?0.64) ? it no longer
reproduces the +0.074
599:(?F1 +0.074, **6/6 golden matches, sign-test p=0.031**, CI [+0.042,+0.104]; over-seg
1.68?1.01 ? the
623:**Gate-A (net-crossings ? 2) is a measured win** ? ?F1 +0.074, 6/6 matches, sign-test **p =
0.031**, CI

=== VERIFY edited test passes against main-checkout corpus ===
..... [100%]
5 passed in 21.39s
```

73. assistant (2026-06-22T12:36:41.046Z)

The edited test passes (**5 passed**). Now `QUALITY\_ITERATIONS.md` has the 2026-06-18 re-grounding note (lines ~590?593) with the same stale numbers (`9 multicourt clips`, `?0.034`, `?0.001`, `1.16?0.64`). The other matches are historical original-6/W1/W2 records and must stay. Let me read and update just that note.

74. user (2026-06-22T12:36:42.005Z)

```
585 the proposal operating point** and re-run the #148 litmus to confirm the lift transfers.
The #148 litmus is
586 now the standing guard: it pins the honest measured behaviour on the **present golden
corpus** and trips if a
587 future windowing/corpus change silently shifts it.
588
589 > **Re-grounded 2026-06-18.** The original-6 single-court golden clips this litmus was
calibrated on have been
590 > retired from the box (their labels are gone); the present golden set is **9 multicourt
clips**. On that
591 > recall-starved corpus (WASB tracks one shuttle while several courts rally ? served
over-seg ratio < 1) the
592 > served Gate-A ?F1 is now **?0.034** (was ?0.008), and under the permissive proposal
windowing the gate is
593 > **~F1-neutral** (?F1 ? ?0.001) while still cutting over-seg (1.16?0.64) ? it no longer
reproduces the +0.074
594 > live. The canonical **+0.074 / 6-of-6 / p=0.031** win remains a real result of the
original-6 corpus,
595 > preserved in the [first-A/B archive](archives/quality-iterations/2026-06-first-ab-
experiment/README.md).
596
597 ### Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125
```

corroboration *(2026-06-14)*
598 **Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness).** The
measured win
599 (?F1 +0.074, **6/6 golden matches, sign-test p=0.031**, CI [+0.042,+0.104]; over-seg
1.68?1.01 ? the

75. user (2026-06-22T12:36:52.152Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\gate-a-reground\docs\QUALITY_ITERATIONS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

76. assistant (2026-06-22T12:37:09.485Z)

Test passes and the doc note is updated. Now lint the edited test and run the full suite in the worktree (with the manifest env so the gate-A test actually runs, not skips) to confirm everything is green.

77. user (2026-06-22T12:38:55.577Z)

```
=== ruff on edited test ===
All checks passed!
=== FULL SUITE (with corpus manifest so gate-A runs) ===
..... [ 86%]
..... [ 91%]
..... [ 96%]
..... [100%]
1418 passed, 2 skipped in 101.51s (0:01:41)
=====
RESULT: PASS - all tests green
=====
```

78. assistant (2026-06-22T12:39:10.842Z)

Full suite **1418 passed, 0 failed** (2 skipped are unrelated data-gated ones), ruff clean. Task 2 is verified. Now commit (explicit paths, re-verifying HEAD per shared-checkout discipline), push, and open the PR.

79. user (2026-06-22T12:39:21.774Z)

```
=== pre-commit verify: branch + base clean ===
fix/gate-a-litmus-reground-n15
(empty = base is origin/master)
=== staged ===
docs/QUALITY_ITERATIONS.md      | 16 ++++++-----
tests/test_served_gate_a_regression.py | 41 ++++++-----
2 files changed, 31 insertions(+), 26 deletions(-)
[fix/gate-a-litmus-reground-n15 7f15a1b] test(eval): re-ground served Gate-A litmus +
QUALITY_ITERATIONS on the 15-clip golden corpus
2 files changed, 31 insertions(+), 26 deletions(-)
=== push ===
remote:
branch 'fix/gate-a-litmus-reground-n15' set up to track 'origin/fix/gate-a-litmus-reground-n15'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
* [new branch]      fix/gate-a-litmus-reground-n15 -> fix/gate-a-litmus-reground-n15
```

80. assistant (2026-06-22T12:39:28.941Z)

Pushed. Now open the PR for the gate-A re-grounding.

81. user (2026-06-22T12:39:41.797Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/322>

82. assistant (2026-06-22T12:39:49.267Z)

PR [#322](https://github.com/Khelsutra/badminton-highlight-indexer/pull/322) opened for task 2. Now let me check status of both #322 (re-grounding) and #319 (worker fix ? did the sibling's ruff fix land + go green?).

83. user (2026-06-22T12:39:57.193Z)

```
=== #319 (worker) ===
mergeable=UNKNOWN/UNKNOWN head=56c1415299e40f90c7d8d3ef3825b73a50a82f77
Built wheel smoke (install + launch)    pass    1m11s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952756124/job/82713460846
Dependency license scan (no copleft in runtime deps)    pass    34s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952756124/job/82713460819
Full test suite (offline, mocked)        pass    2m39s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952756124/job/82713460892
Golden fixtures cross-OS (macos-latest)    pass    22s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952756124/job/82713460947
Golden fixtures cross-OS (ubuntu-latest)    pass    26s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952756124/job/82713460830
Golden fixtures cross-OS (windows-latest)    pass    47s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952756124/job/82713460993
Leak scan (PII / attribution guard)        pass    9s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952756124/job/82713460806
Mypy (lenient baseline)    pass    37s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952756124/job/82713460768
Ruff lint    pass    10s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952756124/job/82713460813
Syntax sweep + import smoke (no GPU stack)    pass    21s
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27952756124/job/82713460750
```

```
=== #322 (gate-A reground) ===
mergeable=MERGEABLE/UNSTABLE
Dependency license scan (no copleft in runtime deps)    pending 0
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953290898/job/82715247393
Full test suite (offline, mocked)        pending 0    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953290898/job/82715247265
Golden fixtures cross-OS (macos-latest)    pending 0    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953290898/job/82715247216
Golden fixtures cross-OS (ubuntu-latest)    pending 0
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953290898/job/82715247213
Golden fixtures cross-OS (windows-latest)    pending 0
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953290898/job/82715247210
Ruff lint    pending 0    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953290898/job/82715247278
Leak scan (PII / attribution guard)        pass    7s    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953290898/job/82715247302
Built wheel smoke (install + launch)    pending 0    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953290898/job/82715247267
Mypy (lenient baseline)    pending 0    https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953290898/job/82715247219
Syntax sweep + import smoke (no GPU stack)    pending 0
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953290898/job/82715247214
```

84. assistant (2026-06-22T12:40:30.216Z)

***#319 is green** ? the sibling pushed the ruff fix (head `56c1415`), all 10 checks pass;

`mergeable` is just GitHub recomputing (transient). ****#322**** CI is running (will pass ? only a test+doc change; the litmus skips in CI).

While CI settles, let me re-check #319's mergeability and survey the docs registers (read-only) to plan tasks 3 (backlog) and 4 (fix/archive docs).

85. user (2026-06-22T12:40:35.486Z)

=== #319 mergeable now? ===
UNKNOWN/UNKNOWN
=== BACKLOG.md (head) ===

Backlog ? deferred items to revisit

Single tracker for work we consciously deferred, so nothing gets lost between sessions. Each item records ****what****, ****why it's deferred****, and ****the trigger to revisit****. Every entry traces to a real decision/PR/doc (no speculative work).

> Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all of these to ****GitHub issues**** instead (or in addition).
> Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.

Last updated: 2026-06-10.

Owned-model / multisport (source: `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`, `NEXT\_STEPS.md`)

- ? ****Rename the repo off "badminton"***** ? owner-emphatic "ASAP" action; the project is multisport. Touches remote name + README/CLAUDE.md framing + any hardcoded URLs. ****Revisit:**** owner action, before broad external sharing.
- ? ****Quantify cost-to-Gen-2 (the largest open risk)**** ? benchmark real VRAM/token/\$ for the VLM SFT+GRPO on a rented A100/L40S (ms-swift multimodal-GRPO examples use 678 GPUs) ****before**** standing up any owned serving stack. ****Revisit:**** before Phase-2 commitment.
- ? ****Define the ship-gate F1@tIoU target**** ? "very high P/R" must be operationalized (floor = beat heuristic LOVO 0.480; learned is now 0.626 post metric-fix). ****Revisit:**** at M0.4 harness.

Engineering hardening deferred from the 2026-06-10 audit sweep

- ? ****App-factory / lifespan + lazy torch-cv2 imports**** ? `uvicorn backend.main:app` leaves globals None, and importing the API pulls the GPU stack. Belongs with the async-job slice (#16). ****Revisit:**** when the async slice starts.
- ? ****WSL temp/frame/cache + Gemini remote-file GC**** ? staged videos, PNG frame dumps, `\_wasbcache`, and orphaned Gemini uploads are never cleaned. Needs a retention policy. ****Revisit:**** before any volume/batch processing.
- ? ****shlex-quote WSL config values**** ? ****DONE 2026-06-11 (#97):****
`detectors/base.py::wsl\_tilde\_quote` quotes everything after a leading `~/`, so quoting and tilde expansion coexist; applied to every `~`-capable arg in both runners;
`tests/test\_wsl\_tilde\_quoting.py` pins the command shapes (no quoted tilde can ever reach `\_wsl\_bash`). Found live: the naive quoting had silently broken wasb_hybrid staging for the default `~/clips`. Residual for tenancy-era: quote the remaining trusted-config interpolations (repo_dir/conda_env) if config ever becomes per-tenant.

Monetization / licensing

(source: `archives/MONETIZATION\_AUDIT.md` §7, ADR-005)

- ? ****Ultralytics Enterprise pricing**** ? get a quote ****only if**** the torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently ***not needed*** (swap done). Quote-only pricing. ****Revisit:**** if detector quality forces reconsidering YOLOv8.
- ? ****Codec patent thresholds (AVC/HEVC/AAC)**** ? confirm exact Via-LA (AVC) / Access Advance (HEVC) royalty rates & free thresholds ****with legal counsel**** before relying on the "small SaaS is under threshold" reading. Mitigation already chosen: emit royalty-free AV1/Opus output. ****Revisit:**** before commercial launch.
- ? ****Per-checkpoint weight/dataset licenses**** ? if we borrow pretrained weights (YOLOX,

ActionFormer, WASB, TrackNetV3), the `*code*` license ? the distributed `*weights/dataset*` terms. Either train our own or confirm each checkpoint. ****Revisit:**** when adopting any borrowed weights.

- ? ****Real GPU \$/video & latency**** ? benchmark the three LLM-step options (offline ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing the product.

****Revisit:**** when choosing the production LLM step.

Detector swap follow-ups

`*(source: PR #8, ADR-005)*`

- ? ****`supervision` ByteTrack deprecation**** ? ``sv.ByteTrack`` is deprecated as of supervision 0.28.0 and ****removed in 0.30.0****; ``requirements.txt`` is pinned `<0.30.0`` as a stopgap. Migrate to ``ByteTrackTracker`` from the separate ``trackers`` package before lifting the cap. ****Revisit:**** before bumping supervision past 0.30.
- ? ****Real-video validation of the swap**** ? the torchvision + ByteTrack Stage-1 is unit-tested with mocks but ****not yet validated on actual footage**** for detection/tracking quality. (ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so expect tuning.) ****Revisit:**** when a real test clip + the pipeline are run together.
- ? ****`yolo\_hybrid` naming**** ? the registry key, ``yolo_*`` config keys, and ``yolo-hybrid-`` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining. Optional rename. ****Revisit:**** if/when a clean break is worth the migration.

Golden-set subproject

`*(source: `docs/GOLDEN_SET_IMPLEMENTATION.md`, `docs/GOLDEN_SET_VENDORING.md`)*`

=== DOC_STATUS.md: any docs marked DONE/COMPLETE/archive-ready? ===

4:> ****Update cadence:**** whenever a doc changes lifecycle state (created / shipped a milestone / delivered / archived), and re-review at each clean-slate.

5:> ****Why this exists:**** so we can (a) ****review completeness**** at a glance, (b) ****update it as work finishes**** to prove we're done, and (c) ****never archive a foundational doc without honestly updating it first**** (owner directive, 2026-06-21).

11:- ****Marking something complete?**** Update the doc's own ****\$7 progress tracker**** (the source of truth for ***its* completeness**), then flip its row here, then ? if terminal ? run the \$2 archival checklist.

12:- ****This register does NOT duplicate completeness detail**** ? it indexes lifecycle + archive-readiness and points to each tracked doc's \$7.

15: DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: ****LIVING**** (permanent reference, never "done"), ****PROPOSAL/DESIGN**** (owner-gated, not started), ****REPORT**** (a terminal record of a run/launch), ****CONVENTION**** (a standing rule/guide).

18: Before a foundational/tracked doc moves to ``docs/archives/``:

19: 1. ****VERIFY claims against the code**** ? no stale assertions; tag load-bearing facts ``[verified]``. (A doc that says "nothing built" when code shipped is ***not*** archive-ready ? fix it first.)

20: 2. ****Flip Status ? DELIVERED / DONE / REJECTED + a completion note**** ? what shipped, where (PRs, paths, artifacts), and what was ****deferred / carried forward****.

22: 4. ****Keep the lineage**** ? ADR forward/back-links + "supersedes / superseded-by" so the evolution stays traceable.

23: 5. ****`git mv` the whole dir/file ? `docs/archives/past\_projects/`**** (terminal-state ONLY; live docs stay at ``docs/`` top level). Relative links inside become historical (note this in the doc's header).

26:> ****Worked example (the template):**** ****DECOUPLED_COMPUTE ? #270**** ? verified M0?M2+M3.5 delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward WASB-C3/DPA/`md5`), reindexed ``docs/README`` + ``past_projects/README``, kept the ADR-010?014 lineage, then ``git mv`` to ``docs/archives/past_projects/DECOUPLED_COMPUTE/``.

30: ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's \$7 is the completeness source-of-truth)

31: | Doc | Status | Archive-ready? | Next action |

34: | ``RALLY_QUALITY_RESEARCH.md`` | Foundation, ****tracked & ACTIVE**** (\$7) | No | Drives the R-series; complete when \$7.1 single quality number is attributed |

35: | ``R2_EVAL_METRIC_DESIGN.md`` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation pending) | Promote-or-reject toLF1 as the gate (ablation) ? then archive with the R-cluster |

36: | ``R1_MILESTONE_LAUNCH_REPORT.md`` | LAUNCHED (#204) ? terminal record | Soon (with the R-cluster) | Keep with R-series for narrative; archive when the track wraps |

38: | ``PER_VIDEO_WORKSPACE.md`` | DRAFT, IN PROGRESS (P0 = #264) | No | ? supersede-reconcile

(finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once) |

39:| `MULTI\_SIGNAL\_FUSION\_PLAN.md` | P1 DONE; P2+ owner-gated | No | P3/P4 (learned fusion, audio) ? needs GPU golden features |

44:| ****Data layer**** ? `docs/data\_pipeline/` (`GOLDEN\_SET\_IMPLEMENTATION` . `GOLDEN\_SET\_PHASE1\_VIDEOS` . `GOLDEN\_VIDEOS` . `VIDEO\_RECORDING\_GUIDELINES` . `GOLDEN\_DATA\_SHARING`) | LIVING (data, ****no-ML****) | n/a | Relocated 2026-06-21 to the isolated data pillar; tracked via `data\_pipeline/README.md`. Roora ***vendor*** docs archived ? `archives/roora-vendor/`. |

52:### 3b. Foundational / strategy references (LIVING ? archive only if ****superseded****, not when "done")

62:| `ROADMAP.md` / `CODE\_MAP.md` / `HOW\_RALLY\_DETECTION\_WORKS.md` / `KHELSUTRA\_PRODUCT\_OVERVIEW.md` | LIVING references | Keep current; not archive targets |

64:### 3c. Reports & terminal records (archive candidates once their track wraps)

67:| `REAL\_FOOTAGE\_VALIDATION\_REPORT.md` | validation complete | Motivates R2; archive with the R-cluster |

70:| `CODE\_AUDIT\_AND\_TEST\_HARDENING.md` . `HARDENING\_LOOP\_HANDOFF.md` . `DEFERRED\_HARDENING\_PLAN.md` | COMPLETED ? ****top-level pointer stubs**** (archived copies exist) | ****?** confirm intentional stubs (per #165) vs finish-archiving (finding #4)****** |

72:### 3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)

73:`PROJECT\_DOC\_TEMPLATE.md` . `NEXT\_STEPS.md` (? refreshed 2026-06-21) . `BACKLOG.md` . `EVALUATION.md` . `ABLATION\_LAYER.md` . `SETUP\_NEW\_MACHINE.md` . `TRACKNET\_WSL\_SETUP.md` . `README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data\_pipeline/`; the field-ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*

75:### 3e. Already archived (`docs/archives/`)

76:`past\_projects/DECOUPLED\_COMPUTE/` (? 2026-06-21, #270) . `past\_projects/low-regret-moves/` . ****`field-pmf/`** (field-ops + customer-feedback, 2026-06-21) . ****`roora-vendor/`** (annotation-vendor engagement, 2026-06-21) . `PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md` . `CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` . `DEFERRED\_HARDENING\_PLAN-COMPLETED-2026-06-14.md` . `HARDENING\_LOOP\_HANDOFF-COMPLETED-2026-06-14.md` . `COMMERCIAL\_READINESS\_REVIEW.md` . `MONETIZATION\_AUDIT.md` . `TEST\_RUN\_SUMMARY.md` . `research/` . `checkpoints/` . `quality-iterations/`. |

78:### 3f. ADR log ? `docs/archives/decisions/DECISIONS.md`

83:> ****?** Doc-restructuring done (2026-06-21, owner-directed):****** field-ops/customer-feedback ? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML ****data layer**** carved into `docs/data\_pipeline/`. Inbound + outbound links rewired; three dir READMEs added (the `data\_pipeline` one sets the "no segmentation/stitching" boundary).

86. assistant (2026-06-22T12:41:30.878Z)

****#319 is green**** (all 10 checks pass; `mergeable` is GitHub's transient recompute). I'll merge it server-side ****without**** `--delete-branch` (the main checkout is on that branch). Let me merge #319 and check #322's CI.

87. user (2026-06-22T12:41:40.360Z)

```
=== merge #319 (worker fix, all checks green) ===
! Pull request Khelsutra/badminton-highlight-indexer#319 was already merged
=== origin/master now ===
  * branch          master      -> FETCH_HEAD
db74f9b docs(data): mark DATA_IN_GCS §7b blocker #1 resolved by #319 (#321)
=== #322 CI ===
9
checks passing ^
Full test suite (offline, mocked)      pending 0      https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953290898/job/82715247265
MERGEABLE/UNSTABLE
```

88. assistant (2026-06-22T12:42:01.340Z)

****Task 1 ???** ? #319 was already merged (by the sibling who fixed the ruff error), plus a sibling follow-up #321 "mark DATA_IN_GCS §7b blocker #1 resolved by #319" also merged. `origin/master` is now `db74f9b` with all of #316/#319/#320/#321.

***322** (gate-A): 9/10 checks pass, only the full suite still running ? will merge shortly.

Now to tasks 3+4. Let me set up a docs worktree off the latest master and survey what's already in BACKLOG.md (re: golden regression fixtures) and whether anything is a clean archival candidate.

89. user (2026-06-22T12:42:06.311Z)

```
Preparing worktree (new branch 'docs/backlog-golden-fixtures-followups')
branch 'docs/backlog-golden-fixtures-followups' set up to track 'origin/master'.
HEAD is now at db74f9b docs(data): mark DATA_IN_GCS $7b blocker #1 resolved by #319 (#321)
=== BACKLOG: any golden-REGRESSION-fixtures / M4 / quality-floor entries already? ===

=== GOLDEN_REGRESSION_FIXTURES.md $7 tracker status ===
1:# Golden Regression Fixtures ? video-free, non-attributable regression suite
3:> **Status:** `IN PROGRESS` ? owner-gated . **Owner:** Avi . **Created:** 2026-06-16 . **Last
updated:** 2026-06-16 . **Legal:** owner **accepted the researched risk** 2026-06-16 (conditions
$3 stand; this is the owner's informed acceptance, **not** a retained-counsel opinion)
11:## 0. TL;DR
26:## 1. Problem & goal
38:## 2. Decisions locked
52:## 3. Legal foundation (researched ? gates on counsel, $8)
76:## 4. Architecture (design)
78:### 4a. Freeze seam + reuse map
82:> ? **Correction (audit 2026-06-16):** the replay does **NOT** import `windowing.py` ? there
is no `windowing.build_windows(preset)` / `WindowingPreset` / "default SERVED" call. The
windowing params travel as a **raw 3-tuple `(vt, mg, mwd)`** pinned in each fixture's
`provenance.json` and passed straight to `build_candidates`; the harness deliberately avoids the
named `windowing.SERVED` constant so a future SERVED retune can't silently churn fixtures.
(`segmentation_metrics`, `experiment.compare/Variant.spec_hash`, `regression.save_baseline`,
`eval_stats` are **not** on the M1 replay path either ? the quality-floor gate M4 will add the
baseline/stats pieces.)
84:### 4b. Tiered model (SEALED-FIXTURES)
92:### 4c. Anti-attribution measures (Tier-0)
105:### 4d. The two gates
109:### 4e. Serialization & optional protobuf
110:JSON with `sort_keys=True` + pre-quantized floats + LF is sufficient. **Protobuf is optional
(01)** ? adopt only if cross-OS byte-stable snapshots are wanted (proto3 `rally.fixture.v1`,
codegen + `protoc && git diff --exit-code` guard, `*.pb` binary -text`). Protobuf is **encoding,
not secrecy** (trivially `--decode_raw`).
112:### 4f. Encryption & keys (Tier-1 only)
117:## 5. Threat model (who's protected vs not)
130:## 6. Honest limits ? what a green suite does NOT prove
145:## 7. Deliverables & progress tracker ? **source of truth**
147:Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. One **small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
149:| ID | Deliverable | Depends on | Counsel-gated? | Status | PR |
151:| **P0** | **PII leak hygiene (partial ? paths + scan).** Scrubbed personal `C:\Users\?`
paths from code docstrings (`eval_partial_labels.py`, `gemini_refine.py`, `README.md`) + added
`tools/leak_scan.py` (32-hex MD5 + personal-path **structural** guard, optional gitignored
`.leakscan_private` name-list) + a CI `leak-scan` job + tests. ? The **functional video-name
rename** (`eval_baselines/*.json` keys / `rally_seg_eval.MULTICOURT_CLIPS` / quality-log names)
is **DEFERRED** per the owner's safe-scope decision (2026-06-16) ? it's coupled to the
gitignored `output/human_lovo_manifest.json` + data files + the training floor and collides with
the active quality track; run the private-list scan + a coordinated rename before open-sourcing.
Git-history rewrite (PR #77) also deferred. | ? | No | ? | _this PR_ |
152:| **M1** | **Fixture framework + synthetic Tier-0 + shared replay helper** (also delivered
M2-core + M3 ? see below). `backend/eval/golden_fixtures.py` (`replay_fixture`, schema dispatch,
`Path(__file__)`-relative) + 3 **tool-generated** synthetic fixtures + manifest + README +
`.gitattributes` LF pin + `tests/test_golden_fixtures.py`. CONTROL-only (no numpy execution on
the replay path); cross-OS verified. | ? | No (synthetic) | ? |
[#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) |
153:| **M2** | **Freeze/refresh extractor.** Core (`freeze_synthetic` / `refresh_snapshot` /
`--check` CLI + idempotence test) shipped in M1 (#189); the `--license`/`--minors-
free`/`--owned` **refusal gate** shipped with P1 (this PR). | M1 | No | ? |
```

```
[#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) · _this PR_ |
154:| **M3** | **Characterization gate** ? `tests/test_golden_fixtures.py`: parse-compared
(exact-int / approx-float 1e-6), **CONTROL-only**, positive no-person/audio assertion,
perturbation + idempotence. **Shipped in M1 (#189).** (Windowing/label fingerprint guard + row-
count cap ? fold into M4.) | M1 | No | ? | [#189](https://github.com/avidullu/badminton-
highlight-indexer/pull/189) |
155:| **MX** | **Cross-OS determinism guard (CI).** Lightweight `golden-crossos` CI job
(ubuntu/windows/macOS) running `--check` + the fixtures test ? locks the cross-OS guarantee
(CONTROL replay is numpy-only, no GPU stack; a *separate fast job*, not full-suite×3). | M1 | No
| ? | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) |
156:| **M4** | **Quality-floor gate + first `regression_baseline.json`.**
`tests/test_golden_quality_floor.py`; create `eval_baselines/regression_baseline.json` with
**windowing fingerprint**; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for
runtime. | M1, M2 | No | ? | ? |
157:| **P1** | **De-attribution + minimization** ? `golden_fixtures.freeze_real_fixture()`
(extends the M1 module, not a separate script): **opaque random `fx_hex` surrogate slug**,
metadata strip, metric-invariant **time-origin reset**, strict GT loader, **positive no-
person/audio/MD5/source-path assertion**, manifest non-clobber; `--freeze-real` CLI with the
**refusal gate** (exit 2 unless owned + minors-free + license). Tooling-only, 12 synthetic
tests; **no real data committed**. | M1 | No (tooling) | ? | _this PR_ |
158:| **P2** | **Owner-accepted (researched-risk), 2026-06-16.** Owner accepted the researched
IP/privacy analysis (§3) to proceed **under its conditions** (owned/Fork-A source · minors
excluded · biometric/person streams excluded · de-attributed). Owner's informed risk acceptance
? **NOT a retained-counsel opinion** ; optional future counsel confirmation per §8. | P1 | Owner-
accepted | ? | ? |
159:| **P3** | **Tier-0 REAL subset ? SHIPPED.** Committed **6 curated, trimmed (212k-frame ?
few-min), de-attributed** real fixtures (`fx_r01`?`fx_r06`: single-court / multi-court×2 /
doubles×2 / continuously-active blob; both 29.97 & 59.94 fps) via `freeze_real_fixture --max-
frames` (opaque slug, time-origin reset, scenario-only labels, **NO** source name/path/video_id
? grep-verified + leak-scan clean). ~1.9 MB total; the characterization + cross-OS gates now run
on real data too. Owner cleared ALL owned golden videos (owned + minors-free) + re-confirmed
option (a) (2026-06-22). The surrogate?source map stays **off-git** (owner-held). | P1, P2 |
conditions §3 ? | ? | _this PR_ |
160:| **O1** | *(optional)* **Protobuf + deterministic serialization.** `rally.fixture.v1`
proto3, codegen + `protoc`/`git diff --exit-code` guard, canonical writer, `.gitattributes`
binary. Only if cross-OS byte-stable snapshots are wanted. | M1 | No | ? | ? |
161:| **O2** | *(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive
token, default) and/or SOPS+age; `_tier1_present()`; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job
gated on secret, never on fork runners. | M1 | No | ? | ? |
162:| **DOC** | **Discoverability cross-links** ? added the `eval_baselines/fixtures/` row to
`CODE_MAP §0` + a `golden_fixtures.py` entry to `CODE_MAP §2.3`; cross-linked
`GOLDEN_DATA_SHARING.md` (committed subset ? private OneDrive corpus). README indexed in #186;
memory `current-state` kept live. (EVALUATION/QUALITY_ITERATIONS cross-links **deferred** ? that
doc surface is hot in the concurrent quality track; avoid collision.) | M1 | No | ? | _this PR_
|
164:**Must-fix-before-any-public-commit (red-team), folded into the rows above:** ? numeric-
tolerance not byte-equality + multi-OS matrix + CONTROL-only (M3/D5/MX ?); ? scrub names in
**all** files + history decision (P0); ? leak-scan a **required** check, hooks are convenience
only (P0); ? person-optional loader + **positive** no-person assertion, don't trust the silent
`0.0` default (P1 ?); ? random surrogate id not HMAC-over-roster (P1 ?); ? quantize/drop
fps/frame_width (P1 ? fps/fw retained as scale constants; quantize is a P3 option); ? document
Tier-0 omits the fusion path (DOC); ? counsel + Fork-A + minors/biometric exclusion as hard
preconditions (P2 owner-accepted; enforced in code by P1's refusal gate ?).
166:### Progress log (quantitative ? coverage must hold/improve, CI incl. the 3-OS cross-OS job
green)
178:## 8. Open owner / counsel questions
180:**Counsel questions ? documented basis (the owner accepted the researched risk on
2026-06-16; see P2). These remain the record, and a retained-counsel confirmation is
optional/recommended before scaling beyond a small owned, minors-free, de-attributed subset:**
194:## 9. Definition of done
196:The project is **DONE** (flip `Status ? DONE`, archive) when:
197:- [ ] **P0, M1?M4, P1, DOC** are ? and merged.
199:- [ ] The **CI leak-scan is a required status check** and passes (no MD5 / real-name / non-
surrogate / plaintext-SOPS in the tree).
```

90. assistant (2026-06-22T12:42:44.520Z)

Now the golden-fixtures status is clear from its §7 tracker / §9 DoD:

- **Done:** M1, M2, M3, MX, P1, P2, P3, DOC ?
- **Open** (not optional ? part of the Definition-of-Done): **M4** (quality-floor gate) ?, **P0** ? (the deferred functional video-name rename)
- **Optional:** O1 (protobuf) ?, O2 (Tier-1 key-gated) ?

So this doc is **not terminal** ? correctly **not** an archival candidate (M4/P0 still open). The deferred/optional items belong in BACKLOG (task 3). Let me read the rest of BACKLOG.md to insert a clean section in the right place and style.

91. user (2026-06-22T12:42:45.983Z)

```
40      *(source: `docs/GOLDEN_SET_IMPLEMENTATION.md`, `docs/GOLDEN_SET_VENDORING.md`)*
41
42      - ? Annotation guideline ? DONE (2026-06-02): authored as the Roora brief +
illustrated guide in the collector repo `docs/annotation/` ? rally CSV schema; ignore dead time
/ include every rally (incl. faults & lets) / shot = one contact; `ending_reason`
(forced/unforced-error taxonomy, shared across net-separated sports); singles & doubles for
badminton & pickleball.
43      - ? Internal gold key ? hand-annotate 5?10 licensed clips to near-perfect as the
vendor answer key. Unblocked. Revisit: anytime.
44      - ? GS-3 ? HumanVendorProvider ? concrete adapter for the chosen annotation vendor
(submit/poll/fetch + DPA/secure-link controls). Vendor selected for the initial pilot: Roora
(Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). Revisit: once the
Roora pilot confirms quality, then build the adapter.
45      - ? GS-4 oracle model ? AutomatedProvider interface is stubbed (this work) with a
deterministic fake. Wire a concrete fast-tier auto-labeler later. Oracle rule: it MUST be
a different model lineage than production, or the regression test is blind to shared failures.
Revisit: when a production LLM step is chosen (pick the oracle's lineage to differ).
46      - ? GS-5 ? regression triggers ? wire `regress()` into CI (pre-merge gate), on-
demand CLI (the "surprise" case), and a scheduled run; include a deliberate re-baseline path.
Revisit: after GS-4.
47      - ? Vendor pilot (Roora) ? run the paid pilot with Roora as a single-vendor
pilot first (not a multi-vendor bake-off): assess quality-per-rupee against the §4 bar;
escalate to the iMerit/Taskmonk fallback only if it misses. Revisit: the pilot doubles as
calibration once the gold key exists.
48      - ? Phase-1 video collection ? gather the round-1 golden corpus per
`docs/GOLDEN_SET_PHASE1_VIDEOS.md`: static self-recorded footage (mirror deployment, not
broadcast), 3 sports incl. singles & doubles, across the diversity matrix; ~12 clips/sport with
? held out by match. Unblocked.
49      - ? Train/eval separation guardrails (was DEFERRED_HARDENING_PLAN #13; design
preserved here on archival 2026-06-14) ? train & eval videos live in separate catalogs
today, kept apart by convention only; nothing enforces that a video used to train the learned
segmenter (`backend/eval/training.py::build_training_set` ? `distill_local`) is absent from the
eval/regression set, so leakage would silently inflate the headline metrics. Concrete design
(from the archived DEFERRED plan §2): split unit = match, never clip (hold ~? of each sport
as eval, per `GOLDEN_SET_PHASE1_VIDEOS.md`); add a `split: "train"|"eval"` field to the
collector's `curate export` manifest (cross-repo contract, ADR-002); [DONE ? pure-logic core,
PR #171] `backend/eval/splits.py` now exists with `load_split(manifest) -> (train_ids,
`eval_ids)` keyed by match id (not file MD5 ? re-encodes share a match but differ in
`compute_video_id`) + `assert_disjoint(train_ids, eval_ids)` that raises on overlap, with
`tests/test_splits.py` (overlapping ids ? `assert_disjoint` raises). STILL DEFERRED: wire
the guard into the training entrypoint (`distill_local`/`training`) and into `run_regression.py`
(baseline computed strictly on the eval split); add the collector's `curate export` manifest
`split` field (above); the integration test (a deliberately-leaked manifest fails the training
entrypoint). Full rationale/risks: [`archives/DEFERRED_HARDENING_PLAN-
COMPLETED-2026-06-14.md`](`archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md`) §2.
Revisit: dedicated design session with the user (needs a collector-side manifest change).
50
51      ## Recording best practices
52      *(source: `docs/VIDEO_RECORDING_GUIDELINES.md`)*
53
54      - ? Validate recording hypotheses ? empirically test which capture conditions
```

(resolution, fps, camera angle/height, lighting) actually move precision/recall, then promote the winners into user-facing "how to record" guidance. Only the static-camera rule is currently evidence-backed (ADR-004). **Revisit:** once a controlled corpus + the eval loop can measure per-condition deltas.

55

56 **## Code & test organization**

57 *(source: owner review comments on PR #35, 2026-06-07)*

58

59 - ? ****`tests/` reorganization to reduce fragmentation**** ? owner wants to isolate test components on a grouping

60 metric (by subsystem/component) so `tests/` is less fragmented, mirroring the new

61 `backend/pipeline/audio/` +

62 `backend/eval/audio/` layout (e.g. audio tests grouped under `tests/audio/`, eval tests under `tests/eval/`).

63 The exact grouping ****metric is the owner's design choice**** (consolidate small files vs. add per-area dirs ? the

64 "lesser fragmentation" goal slightly tensions with adding dirs), so this is left as a ****dedicated refactor PR****,

65 not guessed here. ****Revisit:**** when the owner specifies the grouping scheme. (The

66 `backend/eval/audio/` half of

67 the request ? comment 1 ? was done in PR #35.)

68

69 **## Commercial Readiness & Low-Regret Moves ? Approved Decisions Checkpoint (2026-06-07)**

70

71 *(source: Commercial Readiness Review + structure/sequencing discussion with Grok as second-in-command)*

72

73 - ? ****Docs-only checkpoint PR**** ? This update + the refreshed

74 `docs/archives/past\_projects/low-regret-moves/LOW\_REGRET\_MOVES\_PLAN.md` (since archived)

75 formally record the approved path before any code changes. The initial review docs (PR #37)

76 introduced the findings; this entry locks the execution decisions.

77 - ? ****Approved directory structure**** ? Adopt a layered approach under `backend/` for

78 professionalism and future growth (no flat dumping of new modules, no explosion of 15+ single-

79 file dirs):

80 - `backend/config/` (package ? start with this)

81 - `backend/api/` (FastAPI app, routes, static UI)

82 - `backend/pipeline/` (validators + segmenters + stitcher + audio + detectors)

83 - `backend/infrastructure/` (database + external adapters)

84 - Keep `backend/eval/` prominent as a first-class concern.

85 - Logical grouping for providers, sports, annotations, utils, and a small `core/` only

86 when needed.

87 - ? ****Sequencing (Config first, then atomic refactor)**** ? Implement Deliverable 1

88 (Config) ****first****, introducing the `backend/config/` package immediately. Perform the full

89 directory structure refactor as a ****separate high-pri atomic PR**** afterwards. This minimizes

90 conflicts, keeps PRs focused, and gives Claude a clean window with no concurrent moves.

91 - ? ****Deliverable 1 (Config)**** ? `backend/config/` package (`models.py` + `loader.py`)

92 instead of a single `config.py`. Typed `AppConfig`, central loader with built-in defaults + file

93 override, `extra="allow"` for migration. Update `setup.py` + `main.py`. Add tests. Land this

94 before the layout move.

95 - ? ****Subsequent items**** ? See the refreshed `docs/archives/past_projects/low-regret-

96 moves/LOW_REGRET_MOVES_PLAN.md` (archived) for the full ordered list (Error contracts, logging,

97 packaging, detector abstraction, etc.). The structure refactor is now explicitly tracked as its

98 own high-pri step right after Config.

99

100 ****Status legend reminder:**** ? = approved / in progress tracking, ? = do when triggered.

101

102 **## Observability reports ? shipped; intersections with the low-regret items (2026-06-07)**

103 *(source: per-video observability feature ? indexer PR #41/#42/#43, collector PR

104 #11/#12; shared lib `obsreport`)*

105

106 - ? ****Low-regret items 1 (Config pkg, PR #39) + 2 (directory restructure, PR #40) are**

107 **DONE.**** Remaining = items 3?9 in `docs/archives/past_projects/low-regret-

108 moves/LOW_REGRET_MOVES_PLAN.md` (archived).

109 - ? ****Per-video observability reports shipped in BOTH tools**** (a TECHNICAL report + a

110 CUSTOMER-friendly summary), via the ****soft-dependency**** lib `obsreport`

(github.com/avidullu/sports-obsreport, MIT, v0.1.3). Each run writes ``<video>.report.json / .report.md / .summary.md`` next to the video.

90 - ? ****Item 8 (Input-Quality UX) is LARGELY SUBSUMED**** by these reports (validation stage + footgun flags + verdict already surface quality; a clean customer summary already exists). ****Do NOT build a parallel mechanism**** ? instead **EXTEND** the report: add an audio-readiness stage/flag to ``backend/reporting/`` (harvest) + the obsreport spec. ****Revisit:**** when item 8 is picked up.

91 - ? ****Item 7 (unify hybrid segmenters) should land before the report's real AI-provider timing.**** The report records AI timing as ``not_captured``; once ``wasb_hybrid/`tracknet_hybrid`` are unified there is ONE place to thread a recorder callback through ``provider.analyze_video`` ? the report. ****Revisit:**** with item 7 (Claude to add the timing hook there).

92 - ? ****Item 3 (Error/Result contract) synergizes with the report.**** Today the report INFERS failures (e.g. ``silent_provider_noop`` = 0 segments + no API key). A structured Result/error from providers/segmenters would let the report read the real reason as a first-class flag. ****Revisit:**** with item 3.

93 - ? ****`obsreport` is a SOFT/commented dep**** in ``requirements.txt`` (works locally via editable install; pipeline no-ops if absent). ****Pin the `git+ssh@v0.1.3` tag**** in both repos once SSH/CI access to the private lib repo exists.

94 - ? ****`run\_all\_tests.py`** added at repo root ? run the full suite (clear PASS/FAIL, pytest exit code) before sending PRs.

95 - ? ****Config-migration follow-ups (DONE ? config & identity consistency pass)**** ? the two typed-config bugs from

96 #39/#40 are fixed: (1) ****`/api/compile`** now honors ``--output-dir`` and (2) ****`output\_dir`** is a real field on

97 ``OutputConfig`` (default ``"output"``); the CLI overrides it on the typed model in ``main()``, retiring the

98 write-only ``_runtime_overrides`` hack. Same pass also unified ``video_id`` MD5 derivation into

99 ``backend/utils/hashing.py::compute_video_id`` and migrated all four ``run_*.py`` scripts off raw ``json.load`` onto

92. assistant (2026-06-22T12:43:05.914Z)

I have what I need. I'll add a focused ****Golden regression fixtures**** section to BACKLOG.md (task 3) with the deferred DoD items (M4, P0 rename) and optional items (O1, O2), traced to ``GOLDEN_REGRESSION_FIXTURES.md`` §7/§9, and refresh the "Last updated" stamp.

93. user (2026-06-22T12:43:17.842Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\docs-hygiene\docs\BACKLOG.md has been updated successfully. (file state is current in your context ? no need to Read it back)

94. user (2026-06-22T12:43:22.471Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\docs-hygiene\docs\BACKLOG.md has been updated successfully. (file state is current in your context ? no need to Read it back)

95. assistant (2026-06-22T12:43:46.592Z)

Now task 4 (archival): I've verified there's ****no clean archival candidate**** right now ? ``GOLDEN_REGRESSION_FIXTURES.md`` is non-terminal (M4/P0 open per its §9 DoD), and the only flagged candidate in ``DOC_STATUS.md`` (the hardening pointer-stubs, finding #4) explicitly needs owner confirmation ("intentional stubs vs finish-archiving"). Archiving a non-terminal doc would violate the project's own archive policy, so I'm correctly ****not**** forcing one. Let me commit + push + open the backlog PR.

96. user (2026-06-22T12:43:53.813Z)

=== verify branch + clean base ===
docs/backlog-golden-fixtures-followups
(empty=base origin/master)
[docs/backlog-golden-fixtures-followups 85b86a8] docs(backlog): track golden-regression-fixtures

DoD remainder (M4, P0 rename) + optional 01/02
1 file changed, 11 insertions(+), 1 deletion(-)
branch 'docs/backlog-golden-fixtures-followups' set up to track 'origin/docs/backlog-golden-fixtures-followups'.
To <https://github.com/Khelsutra/badminton-highlight-indexer.git>
* [new branch] docs/backlog-golden-fixtures-followups -> docs/backlog-golden-fixtures-followups

97. user (2026-06-22T12:44:08.865Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/323>

98. assistant (2026-06-22T12:44:19.640Z)

PRs [#322](<https://github.com/Khelsutra/badminton-highlight-indexer/pull/322>) (gate-A) and [#323](<https://github.com/Khelsutra/badminton-highlight-indexer/pull/323>) (backlog) are both up. Let me check their CI and merge whichever are green.

99. user (2026-06-22T12:44:32.217Z)

```
=== #322 ===
MERGEABLE/CLEAN
(all non-fail/non-pending ? green)
=== #323 ===
MERGEABLE/UNSTABLE
Built wheel smoke (install + launch)    pending 0      https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953552946/job/82716119830
Golden fixtures cross-OS (macos-latest) pending 0      https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953552946/job/82716119790
Golden fixtures cross-OS (ubuntu-latest) pending 0
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953552946/job/82716120140
Golden fixtures cross-OS (windows-latest) pending 0
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953552946/job/82716119905
Mypy (lenient baseline) pending 0      https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953552946/job/82716120022
Syntax sweep + import smoke (no GPU stack) pending 0
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953552946/job/82716119820
Dependency license scan (no copyleft in runtime deps) pending 0
https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953552946/job/82716119981
Full test suite (offline, mocked) pending 0      https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/27953552946/job/82716120014
```

100. assistant (2026-06-22T12:44:42.116Z)

#322 is **CLEAN/green** ? merging now (no `--delete-branch`; its branch is in a worktree).
#323's CI is still running.

101. user (2026-06-22T12:44:49.285Z)

```
=== merge #322 (gate-A reground) ===
=== origin/master now ===
db74f9b..32f4ee6 master -> origin/master
32f4ee6 test(eval): re-ground served Gate-A litmus + QUALITY_ITERATIONS on the 15-clip golden corpus (#322)
db74f9b docs(data): mark DATA_IN_GCS $7b blocker #1 resolved by #319 (#321)
b72e6de fix(worker): strip ISO-date prefix when joining labels to videos in train (#319)
```

102. assistant (2026-06-22T12:45:06.847Z)

Task 2 ? merged (#322 ? `origin/master` at `32f4ee6`). Now poll #323 (backlog) and clean up

the now-merged gate-A worktree.

103. user (2026-06-22T12:45:12.499Z)

```
=== cleanup merged gate-A worktree + local branch ===
worktree removed
Deleted branch fix/gate-a-litmus-reground-n15 (was 7f15a1b).
=== #323 status ===
UNKNOWN/UNKNOWN
Golden fixtures cross-OS (windows-latest)      pending 0
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27953552946/job/82716119905
Built wheel smoke (install + launch)    pending 0      https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27953552946/job/82716119830
Full test suite (offline, mocked)        pending 0      https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27953552946/job/82716120014
```

104. assistant (2026-06-22T12:45:23.072Z)

#323's full suite + cross-OS checks need a couple more minutes. I'll watch them to completion, then merge if green.

105. user (2026-06-22T12:46:48.113Z)

```
Leak scan (PII / attribution guard)    pass    7s      https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27953552946/job/82716119839
Mypy (lenient baseline) pass    31s      https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27953552946/job/82716120022
Ruff lint      pass    15s      https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27953552946/job/82716119828
Syntax sweep + import smoke (no GPU stack)    pass    21s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27953552946/job/82716119820
Built wheel smoke (install + launch)    pass    1m6s    https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27953552946/job/82716119830
Dependency license scan (no copyleft in runtime deps)    pass    41s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27953552946/job/82716119981
Full test suite (offline, mocked)        pass    2m29s    https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27953552946/job/82716120014
Golden fixtures cross-OS (macos-latest) pass    24s      https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27953552946/job/82716119790
Golden fixtures cross-OS (ubuntu-latest)    pass    24s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27953552946/job/82716120140
Golden fixtures cross-OS (windows-latest)    pass    1m2s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27953552946/job/82716119905
Leak scan (PII / attribution guard)    pass    7s      https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27953552946/job/82716119839
Mypy (lenient baseline) pass    31s      https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27953552946/job/82716120022
Ruff lint      pass    15s      https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27953552946/job/82716119828
Syntax sweep + import smoke (no GPU stack)    pass    21s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27953552946/job/82716119820
exit=0
```

106. assistant (2026-06-22T12:46:53.045Z)

#323 all green. Merging it (no `--delete-branch`; its branch is in the docs-hygiene worktree), then cleaning up.

107. user (2026-06-22T12:47:05.187Z)

```
=== merge #323 (backlog) ===
=== cleanup docs-hygiene worktree + branch ===
worktree removed
Deleted branch docs/backlog-golden-fixtures-followups (was 85b86a8).
=== final origin/master ===
    32f4ee6..9b37269 master    -> origin/master
9b37269 docs(backlog): track golden-regression-fixtures DoD remainder (M4, P0 rename) + optional
01/02 (#323)
32f4ee6 test(eval): re-ground served Gate-A litmus + QUALITY_ITERATIONS on the 15-clip golden
corpus (#322)
db74f9b docs(data): mark DATA_IN_GCS $7b blocker #1 resolved by #319 (#321)
b72e6de fix(worker): strip ISO-date prefix when joining labels to videos in train (#319)
48f9d20 feat(nightly): make the Cloud Run scheduler path work E2E + turnkey email/trigger (#320)
e221cb9 docs(data): GCS corpus source-of-truth map + Phase-1 metadata migration (#316)
```

108. assistant (2026-06-22T12:47:27.794Z)

All merged. Final check on remaining open PRs and confirm the tree state.

109. user (2026-06-22T12:47:31.027Z)

```
=== remaining OPEN PRs ===
=== my worktrees cleaned? ===
C:/Users/avidu/Projects/badminton-highlight-indexer
db74f9b [fix/served-gate-a-regrounded-15clip]
```

110. assistant (2026-06-22T12:48:13.966Z)

****No open PRs remain ? all merged.**** One concurrency note: the main checkout is now on a sibling branch `fix/served-gate-a-regrounded-15clip` (a parallel gate-A re-grounding a sibling started ***after*** my pre-check; it's now redundant since my #322 landed first ? I left it untouched). My two worktrees are cleaned up.

Let me do a final end-to-end verification: run the full suite against the ****combined**** merged master (each PR's CI ran on its own base; this confirms the union is green) in a throwaway worktree.

111. user (2026-06-22T12:50:00.636Z)

```
Preparing worktree (detached HEAD 9b37269)
HEAD is now at 9b37269 docs(backlog): track golden-regression-fixtures DoD remainder (M4, P0
rename) + optional 01/02 (#323)
HEAD: 9b37269 docs(backlog): track golden-regression-fixtures DoD remainder (M4, P0 rename) +
optional 01/02 (#323)
..... [ 95%]
..... [100%]
1428 passed, 2 skipped in 101.14s (0:01:41)
=====
RESULT: PASS - all tests green
=====
```

112. assistant (2026-06-22T12:50:06.417Z)

Combined merged master (`9b37269`): ****1428 passed, 2 skipped, 0 failed**** with the corpus (gate-A test runs and passes). Note the count rose from 1418 ? 1428 ? that's #319's new worker coverage tests. Cleaning up the verify worktree.

113. user (2026-06-22T12:50:08.878Z)

```
error: failed to delete 'C:/Users/avidu/Projects/badminton-highlight-
indexer/.claude/worktrees/verify-master': Permission denied
=== my worktrees all gone? ===
```

```
C:/Users/avidu/Projects/badminton-highlight-indexer
5edd9f8 [fix/served-gate-a-regrounded-15clip]
```

114. assistant (2026-06-22T12:50:19.298Z)

Removal hit a transient Windows file lock (and the grep matched the sibling *branch name* substring, not my worktree). Let me retry with `--force` and list worktrees cleanly.

115. user (2026-06-22T12:50:21.021Z)

```
fatal: '.claude/worktrees/verify-master' is not a working tree
still locked; pruning admin entry
=== current worktrees ===
C:/Users/avidu/Projects/badminton-highlight-indexer
5edd9f8 [fix/served-gate-a-regrounded-15clip]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/agent-a771ff1d85e18f2ed
7e0881a [perf/wasb-fast-frames]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/bhi-execpolicy
5cb7bbf [bhi-execpolicy]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/boxhill17
8736e2d [run/boxhill17-highlights]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/gap-doc
f04f3f3 [docs/rally-gap-closure]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/hardchop
614325d [feat/w1-hard-cap-fallback]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/pr-stitch
a6adcc5 [fix/runner-stitch-watermark]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/recdoc
9fd4215 [docs/recording-single-match]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/resweep
fe34013 [docs/lovo-resweep-n13]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/tier1-windowing
a3b209a [master]
C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/w2-safeguard
5ba6955 [feat/w2-never-empty-safeguard]
```

116. assistant (2026-06-22T12:50:51.517Z)

All my worktrees are cleaned up; the rest belong to other live sessions. Everything is done and verified.

Done ? all four asks complete

****Merged the green PRs**** (each green CI, coverage-gated full-suite passing; none touching product/strategy/licensing). Final `master` = `9b37269`, and I verified the ****combined**** merged tree end-to-end: ****1428 passed, 2 skipped, 0 failed****.

```
| PR | What | Coverage |
|---|---|---|
| [#316](https://github.com/Khelsutra/badminton-highlight-indexer/pull/316) | docs:
`DATA_IN_GCS.md` corpus map (merged first so #319's `S7` refs resolve) | docs-only |
| [#320](https://github.com/Khelsutra/badminton-highlight-indexer/pull/320) | nightly Cloud-Run
scheduler scripts (`--run-now`, dry-run, `setup_email.sh`) | shell-only |
| [#319](https://github.com/Khelsutra/badminton-highlight-indexer/pull/319) | **(task 1)**
worker date-prefix-strip fix | *** new tests** (suite 1418?1428) |
| [#322](https://github.com/Khelsutra/badminton-highlight-indexer/pull/322) | **(task 2)**
gate-A litmus re-grounding | test+doc |
| [#323](https://github.com/Khelsutra/badminton-highlight-indexer/pull/323) | **(task 3)**
golden-fixtures backlog | docs |
```

****1. Worker fix**** ? merged via #319. A sibling session had pushed two extra skip-branch coverage tests onto the same branch and merged it while I was mid-flight; I confirmed it green (all 10 checks) rather than racing the merge.

****2. Gate-A re-grounding**** ? the litmus was stale **again**: #236 pinned the 9-clip corpus, which has since grown to ****15 clips**** (``mean_seg_off`` drifted 1.16?1.36, failing locally). I re-measured on the current corpus and re-pinned the test + ``QUALITY_ITERATIONS.md`` (proposal ``1.36?0.79``, served `?F1`-0.023``, ratio ``0.90``), preserving the durable ``seg_on < seg_off`` invariant. ?? This test ****skips in CI by design**** (no corpus there), so I verified locally: full suite ****1418 passed**** with the manifest. Merged via #322.

****3. Backlog**** ? added a **Golden regression fixtures** section to ``BACKLOG.md`` for the §9 definition-of-done remainder (****M4**** quality-floor gate, ****P0**** name-rename) plus optional ****01/02****. Merged via #323.

****4. Docs fix + archive**** ? "fix" done (the stale gate-A numbers in ``QUALITY_ITERATIONS.md``, via #322). ****Archive: nothing was archive-ready.**** ``GOLDEN_REGRESSION_FIXTURES.md`` is non-terminal (M4/P0 open in its own §9), and the only flagged candidate in ``DOC_STATUS.md`` (the hardening pointer-stubs, finding #4) explicitly needs **your** call ("intentional stubs vs finish-archiving"). Archiving a non-terminal doc would violate the repo's own archive policy, so I deliberately did not force one.

Two things worth flagging

- ****Heavy concurrency:**** this checkout is shared by ~10 live sessions. I branched my work in isolated worktrees off ``origin/master``, staged explicit paths, and cleaned up after myself.
- ****A sibling started a *parallel* gate-A re-grounding**** (``fix/served-gate-a-regrounded-15clip``, currently checked out in the main checkout and still being committed) ****after**** my pre-check found none in progress. My #322 landed first, so that branch is now redundant ? I left it completely untouched; you may want to close/discard it.

Want me to update the ``current-state`/`session-handoff`` memory with this checkpoint? I held off because those files are shared with the active sessions and I didn't want to clobber a sibling's update.

117. user (2026-06-22T13:24:11.061Z)

yes please update the handoff and the memory and I will trim the concurrency before starting anything new

118. assistant (2026-06-22T13:24:22.618Z)

Concurrency will be trimmed first, so it's safe to write the shared memory now. Let me read the current handoff + current-state files so I update them honestly rather than clobber.

119. user (2026-06-22T13:24:23.014Z)

```
1      ---
2      name: session-handoff
3      description: "Living \"you are here\" handoff ? current state, next steps, open
decisions, ramp-up kit. AUTO-REFRESHED at meaningful checkpoints (PR merged, key decision, phase
shift, session wrap). \"update the handoff\" forces a full rewrite."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: 45ea3a91-654d-4972-9cd3-65f35db54caf
8      ---
9
10     # Session handoff ? badminton-highlight-indexer (rename PENDING) + siblings: sports-
data-collector, rally-annotator, sports-obsreport
11
12     **Last updated:** 2026-06-21 (**ALPHA-SHAREABLE session** ? Claude's half of "make it
UI-driven + alpha-shareable" is **DONE**: 3 PRs merged ? engine **#294** exposure-safety boot
posture + khelsutra **#64** SPA `VITE_API_BASE` + **#65** deploy kit. **KEY CORRECTION:** B-P2
(SPA ingest screen) was ALREADY shipped (#56) ? alpha-shareable was a DEPLOY/AUTH task, not an
SPA build. Owner chose the **SPLIT** topology (SPA on CF Pages + `api.*` cloudflared tunnel + CF
Access). Remaining = OWNER stand-up via `khelsutra/deploy/DEPLOY_ALPHA.md`; then engine M11/M12.
Full record [[ui-driven-cloud-serving-gap]]). Prior: HUGE serving session ? M1?M9-auth shipped
(#279?#291) + real M7 L4 run. **Repo is under the `Khelsutra` GitHub org**.
```

13

14 > **Read order:** this file ? `[[current-state]]` top (dense per-checkpoint detail) ?
``docs/NEXT_STEPS.md`` (the prioritized cross-track plan, refreshed 2026-06-21) ?
``docs/DOC_STATUS.md`` (doc lifecycle/hygiene).

15

16 **## ? You are here (2026-06-21)**

17 engine ``origin/master`` = ``0c0e064`` + (past #292/#293/#294) · khelsutra ``origin/master`` =
``4627830`` (past #63/#64/#65) · **no GCP VMs (audited \$0)** · engine ~1259 tests green.

18 *****? ALPHA-SHAREABLE SESSION (2026-06-21) ? 3 PRs MERGED; Claude's half of the owner's**
top priority is **DONE**:

19 - **KEY CORRECTION** (anti-spiral `[[lesson-verify-engine-surface-before-scoping]]`): **the**
handoff said "do the B-P2 SPA ingest screen + M7 Cloud Run `*service*`" ? but **B-P2** was already
shipped* (khelsutra #56, live e2e in ``DRY_RUN_CUJ.md`` §9) and recon showed alpha-shareable is a
DEPLOY/AUTH task, not an SPA build. CORS is already env-driven; the engine needed no CORS
code.

20 - **Owner decisions (this session):** (1) pursue alpha-shareable via topology **(i)**
Tunnel-to-a-box*; (2) sub-topology = **SPLIT** (SPA on CF **Pages** + engine on ``api.``)
cloudflared tunnel + CF Access over both ? engine ``HOSTING_PLAN.md`` §2), single-origin Caddy
kept as the alternative.

21 - **engine** [#294](https://github.com/Khelsutra/badminton-highlight-indexer/pull/294)
(``0c0e064``) ? exposure-safety boot posture: **backend/config/startup.py** fails fast at boot
(server-only ``include_exposure=True``) when ``security.edge_auth_enabled`` is ON but half-
configured ? ``EXPOSED_NO_VIDEO_ROOT`` (fatal, jail unset), ``EXPOSED_EDGE_AUTH_INCOMPLETE`` (fatal,
CF team/AUD), ``EXPOSED_AUTH_EXTRA_MISSING`` (warn). **Default-OFF** (edge_auth off ? no-op;
worker never opts in). Aligned ``auth.resolve_tenant`` whitespace-strip. Adversarial 3-lens review
(``wf_35b735a9``) ? no bypass. +16 tests; 1259 green cross-OS.

22 - **khelsutra** [#64](https://github.com/avidullu/khelsutra/pull/64) (``8bbc557``) ? SPA
``VITE_API_BASE``: ``apiBase()`` env-driven (default same-origin ``/api``) + ``vite-env.d.ts`` + 3
tests; build-smoke proved the override bakes in.

23 - **khelsutra** [#65](https://github.com/avidullu/khelsutra/pull/65) (``4627830``) ? alpha
deploy kit: ``deploy/DEPLOY_ALPHA.md`` (full split runbook + CORS-preflight gotcha + posture
test + deferred notes) + ``deploy/cloudflared.config.example.yml`` + README/tracker pointers.

24 - **Discipline:** isolated PRs off ``origin/master`` (khelsutra in a dedicated
worktree ? main tree had sibling A8 WIP, since merged as #63), default-OFF, adversarial
review, full CI green cross-OS, merged own green PRs, tracker+changelog in-PR. **REMAINING** for
alpha = **OWNER** stand-up only* (tunnel + Pages + CF Access + key rotation per
``khelsutra/deploy/DEPLOY_ALPHA.md``). **M1?M9-auth** (prior sessions) detail below.

25 *****? HUGE SERVING SESSION (2026-06-21) ? 6 PRs MERGED + a REAL cloud-GPU run, owner**
BATCH-AUTHORIZED the default-OFF code path + \$100/?10k spend: **285** (docs §8 D16/D17 + P3
reconcile) · **286 M5** (``backend/config/startup.py`` per-profile fail/warn-fast checks + the
L4 profile-parity proof; worker stays torch-free) · **287 M6** (``backend/compute/``
``deploy/cloudrun/{Dockerfile,README}``; mocked) · **288 M8** (per-job **cost ledger** + v6
migration + versioned ``pricebook`` \$0 + ``/cost`` API) · **290 M9-auth** (CF-Access JWT verify
RS256 + ``owner_id`` tenant tag; security review found NO bypass; alg=none + HS256-confusion
locked) · **291 ops-agent** codify + ``docs/CLOUD_GPU_DRYRUN_GUIDE.md``
(``deploy/gcp/create_gpu_vm.sh`` always bakes the Ops Agent). All **default-OFF**, isolated,
adversarial-reviewed (a workflow per PR), full-suite+cross-OS CI green, tracker+changelog
in-PR. **REAL M7 L4 RUN DONE + VERIFIED:** native WASB on a real L4 ? **22 rallies** on the
smoke clip ? GCS out; VM deleted (\$0, ~\$0.21). ? the no-Gemini cloud run needs
``skip_ai_handoff=true`` (the long-flagged "0 rallies" cold-start fix). A 2nd dry run via
``create_gpu_vm.sh`` **confirmed** Ops-Agent auto-installs* + found the **watermark-font gap**
(fixed in #292: ``bootstrap.sh`` now installs ``fonts-dejavu-core``). **M1?M4** (prior session) detail
below.

26 - **M1** ([#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279),
``ce22791``): ``deployment.profile`` + ``compute_target`` on ``AppConfig`` + ``backend/config/presets.py``
(``PROFILE_PRESETS`` + ``suggest_profile``) + loader preset/precedence
(defaults<preset<config.json<env via ``_deep_merge``; no-profile path = exact old shallow merge) +
auto-detect-as-SUGGESTION (D15). Pure/additive, profile=' == today. 3 minor review findings
folded.

27 - **M2** ([#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280),
``e111add``): ``StorageConfig.input_backend`` / ``input_root`` +
``backend/storage/{ref.resolve_input_uri,`
`__init__.resolve_input_ref/acquire_local_input/input_is_local}` + `main.py` API/CLI wired
(acquire local; keep source URI for DB/report). local = identity passthrough (byte-for-byte`

unchanged)**; bucket/Drive fetched to scratch (mirrors worker `cmd\_infer`). Unknown scheme?clean 400 (was a 500 regression ? caught by review); `local://` accepted on resolved key; hosted-mode rejects non-local; OS-independent anchored-path detection (os.path.isabs differs Win+py3.13). 1 major + 6 minor review findings folded. Suite 1105 green, cov 84.84%.

28 - **M3 ([#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282), `2afd54f`):** configurable output/scratch base ? `backend/utils/paths.output\_base(config)` (returns `output.output\_dir`, default `"output"`); routed all 6 hardcoded `os.path.join("output", ?)` scratch sites in `trajectory\_hybrid`/`yolo\_hybrid`/`gemini` through it (create+cleanup same var). **DEFAULT `"output"` ? byte-identical** (golden + cross-OS green); a custom `--output-dir` now actually moves scratch (latent bug fixed). Default-OFF ? no Launch Report. `tests/test\_output\_base.py` + a yolo call-site capture; clean-bill focused review (yolo-coverage minor folded). Suite 1114 green.

29 - **M4 ([#283](https://github.com/Khelsutra/badminton-highlight-indexer/pull/283), `cee4baa`):** `DeviceContext` (`backend/pipeline/detectors/device.py`) + native WASB **CPU-fallback** via a new `device="auto"` (cuda-if-available-else-cpu). **Explicit `"cuda"` stays STRICT** (healthcheck still hard-fails without a GPU ? no silent slow-CPU downgrade; "cuda default unchanged"). CUDA probe **cached** (healthcheck?run_predict reuse); `\_effective\_device()` resolves before argv so "auto" NEVER reaches `wasb\_infer.py`; fail-fast on a typo'd device. `wasb\_infer.py` already supports cpu ? no change there. Device-selection tested offline (mocked subprocess); **real CPU-inference run = owner/GPU-verified** `tests/test\_device\_context.py` + native-runner additions; clean-bill focused review (2 nits folded). Suite 1126 green, cov 84.96%.

30 ***? NEXT (priority order):**

31 1. **? ALPHA-SHAREABLE ? Claude's half DONE this session (#294/#64/#65).** Engine is exposure-safe by construction; SPA targets a split origin; the deploy kit + runbook are merged. **? REMAINING = OWNER stand-up** (not Claude-doable): on the box `pip install -e .[auth]` + the safe-exposed `config.json` + `RALLY\_CORS\_ALLOW\_ORIGINS` + rotate `GEMINI\_API\_KEY` + run `backend.main --server`; CF ? cloudflared tunnel (`api.khelsutra.guru`) + CF Pages (`app.khelsutra.guru`, `VITE\_API\_BASE`) + ONE CF Access app over both + the OPTIONS-preflight bypass. **Runbook = `khelsutra/deploy/DEPLOY\_ALPHA.md`.** (A later UI polish, not the alpha blocker: a compute PICKER in the UI, D13/D15.)* Full record: [[ui-driven-cloud-serving-gap]].

32 2. **M11** (data-flywheel plumbing) ? ? privacy-sensitive (consent/data), give it FRESH focus: `backend/flywheel.py` capture?store?shadow-eval?goldenize reusing `workspace.py::for\_video` + `backend/eval/experiment.compare` + `backend/eval/promotion.py::promote\_if\_served\_safe`, behind `flywheel.\*`=OFF + a faked `consent\_attested`. ? depends on M9's not-yet-landed **v7 consent migration** (M8 took v6) + counsel ToS/DPA; only a default-OFF skeleton is Claude-doable.

33 3. **M12** (`gdrive-api` OAuth StorageBackend) ? `backend/storage/gdrive\_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme (regex allows the hyphen; add to `\_KNOWN`+`\_backend\_class`+the StorageConfig Literal) vs an INJECTED fake Drive (mirror `gcs.py` DI); keep mount-only `gdrive://`. ? M9-consent cols + per-video P1 = a new **v7** migration (M8 took v6).

34 ***? owner-only residual:** real M5 truly-local runlog (template pre-staged) · GCP spend for the M7 *service* deploy + a real e2e · counsel ToS/DPA (M9-consent) · real GCP prices+FX + the M8 usage-emission follow-up · Google OAuth app (M12 real) · CF team-domain/AUD+ingress-lock · repo rename · DO droplet/token retire.

35

36 ## ? Icebreaker for the NEXT session (copy-paste)

37 > Continue the badminton-highlight-indexer ENGINE serving + UX track. FIRST: `git pull --ff-only`, then read memory `session-handoff.md` + `current-state.md` (top) + [[ui-driven-cloud-serving-gap]] + `docs/NEXT\_STEPS.md`.

38 > STATE: **Claude's half of "make it UI-driven + alpha-shareable" is DONE** ? engine #294 (exposure-safety boot posture, default-OFF) + khelsutra #64 (SPA `VITE\_API\_BASE`) + #65 (the split-topology deploy kit `khelsutra/deploy/DEPLOY\_ALPHA.md`). The remaining alpha step is the **OWNER stand-up** (cloudflared tunnel + CF Pages + CF Access + key rotation ? the runbook is written). **NEXT engine code (Claude-doable, default-OFF, owner BATCH-authorized #285):**

M12 (`gdrive-api` OAuth StorageBackend ? clean, ~300 LOC mirroring `gcs.py` DI + fake, **no DB migration**): regex allows the hyphen, add to `\_KNOWN`+`\_backend\_class`+StorageConfig Literal; keep mount-only `gdrive://`) ? the cleanest pick. Then **M11** (flywheel skeleton ? ? privacy-sensitive, depends on M9's unlanded **v7** consent migration + counsel, so only a default-OFF skeleton lands). Same discipline: ONE isolated PR per item off `origin/master`, default-OFF + golden/cross-OS-gated, adversarial review before commit, full `run\_all\_tests.py`+ruff+mypy green, merge own green PRs, tracker+changelog in-PR, checkpoint memory. ? **shared checkout** (engine AND khelsutra ? sibling sessions commit concurrently; khelsutra needed a dedicated **worktree** this session): branch from `origin/master` explicitly, re-verify HEAD, stage

```

explicit paths; for a node repo worktree, junction `web/node_modules` then **`rmkdir` the
junction BEFORE `git worktree remove` (else it follows into the real node_modules). ? any GCP
VM: `bash deploy/gcp/create_gpu_vm.sh` (auto Ops-Agent) + **DELETE after** ([[gcp-vm-always-
delete]]); $100/?10k cap.
39
40     Prior session delivered the **engine compute/serving + doc-hygiene** track (10 PRs
merged: #263?#267, #270, #272?#274, #276):
41     - **M2 FINALIZED on a real GPU** ? `worker train --trajectory-source detector` on a GCP
**L4 (Singapore; Mumbai was stocked-out)** ? Gen-0 bundle `gs://khelsutra-rally-
corpus/weights/0.1.0-l4/` (honest **n=2 plumbing** result, not a quality model). Both VMs
**deleted**.
42     - **DigitalOcean DROPPED for compute ? GCP is the sole stack (ADR-013; DO GPU not
enabled/credit-funded, Paperspace subscription-gated). [[compute-stack-gcp-only]] · [[gcp-vm-
always-delete]] · [[gcp-ops-agent-enable]].
43     - **DECOUPLED_COMPUTE delivered + ARCHIVED** ?
`docs/archives/past_projects/DECOUPLED_COMPUTE/`.
44     - **NEW tracked subproject: `docs/COMPUTE_DECOUPLED_SERVING/` + ADR-014** ? one pure
core (DETECT?WINDOW?STITCH, never branches on profile) + `ComputeBackend`/`StorageBackend`
profiles (**truly-local**, **cloud-serving** [Cloud Run GPU runs detect AND stitch, metered],
**cpu-mock**). **? AWAITING OWNER M0 DESIGN SIGN-OFF before any code.**
45     - **Doc governance:** `docs/DOC_STATUS.md` register + **archival due-diligence**
(honestly-update?then-archive; codified in CLAUDE.md; [[doc-status-register]]); per-video docs
reconciled; **data layer** carved into `docs/data_pipeline/` (no-ML), field-
ops/PMF?`archives/field-pmf/`, Roora?`archives/roora-vendor/`.
46
47     ? **Shared checkout** ? sibling sessions commit + flip branches here mid-session (hit
twice this session). **Branch from `origin/master` explicitly, re-verify HEAD before commit,
stage explicit paths, never `git add -A`. **[[gotcha-shared-checkout-branch-collisions]]. ?
**`sed -i` over a glob churns CRLF on all matched files** ? separate real changes with `git diff
--ignore-cr-at-eol` and revert the noise.
48
49     ## ? Next steps (resume here)
50     1. **? M0?M4 SHIPPED (#279/#280/#282/#283) ? the config/input/output/device SEAM LAYER
is COMPLETE ? NOW M5+ (mostly owner-gated). **Decisions locked in README **$2 D7?D15** (D7
scale-to-zero · D8 versioned DB pricebook USD/INR · D9 Cloudflare Access · D10 data-for-reels
trade adults-only · D11 Gemini-ON-until-floor + D12/M11 flywheel · D13 user-switchable profiles
· D14 NORTHSTAR gdrive-api OAuth = M12 · D15 suggest?confirm). **Engine build sequence:** ~~M1
deployment.profile~~ ? ~~M2 input_backend~~ ? ~~M3 output-base~~ ? ~~M4 DeviceContext + CPU-
fallback~~ **ALL DONE.** **? M5** = truly-local profile e2e (preset + startup checks + first
committed runlog) ? **? owner real-GPU** (needs a GPU box / owner's WSL machine to run
detect+stitch locally, then commit the runlog). **M6/M7** Cloud Run dispatch + container image
(? owner GCP spend), **M8** cost ledger + pricebook, **M9** edge-auth (Cf-Access), **M11** data-
flywheel harness, **M12** gdrive-api OAuth. **Per-video-workspace P3 (`output/chunks_*)` is now
subsumed by M3 ? reconcile/close it. **Claude-doable WITHOUT owner gating:** the M5
preset+startup-check code (the actual run is owner's); the M8 cost-ledger DB migration +
pricebook + aggregation (pure code, owner does the real-cost calibration); the M11 harness
plumbing. M6/M7/M9/M12 need owner cloud/counsel sign-off.
51     2. **Per-video workspace P1?P6** (#264 shipped P0): P1 v6 DB migration `workspace_uri` ?
P2 worker ? P3 `output/chunks_*)` de-hardcode ? ? (Claude-doable, default-OFF).
52     3. **Ops-Agent codification into `deploy/gcp`** (auto-enable observability on every VM)
? small, [[gcp-ops-agent-enable]].
53     4. **[owner]** Grow the golden corpus (highest-leverage) · set ship-gate target #172 ·
full-corpus training (n?5) for real weights.
54     5. Full prioritized P0/P1/P2 list = `docs/NEXT_STEPS.md` top.
55
56     ## ? Open decisions / owner gates
57     - **M0 design sign-off** (the blocker above) + the COMPUTE_DECOUPLED_SERVING $8
questions.
58     - Doc-hygiene leftovers (`DOC_STATUS.md` $4): confirm the 3 COMPLETED top-level pointer-
stubs; `HOSTING_PLAN`/`UI_PROPOSAL` currency.
59     - Housekeeping: retire the shared DO droplet `claude-do-rally-test` (~$24/mo) + the DO
token; rename repo off "badminton".
60
61     ## ? Ramp-up kit
62     - **Memory:** [[current-state]] (read first) · [[compute-stack-gcp-only]] · [[gcp-vm-

```

```

always-delete]] · [[gcp-ops-agent-enable]] · [[doc-status-register]] · [[gotcha-shared-checkout-branch-collisions]].
63     - **Docs:**
`docs/COMPUTE_DECOUPLED_SERVING/{README, TESTING_STRATEGY, architecture.svg, runlogs/}` ·
ADR-012/013/**014** in `docs/archives/decisions/DECISIONS.md` · `docs/NEXT_STEPS.md` ·
`docs/DOC_STATUS.md` · `docs/PER_VIDEO_WORKSPACE.md` · `deploy/gcp/README.md` (GPU runbook;
teardown defaults to `delete`).
64     - **Code anchors (the pure core the serving project builds on):**
`backend/pipeline/detectors/base.py` (DetectorRunner CSV contract) ·
`tracknet_runner.py::trajectory_to_action_windows` ·
`backend/pipeline/stitcher/compiler.py::compile_highlights` ·
`backend/pipeline/segmenters/trajectory_hybrid.py::_resolve_runner` (the compute seam) ·
`backend/storage/*` · `backend/worker/__main__.py` · `backend/utils/workspace.py`.
65
66     ## ? Parallel track (NOT this session ? sibling sessions)
67     **KhelSutra brand-site + product UX/Hindi-Kannada localization** (repo
`Khelsutra/khelsutra` + #275 here) ? [[next-session-ux-localization]] / [[khelsutra-domain-launch]]. The **rally-quality / owned-model** track is live too (R-series,
`RALLY_QUALITY_RESEARCH.md`; gated on corpus growth). These are background to the
compute/serving resume above.
68

```

120. assistant (2026-06-22T13:24:59.749Z)

This handoff is centered on the 2026-06-21 serving/alpha track. My 2026-06-22 session was a different slice (PR-merge sweep + quality housekeeping), so I'll refresh "You are here"/"Last updated" and the relevant next-steps/open-decisions, without discarding the still-valid serving plan. Let me read current-state.md first.

121. user (2026-06-22T13:25:01.153Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-44 of 156 total (75139 tokens, cap 25000). Call Read with offset=45 limit=44 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```

1     ---
2     name: current-state
3     description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4     metadata:
5         node_type: memory
6         type: project
7         originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8     ---
9
10    # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12    > **Scope of this file = LIVE only:** the current handoff + *today's* checkpoints + open
PRs/decisions. Keep it read-first-short (?1 page). Older checkpoints are archived in [[current-
state-archive]] ? when today's checkpoints age out (next clean-slate day), move them there,
don't let them pile up here.
13
14    **Checkpoint:** 2026-06-22 (**`worker train` dated-label join FIX [#319] + doc [#321]
MERGED; eval litmus re-grounded [#324] OPEN for owner**). master `db74f9b`. (1) **[#319]** ?
`cmd_train --trajectory-source detector` couldn't match canonical `labels/<YYYY-MM-
DD>.<name>.rallies.csv` to `videos/<name>.mp4` (date prefix ? every clip skipped ? `<2 videos`
abort; `DATA_IN_GCS $7b#1`). Fix = new helper `_label_clip_name()` in
`backend/worker/__main__.py` strips the `.rallies.csv` suffix + a leading
`^\\d{4}-\\d{2}-\\d{2}`; +6 tests (worker cov 82?84%). (2) **[#321]** ? flipped `DATA_IN_GCS
$7b#1` to ? RESOLVED + $8 tracker row. Both squash-merged (3-dim adversarial verify workflow:
all_ok). (3) **master "broken" locally on `test_served_gate_a_regression`** (`mean_seg_off`
1.16?1.36) ? NOT a code regression: the **golden corpus grew 9?15** (#316 restored the
original-6 single-court labels), see [[golden-corpus-grew-9-to-15]]. I re-measured + re-grounded
it (PR #324) but a **parallel session had already merged the IDENTICAL fix as [#322]**
(`32f4ee6`, same pins 1.36/0.79/+0.03) ? **master already fixed; my #324 closed as a duplicate**

```

(cross-validated ? independent re-measurement, same numbers). Verified `origin/master` litmus = 5 passed locally. master now `9b37269`. `[[gotcha-shared-checkout-branch-collisions]]` `[[decision-rigor-norm]]`

15

16 ****Checkpoint:**** 2026-06-22 (****NIGHTLY E2E ? Cloud Run scheduler path FIXED + cloud-validated, [#320] MERGED****). master `48f9d20`. Owner asked Option A (make `register\_scheduler.sh` actually work) + a trigger command + email setup. ****Two bugs caught by an in-cloud dry-run validation**** (build the launcher image ? temp dry-run Cloud Run job ? exec): (1) the launcher container is `COPY . /app` but Cloud Build excludes `.git` ? `launch.sh`'s `git archive` failed ? ****fixed****: `launch.sh` now tars the baked tree when no `.git` + takes the SHA from `RALLY\_GIT\_SHA` (+ a `RALLY\_DRY\_RUN` smoke mode); (2) `gcloud builds submit --tag` has ****no -f flag**** ? added `cloudbuild.launcher.yaml` + `--config`. Also: `register\_scheduler.sh` auto-grants the SA IAM (compute.instanceAdmin.v1+iam.serviceAccountUser+run.invoker) + bakes the SHA + `--run-now`; new ****setup_email.sh**** (Gmail app-pw via STDIN ? Secret Manager + SA secretAccessor). ****VALIDATED in-cloud****: image built, the container ran launch.sh (no .git ? tar, SHA=4051ed5), staged the tarball, hit "Skipping VM create" (dry-run) ? with the earlier real-L4 run, the FULL Cloud Scheduler?Cloud Run job?launch.sh?ephemeral-GPU-VM chain is proven. Temp job+image deleted; nightly bucket kept; GCP \$0. ****? Owner go-live (all from a LOCAL master checkout, NOT the droplet): `create\_nightly\_bucket.sh` (done) ? upload clips/labels to `gs://?/nightly-inputs/` ? `setup\_email.sh` ? `register\_scheduler.sh --run-now`. Start on demand: `gcloud run jobs execute nightly-regression-launcher --region asia-south1`.**** `[[gcp-vm-always-delete]]` `[[working-agreement-merging]]`

17

18 ****Checkpoint:**** 2026-06-22 (****NIGHTLY E2E ? FAITHFUL WASB run VALIDATED on a real GCP L4 GPU; follow-up [#318]****). Owner: "use GCP GPU VMs (auth/tooling available)." Spun up an ****L4**** (`rally-nightly-val`, asia-south1-b) via `create\_gpu\_vm.sh`, set up the real WASB-native conda env + weights (`gs://khelsutra-rally-corpus/models/wasb\_badminton\_best.pth.tar`), ran `nightly\_reelcount.py` with `wasb\_hybrid`/`detector\_impl=native`/`skip\_ai\_handoff=true` on `gs://?/videos/mahadevpura\_smoke\_180s.mp4`: ****22 reels, DETERMINISTIC across 3 runs, clean PASS (exit 0), faked regression caught (exit 1, `reel\_count 25?22`), cost from real uptime (\$0.069/?5.77, 352s @ \$0.71/hr)****. ****VM DELETED**** (\$0 idle, no orphan disks ? note `teardown.sh` defaults name `rally-gpu`; set `RALLY\_VM\_NAME` or `gcloud delete` directly). The real gs://+GPU+WASB run caught ****4 deploy bugs the droplet/mocks couldn't****, all fixed in ****[#318]**** (CI green, merging): (1) `gs://` fetch needs a `dst` (GCS no-passthrough) ? new `\_fetch\_to` downloads remote into the workdir (+test, also `\_load\_gts`); (2) `run\_on\_vm.sh` must export `WASB\_REPO\_DIR`+`WASB\_PYTHON` (not just `WASB\_WEIGHTS`); (3) weights default `weights/?`models/`; (4) manifest `skip_ai_handoff=true` (pure-engine, no paid Gemini ? deterministic+free). GCP auth = `gcloud` as `avi.dullu@gmail.com`/`gen-lang-client-0306026826`, GPU quota=1, weights in `models/`, clips+labels+cached-trajectories in the bucket. Suite 1413?. **\*\*? Runner+orchestration now GPU-proven; only the OWNER deploy steps (GCS upload + SMTP secret + `register_scheduler.sh`) remain to go live.\*\*** `[[gcp-vm-always-delete]]` `[[compute-stack-gcp-only]]` Owner: "use GCP GPU VMs (auth/tooling available)." Spun up an **\*\*L4\*\*** (`rally-nightly-val`, asia-south1-b) via `create_gpu_vm.sh`, set up the real WASB-native conda env + weights (`gs://khelsutra-rally-corpus/models/wasb_badminton_best.pth.tar`), ran `nightly_reelcount.py` with `wasb_hybrid`/`detector_impl=native`/`skip_ai_handoff=true` on `gs://?/videos/mahadevpura_smoke_180s.mp4`: **\*\*22 reels, DETERMINISTIC across 3 runs, clean PASS (exit 0), faked regression caught (exit 1, `reel_count 25?22`), cost from real uptime (\$0.069/?5.77, 352s @ \$0.71/hr)\*\***. **\*\*VM DELETED\*\*** (\$0 idle, no orphan disks ? note `teardown.sh` defaults name `rally-gpu`; set `RALLY_VM_NAME` or `gcloud delete` directly). The real gs://+GPU+WASB run caught **\*\*4 deploy bugs the droplet/mocks couldn't\*\***, all fixed in **\*\*[#318]\*\*** (CI green, merging): (1) `gs://` fetch needs a `dst` (GCS no-passthrough) ? new `_fetch_to` downloads remote into the workdir (+test, also `_load_gts`); (2) `run_on_vm.sh` must export `WASB_REPO_DIR`+`WASB_PYTHON` (not just `WASB_WEIGHTS`); (3) weights default `weights/?`models/`; (4) manifest `skip\_ai\_handoff=true` (pure-engine, no paid Gemini ? deterministic+free). GCP auth = `gcloud` as `avi.dullu@gmail.com`/`gen-lang-client-0306026826`, GPU quota=1, weights in `models/`, clips+labels+cached-trajectories in the bucket. Suite 1413?. ****? Runner+orchestration now GPU-proven; only the OWNER deploy steps (GCS upload + SMTP secret + `register\_scheduler.sh`) remain to go live.**** `[[gcp-vm-always-delete]]` `[[compute-stack-gcp-only]]`

19

20 ****Checkpoint:**** 2026-06-22 (****NIGHTLY E2E ENGINE REGRESSION ? #315 MERGED + DROPLET-VALIDATED****). master `fa2342c`. Owner asked for a ***pure-engine*** nightly: run the configured+checkpointed pipeline on 2-3 short clips, assert the ****reel count is unchanged**** + quality + ****cost****, ****email**** the report, on a ****GCP GPU VM**** (not local/CI ? GitHub CI can't run WASB). Built `scripts/nightly\_reelcount.py` (reel-count=`len(play\_segments)` EXACT + R2

quality + `backend.billing` cost USD/INR; exit 0/1/2/3/4; re-run-once guard) +
`scripts/nightly\_email.py` (SMTP via Secret Manager/env, verified-TLS, graceful skip) +
`deploy/gcp/nightly\_regression/` (self-deleting `run\_on\_vm.sh` + driver-agnostic `launch.sh` +
`register\_scheduler.sh` Cloud Scheduler?Cloud Run job + `create\_nightly\_bucket.sh`) +
`eval\_baselines/reelcount\_manifest.json` (extensible) + tracked doc
`docs/NIGHTLY\_ENGINE\_REGRESSION.md`. **Complements #202** (cached-trajectory CPU re-score) ?
this runs the REAL model E2E. **TWO-BUCKET model (owner-agreed):** ephemeral `gs://khelsutra-
rally-nightly` (30-day TTL: reports + repo tarball + build staging) vs permanent
`gs://khelsutra-rally-corpus` (clips/labels/weights, no TTL) ? a bucket-wide TTL on the
dedicated bucket can't touch golden data. Owner asked for a *pure-engine* nightly: run the
configured+checkpointed pipeline on 2-3 short clips, assert the **reel count is unchanged** +
quality + **cost**, **email** the report, on a **GCP GPU VM** (not local/CI ? GitHub CI can't
run WASB). Built `scripts/nightly\_reelcount.py` (reel-count=`len(play\_segments)` EXACT + R2
quality + `backend.billing` cost USD/INR; exit 0/1/2/3/4; re-run-once guard) +
`scripts/nightly\_email.py` (SMTP via Secret Manager/env, verified-TLS, graceful skip) +
`deploy/gcp/nightly\_regression/` (self-deleting `run\_on\_vm.sh` + driver-agnostic `launch.sh` +
`register\_scheduler.sh` Cloud Scheduler?Cloud Run job + `create\_nightly\_bucket.sh`) +
`eval\_baselines/reelcount\_manifest.json` (extensible) + tracked doc
`docs/NIGHTLY\_ENGINE\_REGRESSION.md`. **Complements #202** (cached-trajectory CPU re-score) ?
this runs the REAL model E2E. **TWO-BUCKET model (owner-agreed):** ephemeral `gs://khelsutra-
rally-nightly` (30-day TTL: reports + repo tarball + build staging) vs permanent
`gs://khelsutra-rally-corpus` (clips/labels/weights, no TTL) ? a bucket-wide TTL on the
dedicated bucket can't touch golden data.

21 - **RIGOROUSLY TESTED on the D0 droplet** (`claude-do-rally-test` 168.144.126.176,
CPU/motion segmenter ? validates the runner+alert+cost, not WASB-on-GPU): ran a couple of times
clean (deterministic reels=2, exit 0); **faked TWO issues ? alert caught both** ? behaviour
regression (`dead\_time\_merge\_gap` 2?5 ? reels 2?1, exit 1) + a missing clip (pipeline error,
exit 1, run completes). **The real-machine run caught 2 bugs the mocked tests couldn't** (now
fixed + regression-tested): (1) missing `add\_video` ? FK fail ? reels silently 0; (2) one bad
clip crashed the whole run (exit 2) since `storage.fetch` passes through missing local paths ?
now per-clip isolated. Droplet restored to as-found (`~/rally` removed; `/srv/khelsutra-site`
untouched). Adversarially reviewed (45-agent workflow, 41?29 findings, HIGH/MED fixed: SMTP
cert-verify, re-run-fail?inconclusive, VM self-delete robustness). Suite 1387?, ruff+mypy+leak-
scan clean, all 10 CI green.

22 - ? **OWNER to DEPLOY (the nightly code is merged; going live is owner-gated ? needs GCP
secrets/quota I lack):** (1) `bash deploy/gcp/nightly\_regression/create\_nightly\_bucket.sh`; (2)
upload 2-3 short owned clips+labels to `gs://khelsutra-rally-corpus/nightly-inputs/` + WASB
weights to `weights/`; (3) SMTP app-password ? Secret Manager; (4) `launch.sh` dry-run (the
FAITHFUL WASB-on-GPU run + the first `--update-baseline` freeze) ? commit
`reelcount\_baseline.json`; (5) `register\_scheduler.sh`. Two open calls (defaulted, easy to
change): **driver** (Cloud Run job vs cron vs Claude routine) + **which clips**. Runbook:
`deploy/gcp/nightly\_regression/README.md`. [[eval-dual-set-regression]] [[compute-stack-gcp-
only]] [[gcp-vm-always-delete]]

23

24 **Checkpoint:** 2026-06-22 (GOLDEN-FIXTURES **P3 ? #317 MERGED**; the golden-fixtures
project is now IMPLEMENTATION-COMPLETE). master `7a40434`. Owner cleared ALL owned golden videos
(owned+minors-free) + re-confirmed **option (a)** ? minted **6 curated, trimmed (?12k-frame ?
few-min), de-attributed REAL Tier-0 fixtures `fx\_r01`?`fx\_r06`** (single / multi-court?2 /
doubles?2 / continuously-active blob; both 29.97/59.94 fps; **~1.9 MB total**) via a NEW
`freeze\_real\_fixture --max-frames` trim. De-attributed (opaque slug, time-origin reset,
scenario-only labels, **NO** source name/path/video_id ? grep-verified + leak-scan clean); the
characterization + 3-OS cross-OS gates now replay REAL data (F1 0.0?0.59 = real CONTROL-path,
behaviour-locking NOT field quality; hard multi-court/blob collapse to a mega-window F1=0 =
the known over-merge, now frozen). DATA found on this box: `C:\Users\avidu\Projects\Annotat
ion Setup\Collect\{Trajectories,Golden Labelled}` + repo `output*.rallies.csv` (manifest
`output/human\_love\_manifest.json` = 15 videos); F: has the source GoPro videos (not needed).
Surrogate?source map kept OFF-git (given to owner in chat: fx_r01=mahadevpura_single,
r02=kushagra, r03=mahadevpura_1, r04=targetest_doubles, r05=GX010128, r06=Boxhill_Doubles).
Suite **1390?/7s**, cov **85.27%**, all CI green incl. leak-scan + 3-OS. **Golden-fixtures
tracker now M1/M2/M3/MX/P1/P2/DOC/P0/P3 ALL ?** (P0=safe-scope; the FUNCTIONAL video-name rename
DEFERRED to a coordinated pre-open-source effort; M4 quality-floor + 01 protobuf + 02
Tier-1-key-gated remain OPTIONAL). Track PRs = #186/#189/#191/#192/#193/#194 + #314 + #317, all
merged+audited. [[golden-set-vendoring]] [[gotcha-shared-checkout-branch-collisions]]

25

26 **Checkpoint:** 2026-06-22 (**khelsutra brand-repo HYGIENE + DOC-FRESHNESS SWEEP ? 2 PRs

merged, clean slate**). Session = "sync to remote head + low-hanging fruit + quick doc sweep." khelsutra was at master `e2570e4` (#93). ? **[#94] hygiene** ? `.gitignore` now covers `\*.db` (runtime `site/access.db`) + `.claude/` (machine-specific; both were untracked ? committable) + corrected a stale NEXT_STEPS PR count; **caught my own `.gitignore` inline-comment bug via `git check-ignore` BEFORE commit** (`.gitignore` has no trailing-comment syntax ? comments must be own-line). ? **[#95] doc-freshness sweep** ? a **15-agent AUDIT** (one verifier/doc, every falsifiable claim checked at file:line incl. cross-repo vs engine `origin/master`) found **44 stale findings** (the pattern: docs froze at their authoring-PR while the SPA / engine API / localization / alpha-deploy shipped past them) ? a **12-agent FIXER fan-out** applied **63 surgical edits across 13 docs** ; I **independently re-verified every load-bearing engine claim** on engine `origin/master` before merging (`.api/{health:308,capabilities:340,videos:374}` + compile `download\_url:768` + `native\_wasb\_runner.py` on master + `auth.py::CF\_ACCESS\_HEADER` + `\_allowed\_hosts\_from\_env` REPLACE-semantics ? all real; brittle engine line-anchors ? symbol refs like `main.py::compile\_highlights`). Honesty held (P6 stays `` engine-done/SPA-pending, native-WASB "not production-verified", footage shots owner-gated, historical changelogs preserved; the 2 `fresh` docs untouched). Also pruned **6 stale merged local branches**. **All CI green (site+web+layout); 0 open PRs; only `master` locally; khelsutra master now `7dc7de5`.** *(Brand/site track ? distinct from the serving/packaging/UX/golden threads below.)* [[lesson-verify-engine-surface-before-scoping]] [[working-agreement-merging]] [[doc-status-register]]

27

28 ****Checkpoint:**** 2026-06-22 (GOLDEN-FIXTURES **P0 ? #314 MERGED**, master `5c2eec8`; a SEPARATE thread from packaging/UX). Owner: "all owned golden videos are owned+minor-free; pick (a); make P0 now." On scouting, P0 was bigger/riskier than the audit's "4 files": the video names are FUNCTIONAL (eval_baselines keys ? training floor + the gitignored `output/human\_lovo\_manifest.json` + data files; `MULTICOURT\_CLIPS`; hot in QUALITY_ITERATIONS/the quality track) ? surfaced it ? owner picked ****SAFE SCOPE****. Shipped: scrubbed personal `C:\Users\avidu` paths from 3 code docstrings + ****tools/leak_scan.py**** (structural PII guard: 32-hex MD5 + personal paths over backend/training/scripts/tools/eval_baselines; tests/ excluded; optional gitignored **.leakscan_private`** for local name scanning) + CI `leak-scan` job + 10 tests. Clean scan; suite **1365?7s**, cov **85.27%**, all CI green. ****DEFERRED (safe-scope):**** the functional video-name rename + git-history rewrite (need manifest/data/training-floor coordination + quality-track pause; do before open-sourcing). ?? ****NEXT = P3 real Tier-0 fixtures**** ? owner APPROVED (all owned cleared + ****a**** commit de-attributed per-frame CSV); needs the golden ****trajectory CSVs + label CSVs**** (`.Annotation Setup/Collect/Trajectories/` + `Golden Labelled/` / OneDrive `golden-data/`) ? I locate them or owner points me, then **--freeze-real`** per video. Golden-fixtures track = #186/#189/#191/#192/#193/#194 + #314, all merged+audited. [[golden-set-vendoring]]

29

30 ****Checkpoint:**** 2026-06-22 (? ****GCS = corpus source-of-truth ? `docs/DATA\_IN\_GCS.md` + Phase-1 metadata MIGRATED ? PR [#316](https://github.com/Khelsutra/badminton-highlight-indexer/pull/316) OPEN; F: label backup done****). Owner goal: run ALL train+infer from the cloud with ALL data in `gs://khelsutra-rally-corpus` ? remove the local box as a blocker; document where data lives so other sessions pick up from the bucket. [[gcs-data-source-of-truth]]

31 - ? ****Authoritative map `docs/DATA\_IN\_GCS.md`**** (indexed in docs/README.md; branch `work/golden-corpus-hygiene`, commit `8884c9f`): bucket layout . current contents + ****drift**** . how `backend.worker infer|train` consume `gs://` . the gap . phased migration plan + DoD. The ****cloud engine already exists**** (archived DECOUPLED_COMPUTE: `worker infer --video-uri` / `worker train --corpus-uri`, URI backends in `backend/storage/ref.py`, `deploy/gcp/`) ? the doc records WHERE bytes are, it does NOT redesign.

32 - ? ****Gap measured:**** corpus ***metadata*** is ****~25 MB & entirely local**** ? 20 trajectory CSVs (17.5 MB at `C:\?Collect\Trajectories\`), `output/human\_lovo\_manifest.json` (C:-absolute paths), most labels. Bucket has only 3 videos, ****no `trajectories/`**, inconsistent `\_proxy` label naming. Only the VIDEOS are large ? migrate ****Drive?bucket** from a cloud box**, not via the local box.**

33 - ? ****F: backup done:**** the 7 single-copy new-clip labels hash-verified into `F:[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden Labelled\`. ? F: root was RENAMED `GoPro Hero Black 12`?`[Khelsutra] GoPro Hero Black 12` between turns ? live proof of the path-fragility GCS fixes.

34 - ? ****Phase-1 DONE (uploaded to the bucket):**** `trajectories/<name>.csv` (15), `labels/<YYYY-MM-DD>\_<name>.rallies.csv` (15), `manifests/golden.json` (gs://-pathed, 15 clips). ****Naming LOCKED (owner):**** manifest name + authoring-date prefix = earliest mtime across C:F: copies (robust to copy-resets). ? ****These golden traj+label CSVs are exactly what the GOLDEN-FIXTURES P3 thread (#314 checkpoint above) was hunting for ? now in GCS.****

35 - ? ****Owner decisions RESOLVED (DATA_IN_GCS \$7):**** naming=date-prefixed manifest name;

cache trajectories=yes; GCS eval manifest=yes. ****Still OPEN:**** (4) originals migration scope. ? ****#1 cleanup DONE:**** the 7 stale `\_proxy`/bare label objects were MD5-verified byte-identical dupes + deleted ? `labels/` = exactly 15 canonical.

36 - ? ****Phase-2 SET UP + demo'd:**** per-clip video? `videos/<name>.mp4` mapping (7 Drive proxies in `[Done] Video N` + 8 large F: originals) + `scratch/copy\_videos\_to\_gcs.ps1` (skip-existing; temp-stages bracketed paths) + the decoupled `rclone`-on-cloud path. Proved E2E: `videos/mahadevpura\_1.mp4` (523MB) copied Drive2bucket. ****14/15 videos remain.**** GOTCHA: `gcloud storage cp` globs `[...]` ? `[Done]`/`[Khelsutra]` paths need temp-staging (rclone avoids it).

37 - ? ****4-lens adversarial review (`wf\_18dcd7ff`)** FOUND a real BLOCKER (code-verified): ****date-prefixed labels break `worker train --trajectory-source detector` ? `cmd\_train` joins the FULL label stem (`2026-06-22\_GX020094`) vs bare video stems (`backend/worker/\_\_\_main\_\_.py:384` vs `:378/:387`) ? every clip skipped. Fix = strip the date prefix in cmd_train (chip** ****task_53b5e190**, own PR+test). synth-train + eval unaffected. Also: Phase-3 eval-on-GCS is** ****net-new code**** (eval is local-FS only, no gs:// fetch); `manifests/golden.json` consumed by zero code yet. (Risk-agent hallucinated 2 nonexistent scripts ? disregarded per `[[working-agreement-attribution]]`.)

38 - ? Branch `work/golden-corpus-hygiene` off origin/master (`689c738`), ****4 docs(data)** commits, pushed ? PR `#316`(<https://github.com/Khelsutra/badminton-highlight-indexer/pull/316>) OPEN for review** (no self-merge; docs-only). Serving WIP `6fdd1d4` + `scratch/{PACKAGING\_PLAN.serving-wip.md,migrate\_golden\_to\_gcs.py,copy\_videos\_to\_gcs.ps1}` preserved (gitignored ? not in PR; promote to scripts/ next). Prior #236 served-Gate-A MERGED 2026-06-19; re-measuring the litmus on the GCS 15-clip set is a `[[launch-report-convention]]` dual-set event. ****? NEXT-SESSION ORDER:**** (1) merge #316; (2) worker date-strip fix (chip task_53b5e190); (3) Phase-2 videos via rclone-on-cloud, proxies-first; (4) Phase-3 eval-fetch code + #236 re-measure; (5) unblock the golden-fixtures thread (CSVs already in GCS); (6) GPU quota + cloud secrets (owner) + promote scratch scripts. `[[compute-stack-gcp-only]]` `[[gotcha-shared-checkout-branch-collisions]]` `[[lesson-verify-engine-surface-before-scoping]]`

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40 ****Checkpoint:**** 2026-06-22 (****PACKAGING P2b + P2c ? BOTH MERGED; v1 LIGHT engine-complete + wizard'd****). Same session as the P3 checkpoint below; owner: "complete P2b-install" ? "start P2c wizard SPA now" ? "merge #310 now". ****? P2b-install ? engine** `#310`(<https://github.com/Khelsutra/badminton-highlight-indexer/pull/310>) MERGED** (master `f8b2c3b`): `write\_light\_default\_config` seeds a truly-local + ****`motion`**** (gate ****08**** = resolved) torch-free default `config.json` on the first `indexer --app` (minimal, profile-driven, NEVER overwrites; placed before `load\_app\_config`, scoped to `--app`); `scripts/install.py` (stdlib, cross-platform ? `uv tool install` else pip + a per-OS launcher shortcut that ****pins `RALLY\_APP\_HOME`**** ? load-bearing since `load\_dotenv` runs at import + writes an uninstall manifest) + `scripts/uninstall.py` (manifest-driven; KEEPS user data unless `--purge`) + README. ****12-agent adversarial review (`wf\_2be29ecf`)** folded: ****MAJOR PATH-robustness (`uv tool update-shell` + resolve+embed the abs `indexer` path + warn) + uninstall** ****containment guard**** (`\_within(path, app\_home)` refuses rmtree outside app-home ? a corrupted/foreign manifest can't nuke an arbitrary dir) + `--dry-run` fidelity + tests for the destructive path. Suite ****1352?**, ruff+mypy clean, cov **85.17%****. ? ****CI was billing-blocked for hours**** ? the engine repo bills under the ****`Khelsutra` ORG**** (separate from the personal account); owner fixed the ORG billing ? CI green ? merged. ****? P2c first-run wizard ? khelsutra** `#88`(<https://github.com/avidullu/khelsutra/pull/88>) MERGED** (master `7ae9cee`): `web/src/components/SetupWizard.tsx` ? 3 steps (AI-key *guided-setup* since no key-write endpoint yet / compute from `GET /api/capabilities` / videos) rendered before `IngestPanel`, shows once per browser (localStorage), 6 langs (en/kn/hi/es/da/id, brand stays Latin "KheLSutra" per #77), `getCapabilities()`+`EngineCapabilities`. ****223 web tests**** (+ the i18n parity/brand/no-bare guards) + ****browser-verified e2e against a live local engine**** (all 3 steps render real caps: ffmpeg+GPU detected; offline degrades gracefully). ****13-agent review** (`wf\_191a9b86`) folded: ****MAJOR a11y** (the "Get a free Gemini key" link was below the 44px tap floor ? `min-h-11`) + loading-window finish-button (gated to `offline` only) + `role=dialog`?`region` + kn brand spacing + the no-bare-strings guard now covers SetupWizard. ? a sibling reorg-merged ****#87**** (es/da/id catalogs) mid-build ? rebased (auto-merged clean) + re-synced `site/public/locales` via `npm run copy:locales` (a `locales-drift.test` enforces the mirror). Both built in ****isolated worktrees off `origin/master`**** (since removed); sibling worktrees untouched. ****Also this session (owner asks): deployed the real Gemini key (from `G:\My Drive\gemini.api\_key.txt`) to the droplet `/etc/khelsutra/engine.env` + validated live (55 models, \$0) + ran the FULL suite on the droplet (1313 passed); injecting the key auto-started** ****`khelsutra-engine` (enabled unit) on loopback ? owner: "keep it running" (NOT publicly exposed ?** ****`api./app.khelsutra.guru` have no public DNS route yet).**** ? ****LOOP CLOSED ? engine** `#312`(<https://github.com/Khelsutra/badminton-highlight-indexer/pull/312>) MERGED** (master `293e5c0`): re-bundled `backend/ui/` from khelsutra@7ae9cee via `scripts/bundle\_ui.sh` so

****indexer --app` now SERVES the wizard\*\*** (`/api/version` spa\_commit=7ae9cee; `/` serves `index-GSXqL6iY.js`; live-boot proven). Pure vendored-asset refresh; 53 UI/skew contract tests + full suite 1355? + cov 85.25%. ****? v1 LIGHT is now feature-complete ? only OWNER steps remain: a clean-Windows/macOS-box acceptance run + ffmpeg-acquisition UX**** (winget/brew/apt is still a manual dep). North-star (native-WASB-in-default-download P3b/P4a + training P4b + annotation P4c) stays future + counsel(03)/owner-GPU-gated. ?? ****THIS SESSION TALLY: 3 PRs merged (#310 P2b-install, khelsutra #88 P2c wizard, #312 bundle) + P3 GPU validation (\$0) + droplet key/suite. 0 open PRs, 0 VMs, all worktrees/branches cleaned.**** North-star (native-WASB-in-default-download P3b/P4a + training P4b + annotation P4c) stays future + counsel(03)/owner-GPU-gated. [[packaging-app-plan]] [[next-session-packaging-app]] [[gotcha-shared-checkout-branch-collisions]] [[working-agreement-merging]]

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42 ****Checkpoint:**** 2026-06-22 (****PACKAGING P3 ? GPU validation of the LIGHT-tier WHEEL on a real L4, end-to-end, \$0****). Owner greenlit the spend (AskUserQuestion). Provisioned an ****L4**** `rally-gpu` (`g2-standard-4`, ****asia-south1-a**** Mumbai) ? shipped the `origin/master` archive ? ****built the wheel ON the box + installed it NON-editable**** (`badminton\_highlight\_indexer-0.1.0-py3-none-any.whl`, base LIGHT, no extras) ? ****import backend.main` works with `torch installed = False` (the P2e torch-free promise PROVEN on a GPU box; every backend `import torch` is function-local) ? built the \*\*SEPARATE\*\* WASB conda env (py3.8/torch1.11) reached via `\$WASB_PYTHON` (the native path is a pure `subprocess` shell-out ? `native_wasb_runner.py`).** Then booted the **\*\*packaged launcher `indexer --app` from a NON-repo cwd\*\*** (app-home isolated state) ? `/api/health` ok . `/api/version` reports the bundled SPA (khelsutra@46e0244, content\_hash ac2a3d6) . `/api/capabilities` = `ffmpeg:true,ffprobe:true / nvidia_smi_present:true / wasb_weights_present:true / gemini_key_present:false` . **\*\*bundled SPA served at `/*` ? drove the REAL API `POST /api/process(wasb_hybrid) ? poll /api/jobs ? /api/query ? /api/compile ? /api/reels` with the tuned 22-rally recipe (`detector_impl=native` + `skip_ai_handoff=true` + `min_shot_count=0`) ? **\*\*`RALLIES_DETECTED=22`\*\*** (reproduced the golden result via the PACKAGED stack, not `scripts/`) ? **\*\*`p3_highlights.mp4` 73.7 MB\*\*** watermarked reel ("Dynamic highlight reel saved", all 22 segments). **\*\*Proof preserved to `gs://khelsutra-rally-corpus/highlights/p3_{packaging_highlights.mp4,run.log,app.log}`\*\*** (survives teardown). **\*\*VM DELETED + audited `0 instances / 0 disks ? \$0`\*\*** (~14 min ? **\*\*\$0.17\*\***, far under the \$100/??10k cap). Ops-Agent auto-installed + verified (NVML receiver healthy, GPU graphs live; boot-time duplicate-write export errors are transient, settled). ? The ONLY hiccup was cosmetic: my driver's `urlopen(timeout=60)` tripped on the synchronous 22-clip `/api/compile` (~65s) ? ****client-side only; the server completed the stitch**** (sync endpoint runs to completion regardless of client disconnect). The packaging insight WORTH KEEPING: ****a plain torch-free LIGHT wheel + a separately-provisioned WASB conda env = full GPU-quality reel**** ? the dual-env architecture (PACKAGING_PLAN \$4.4) is real on hardware. ? ****P3 DONE**** ? next packaging items = ****P2b-install**** then ****P2c wizard**** (per [[next-session-packaging-app]] / [[packaging-app-plan]] \$7). Screencast = owner records from the preserved reel/logs (or a local LIGHT-tier `indexer --app` browser capture, \$0). Built in a detached worktree off `origin/master` (since removed); the serving checkout is untouched. [[gcp-vm-always-delete]] [[gcp-ops-agent-enable]] [[packaging-app-plan]]**

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44 ****Checkpoint:**** 2026-06-22 (****ALPHA-DEPLOY runbook DE-RISKED ? verified vs engine `origin/master` + repro'd; 2 khelsutra PRs****). Owner: "make T2 deploy real" + "make the PRs, run tests on droplet/GPU within the bill cap." ****T2 itself is owner-gated INFRA**** (CF Pages/Access/`cloudflared` tunnel + GEMINI_API_KEY rotation) ? nothing for Claude to ***execute***; the high-leverage move = ****adversarially verify `deploy/DEPLOY\_ALPHA.md` against the REAL deploy code**** (engine `origin/master`, NOT this stale serving checkout) via an 11-agent verification workflow + a deploy-readiness skeptic, then ****repro the failures hermetically**** (\$0 ? droplet/GPU NOT needed, so the bill stayed ~\$0). ****Found 2 deploy-BREAKERS the original kit shipped:**** (1) ****`RALLY\_ALLOWED\_HOSTS` missing**** ? `edge\_auth` ON ? engine binds `0.0.0.0` + Host-guard stays loopback-only default ? ****400s EVERY tunneled request**** (boots "healthy" then dead; only a WARN). ****Independently CORRECTED the skeptic:**** ****`RALLY\_BIND\_HOST=127.0.0.1` does NOT fix it (the Host-allowlist is decoupled from the bind ? read `backend/main.py:88-128`); `RALLY\_ALLOWED\_HOSTS="api.khelsutra.guru"` is the load-bearing fix. **Repro (real Starlette `TrustedHostMiddleware`, exact engine allowlist): `/api/health` Host=`api.khelsutra.guru` ? 400 unset / 200 set.**** (2) ****gs:// ingest REJECTED**** by the required `allowed\_video\_root` jail (repro'd 400 for `gs:///s3:///gdrive:///` ; pinned by `test\_process\_hosted\_mode\_rejects\_nonlocal\_uri`) ? \$6/\$7 over-promised it; alpha path = ****browser upload****. ****2 PRs MERGED (CI green, squash):**** ****[khelsutra #78](https://github.com/avidullu/khelsutra/pull/78)**** (runbook fixes: +2 env vars, gs:// drop, 401-not-500, config.json `--setup` location, ~95MB chunk ? the runbook is now correct; the owner just deploys from it) + ****[#79](https://github.com/avidullu/khelsutra/pull/79)**** (stale \$7

tracker prose). **Master moved since last handoff: A3(#74)/A6(#75)/A7(#76) MERGED + #77 kn/hi console native-review pass** ? A-workstream effectively DONE bar the es/da/id+watermark native review. Verified in an **isolated `origin/master` worktree (since removed)**; serving checkout (`serving/exposure-safety-preflight`) untouched. **? Engine-side thread to route to the SERVING session:** consider escalating the Host-allowlist WARN?ERROR (the one exposure misconfig that doesn't fail-fast yet 100%-breaks); #78 unblocks the deploy regardless. **Follow-up ? owner started the CF setup:** confirmed **NO \$25/Pro plan needed** (Pages + Tunnel + Access all free, Access ?50 users; enabling Zero Trust may ask for a card ? pick the **Free** plan). **Team domain LOCKED = `https://khelsutra.cloudflareaccess.com`** (`cf\_team\_domain`). Owner does the key rotation + firewall (single user). Shipped the **tailored droplet kit ? [khelsutra #81](https://github.com/avidullu/khelsutra/pull/81) MERGED** (`deploy/droplet/`: `config.json`/`engine.env`/`khelsutra-engine.service`/`README` pre-filled for `app.`/`api.khelsutra.guru` + `avi.dullu@gmail.com` + `/srv/khelsutra` jail; declarative templates, JSON-validated, NOT yet run on the box). **? DEPLOYED + LIVE-VERIFIED ON THE DROPLET (owner: "ssh in with the previous-session key + take control"):** SSH works via **`~/.ssh/khelsutra\_droplet`** ? `root@claudeddo-rally-test` (Ubuntu 24.04, 2-core; ALSO runs `khelsutra-site.service` on :8787 ? left untouched; :8000 free; no git creds on box). Pushed engine `origin/master` (git-archive?tar over SSH, no box creds), `/opt/khelsutra/venv` + `pip install -e .[auth]` (**no torch in core deps**; cv2 needs `libgl1`+`libgl1.0-0`+`libgl1.0-0-t64`). Wrote `/srv/khelsutra/config.json` (AUD placeholder), `/etc/khelsutra/engine.env` (GEMINI placeholder + `RALLY\_ALLOWED\_HOSTS=api.khelsutra.guru,localhost,127.0.0.1` + `RALLY\_BIND\_HOST=127.0.0.1` + CORS), `khelsutra-engine.service` (enabled, NOT started ? correctly **fail-fast until AUD=key**). cloudflared 2026.6.1 installed. **LIVE-PROVEN on the box (4/4):** boot-posture fail-fast ? exit 1 `refusing to start`; Host-guard `api.khelsutra.guru` ? **400 unset / 200 with RALLY_ALLOWED_HOSTS** (the #78 fix observed on the deploy target); real exposed config boots **`127.0.0.1:8000`** (RALLY_BIND_HOST overrides the edge_auth 0.0.0.0). **REMAINING = owner CF-account + key ONLY:** (1) rotated GEMINI key ? env; (2) create CF Access app ? paste AUD into config.json ? `systemctl start khelsutra-engine`; (3) CF Pages for `app.`; (4) `cloudflared tunnel login` (browser) + create/route. **OR** give me a scoped **CF API token + account-id + the rotated key** and I'll do Access/Pages/tunnel via API + run the full upload?reel CUJ. ? `StandaloneUser` Windows path = just different config strings later (nothing hardcoded).

122. assistant (2026-06-22T13:26:48.935Z)

current-state.md already has a sibling-written checkpoint (line 14) covering the worker fix + gate-A race (it records my #322 winning and their #324 closed as duplicate), but it's missing #323 (backlog), the #316/#320 merge confirmation, and the combined-master verification. I'll prepend an authoritative consolidating checkpoint and leave the sibling's intact below it.

123. user (2026-06-22T13:26:58.623Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

124. assistant (2026-06-22T13:27:11.781Z)

The partial (truncated) read didn't register the file as read for editing. Let me read the exact line region, then edit.

125. user (2026-06-22T13:27:13.900Z)

14 **Checkpoint:** 2026-06-22 (**`worker train` dated-label join FIX [#319] + doc [#321] MERGED; eval litmus re-grounded [#324] OPEN for owner**). master `db74f9b`. (1) **[#319]** ? `cmd\_train --trajectory-source detector` couldn't match canonical `labels/<YYYY-MM-DD>\_<name>.rallies.csv` to `videos/<name>.mp4` (date prefix ? every clip skipped ? `<2 videos` abort; `DATA\_IN\_GCS \$7b#1`). Fix = new helper `\_label\_clip\_name()` in `backend/worker/\_\_\_main\_\_.py` strips the `.rallies.csv`/`.csv` suffix + a leading `^\\d{4}-\\d{2}-\\d{2}\_`; +6 tests (worker cov 82?84%). (2) **[#321]** ? flipped `DATA\_IN\_GCS \$7b#1` to ? RESOLVED + \$8 tracker row. Both squash-merged (3-dim adversarial verify workflow: all_ok). (3) **master "broken" locally on `test\_served\_gate\_a\_regression`** (`mean\_seg\_off` 1.16?1.36) ? NOT a code regression: the **golden corpus grew 9?15** (#316 restored the original-6 single-court labels), see [[golden-corpus-grew-9-to-15]]. I re-measured + re-grounded it (PR #324) but a **parallel session had already merged the IDENTICAL fix as [#322]** (`32f4ee6`, same pins 1.36/0.79/+0.03) ? **master already fixed; my #324 closed as a duplicate** (cross-validated ? independent re-measurement, same numbers). Verified `origin/master` litmus =

5 passed locally. master now `9b37269`. [[gotcha-shared-checkout-branch-collisions]] [[decision-rigor-norm]]

126. user (2026-06-22T13:27:32.006Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

127. assistant (2026-06-22T13:27:51.613Z)

current-state.md done. Now the handoff ? refresh the "Last updated" header and the stale master SHA in "You are here", keeping the still-valid serving/alpha plan intact below.

128. user (2026-06-22T13:27:56.069Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\session-handoff.md has been updated successfully. (file state is current in your context ? no need to Read it back)

129. user (2026-06-22T13:28:06.009Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\session-handoff.md has been updated successfully. (file state is current in your context ? no need to Read it back)

130. assistant (2026-06-22T13:28:26.880Z)

Done ? both living memory files are updated.

Memory updated

session-handoff.md

- **"Last updated"** ? 2026-06-22, summarizing this merge-sweep session (the 6 PRs now on master, `9b37269`, 1428?, 0 open PRs, docs-archive owner-gated), with a pointer to `current-state` for detail. The prior 2026-06-21 ALPHA-SHAREABLE narrative is preserved (demoted to "Prior").
- **"? You are here"** ? added a fresh 2026-06-22 block (current master SHA, ~1428 tests, 0 open engine PRs, the concurrency flag) above the still-valid 2026-06-21 serving/alpha plan, which I left unchanged since it remains the forward plan.

current-state.md

- Prepended an authoritative 2026-06-22 checkpoint consolidating the whole session: the 3 ready-green merges (#316?#319?#320 ordering rationale), #322 gate-A re-ground (with the corpus 9?15 drift cause), #323 backlog, the "no archival candidate" finding, and the **combined-master** verification (1428? with the corpus). I left the sibling-written checkpoint below it intact (it correctly records their #324 being closed as a duplicate of my #322).

MEMORY.md index ? already current ([[golden-corpus-grew-9-to-15]] already references "#322 (merged)"), so no change needed.

One thing to trim

When you clean up concurrency: the main checkout is currently sitting on a sibling's branch **fix/served-gate-a-regrounded-15clip** (the redundant parallel gate-A re-grounding ? its PR #324 was closed as a duplicate of the merged #322). It's safe to discard. The other ~10 worktrees under `.claude/worktrees/` are other live sessions ? your call which to keep. All of **my** worktrees from this session are already removed, and there are **0** open PRs.